# MODULAR SENSOR CONCEPTS AND CONFIGURATION TRANSCRIPT


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# L00 - INTRODUCTION AND AGENDA

## L00A-PRES: Modular Sensor Module Lessons Agenda (08m:52s)

**00:00-01:00**
Hi everyone. Welcome to the *Modular Sensor Concepts and Configuration* module. I'm Paul Levasseur of the Stellar Cyber Enablement and Training teams. The lessons in this module cover concepts, configuration, examples, and demonstrations of how the Modular Sensor and its features are used in the Stellar Cyber Open XDR architecture. Throughout this module there'll be lots of new concepts and terms that you may not be familiar with. This is definitely okay and expected. You may have even noticed I've said something similar to this before in other modules. As you progress through the Stellar Cyber Open XDR training curriculum, with repetition, iteration, and examples, you'll be able to understand and apply what you have learned in becoming a cybersecurity superstar. Thank you.

Welcome to the *Stellar Cyber Modular Sensor Concepts and Configuration* module. Here is an agenda of the lessons available in this module.

**01:00-02:00**
In the *Stellar Cyber Modular Sensors Concepts and Configuration* module, the following lessons are covered.

*Modular Sensors Introduction*. This lesson explains how Modular Sensors are deployed within the Stellar Cyber Open XDR architecture as well as the types of Modular Sensors deployed as hardware or virtual appliances.

*Essential Terminology*. This lesson gives definitions of the key terms used within the Modular Sensor Concepts and Configuration training module.

*Collecting Network Traffic*. This lesson explains the flows of network traffic and how Modular Sensors obtain network traffic from switch SPAN ports and TAP devices.

*Functionality Overview*. This lesson provides an overview of each of the core features that will be explained in detail throughout the *Modular Sensors Concepts and Configuration* training module. This will help you see the big picture of how the features work as a whole before diving into each individual feature.

**02:00-03:00**

*Stellar Cyber Platform Data Processing Overview*. This lesson gives an overview from the perspective of the Stellar Cyber Open XDR platform on how it processes the data received from the Modular Sensors.

*Modular Sensor Configuration with Modular Sensor Profiles*. This lesson explains the concepts and tasks needed to configure Modular Sensor profiles used to configure Modular Sensors.

*Modular Sensor Specific Configuration*. This lesson goes through the parameters that are configured on individual Modular Sensors.

*Parsers for Normalization into Interflow*. This lesson provides details for how parsers are used on Modular Sensors to Normalize metadata and telemetry into Interflow records in JSON format.

*Log Forwarder and Log Filter Configuration*. This lesson explains the process and configuration steps to ingest logs from third-party security, identity, and interest structure systems and tools as well as how to optimize ingestion by filtering logs that are not essential for security operations.

**03:00-04:00**

*Log Forwarder Example*. This lesson uses a pfSense firewall as an example to describe the higher end-to-end process of configuring log forwarding for the Modular Sensor to ingest pfSense firewall logs in order to parse and Normalize into Interflow records and send to the Stellar Cyber Open XDR platform.

*Log Source Configuration for Single TCP / UDP Port*. This lesson shows how to configure the Stellar Cyber platform to enable the Modular Sensor to receive logs from systems using a single port such as the default syslog port 514.

*Validate Log Forwarder*. This lesson provides the steps to validate the parsed and Normalized logs are being enlisted by the Modular Sensor as Interflow records in the Stellar Cyber Open XDR platform.

**04:00-05:00**

*Log Forwarder Preserving Raw Logs*. This lesson shows how to configure Modular Sensors to include the original raw log message in the Interflow record that is sent to the Stellar Cyber Open XDR platform.

*Network Traffic Analysis Details and Configuration*. This lesson explains how to configure Modular Sensors to collect network traffic off SPAN ports or TAP devices and provides details of how the Modular Sensor analyzes network traffic to create sessions and metadata and forward it as Interflow records to the Stellar Cyber Open XDR platform.

*Network Detection and Response (NDR) in the Stellar Cyber Platform.* This lesson explains some key concepts of how data collected from Modular Sensors in the Stellar Cyber platform is analyzed and utilized.

*Optimizing Network Traffic Ingestion*. This lesson provides explanations and configuration examples to optimize ingestion of network traffic by excluding network applications and services that are not essential to ingest on the Stellar Cyber platform.

**05:00-06:00**

*Example for Optimizing Network Traffic Ingestion*. This lesson gives an end-to-end example of configuring traffic filters to exclude Modular Sensor configuration management and ingestion communication to the Stellar Cyber Open XDR platform from being ingested by the platform.

*Intrusion Detection System (IDS) Details and Configuration*. This lesson provides the explanation and configuration of the IDS module on Stellar Cyber Modular Sensor.

*Intrusion Detection System (IDS) Stellar Cyber Platform Operations*. This lesson explains how to view the IDS event data received from Modular Sensors and how the Stellar Cyber Open XDR platform uses machine learning to create actionable Alerts from the IDS events events.

*Malware Sandbox Details and Configuration*. This lesson provides the concepts and details on how to configure the malware sandbox functionality on a Modular Sensor.

**06:00-07:00**

*Malware Visibility in the Stellar Cyber Platform*. This lesson explains how to view details about files collected and analyzed by the malware sandbox module in the Stellar Cyber Open XDR platform.

*Aggregator Details and Configuration*. This lesson explains the concepts and configuration of Modular Sensors to be an Aggregator as well as how to configure other Stellar Cyber sensors to use the Modular Sensor as an Aggregator.

*Tenable Nessus Scanner Details and Configuration*. This lesson provides the concepts and configuration of how to enable the ability to initiate scans from the Tenable cloud website to run all the Nessus vulnerability scanner module that can be enabled on Modular Sensors.

*Stellar Cyber Platform Connectors Example with Tenable.io Cloud*. This lesson explains how to configure a Tenable.io Connector on the Stellar Cyber platform to collect scan data and reports as well as collect assets learned by the Tenable.io vulnerability scan.

**07:00-08:00**

*Active Directory Connector Details and Configuration.* This lesson provides details and explanations how to configure an Active Directory Connector on Modular Sensors to collect Windows event logs, user behavior and computer asset information, as well as perform response actions from the Stellar Cyber platform to disable users in Active Directory.

*Active Directory Connector Collection of Asset and User Details*. This lesson shows how to view the asset and user behavior information collected by the Active Directory Connector in the Stellar Cyber Open XDR platform.

*Modular Sensor Active Directory Connector Response Actions to Disable a User*. This lesson provides details on the many ways users can be disabled from the Stellar Cyber platform in Active Directory to help mitigate threats.

**08:00-09:00**

*Buffering*. This lesson provides a summary of the buffering capabilities of the Modular Sensors that are used in an event where connectivity is temporarily lost between a Modular Sensor and the Stellar Cyber Open XDR platform, and how data is trickled back once the connection is restored.

*Modular Sensor Summary*. This lesson provides a brief summary of Modular Sensors within the Stellar Cyber Open XDR architecture and ends the *Stellar Cyber Modular Sensor Concepts and Configuration* module.

The lessons in this module provide explanations and configuration for Modular Sensors and the features that will enable you to take advantage of the capabilities the Stellar Cyber Open XDR platform has to offer for security operations. Happy learning.

This lesson is complete. Thank you for watching.

# L01 - MODULAR SENSORS INTRODUCTION

## L01A-PRES: Modular Sensors Introduction (06m:49s)

**00:00-01:00**
The *Stellar Cyber Modular Sensor Concepts and Configuration* module we'll start with the lesson, *Modular Sensors Introduction*. This lesson, *Modular Sensors Introduction,* explains how Modular Sensors are deployed within the Stellar Cyber Open XDR architecture, as well as the types of Modular Sensors deployed as hardware or virtual appliances.

The focus of the next set of lessons will be on the Stellar Cyber Modular Sensors. Looking at the Modular Sensors in the big picture, which is the big picture of the Stellar Cyber Open XDR architecture, on the left-hand side I see the Stellar Cyber Modular Sensor and how it fits in within an organization.

**01:00-02:00**
We will cover in this set of lessons the Modular Sensor features including log forwarding, network traffic analysis, IDS, Malware Sandbox, Aggregator, Tenable Nessus scanning capabilities, Connectors, and how the sensors can buffer. I'll also describe in more detail how the logs are sent to the Stellar Cyber Modular Sensors so that they can Normalize all the data and send them to the Stellar Cyber platform. As well as how the Stellar Cyber Modular Sensor can connect to Active Directory or firewalls or vulnerability scanners to grab data, but also to perform response actions. Modular Sensors as virtual appliances can be installed on VMware ESXi, AWS, Microsoft Azure, Google Cloud Platform, or GCP, HyperV, KVM, and Oracle Cloud infrastructure, or OCI.

**02:00-03:00**
Stellar Cyber Modular Sensors also come as hardware appliances. The Photon-160 can support up to 500 megabits per second and has 6-gigabit interfaces. The Photon-250 can support up to one gigabit per second and has many different interface options, essentially 6-gigabit interfaces, and two more gigabit-pluggable interfaces that you can choose the type of Connectors on them. The Photon-400 supports up to 10-gigabits per second. It has 6-gigabit RJ45 connections and two 10-gigabit SFP connections. So now for the big question, where are Modular Sensors deployed?

**03:00-04:00**
Well, if your organization has one site, then you would have a Modular Sensor for that site, or you may have multiple depending on you know how many security tools and how much

data you are trying to collect. A Modular Sensor again has different speeds. You can have 10-gig Modular Sensors, 1-gig Modular Sensors, and even smaller than that. The goal here is if you need to collect or analyze traffic that's over a gig throughout your network, then you can either install a larger Modular Sensor, or you can install multiple Modular Sensors. Each Modular Sensor could be for different networks, for example. Maybe some of the VLANs in the organization get spanned off to the Modular Sensor for analysis and for IDS detections, or you have lots of different security tools.

**04:00-05:00**

Maybe you want to determine or break up the load with some security tools going to one Modular Sensor and some going to another. It's all about speeds and feeds. Again, you could have a Modular Sensor as an Aggregator and other Modular Sensors grabbing or collecting data from other networks and then forwarding to that single Modular Sensor then to send to the Stellar Cyber platform. It all depends on what you are trying to collect. But in general, you have at lEast one Modular Sensor per site. The site could be a data center. The site could be in a cloud environment if you are using virtual hosted servers and machines and security tools. But in this picture, as an example, there are multiple sites and I have one Modular Sensor at each site.

**05:00-06:00**

I might also have Windows Server sensors and Linux Server sensors at each site as well, depending on what type of servers I have at each site. Each of these sites could be a separate customer. If you are an MSSP, you would give a customer the Modular Sensor, or if you're managing it for them, you would install a Modular Sensor for them at each site to collect data. At that point, each of these Modular Sensors would be associated with its own tenant for that specific customer. The Modular Sensors again at each site collect network traffic. They collect logs. They have Connectors to connect to local security tools, and they Normalize all that data into Interflow records, which are then sent to the Stellar Cyber platform, either over the internet

**06:00-06:50**

depending on the type of Stellar Cyber Open XDR platform deployment, or through your local network. Alright. Well, that's how Modular Sensors get deployed. The deployment options are very flexible and it's really up to your organizational needs to determine how many you need based on the type of sensors that we have. The lessons in this module go through the concepts and configuration of the core components and features of Modular Sensors. We have other lessons as well that go through the installation, including how to configure the Modular Sensors in cloud environments to get them up and running.

This lesson is complete. Thank you for watching.

# L02 - ESSENTIAL TERMINOLOGY

## L02A-PRES: Essential Terminology (12m:58s)

**00:00-01:00**

The next lesson is *Essential Terminology*. This lesson, *Essential Terminology,* gives definitions of the key terms used within the *Modular Sensor Concepts and Configuration* training module.

Throughout these lessons on Modular Sensors, there's some terminology that will keep on coming up. I want to make sure I cover it ahead of time. For example, I'll be talking about telemetry. Telemetry is collected by Modular Sensors to create Interflow records. Telemetry refers to the data collected about the behavior and actions within a network system. This includes detailed records of events such as system logs, data collected via API Connectors from security, identity and infrastructure systems, and user-activity logs.

**01:00-02:00**

Telemetry data is rich and comprehensive, providing a real-time or near-real-time stream of information that can be analyzed to detect anomalies, track system performance, and identify potential security threats or breaches. Collecting telemetry data allows security analysts to conduct in-depth investigations and respond to incidents with a high degree of precision. Here's a cool note for you. The roots of the word telemetry come from tele, which is remote or distant, plus metria, which is the act of measuring. It literally means collecting measurements about things from a distance. Metadata is another big term we'll be using a lot. Metadata is created by Modular Sensors when creating Interflow records. Metadata is data about data.

**02:00-03:00**

Metadata provides contextual information about other data being analyzed or stored, including information about files, such as creation date, file size and type, network traffic source and destination IP addresses, port numbers, protocol used bytes transmitted and received, TCP connection status, and user activities including log in times, and resource access logs. While metadata does not contain the content of the data itself, which is also great for security reasons, collecting metadata is invaluable for security analysis because it can help in identifying patterns, trends, and anomalies without the need to process the entire content of the data streams. Telemetry and metadata are crucial for cybersecurity, but they serve different roles.

**03:00-04:00**

Telemetry provides a granular view of events and behaviors within systems and networks, enabling detailed analysis and response. Metadata offers contextual insights about data, which can aid in analysis, organization, and sometimes preserving privacy while still monitoring the security threats. Interflow records created by Stellar Cyber consist of both telemetry and metadata from security, identity, and infrastructure systems. Let's test your knowledge a little bit with the definitions I just provided. Modular Sensors create Interflow records from telemetry and metadata. Which of the following is telemetry or metadata? Data gathered through API Connectors from security, identity, and infrastructure systems.

**04:00-05:00**

You got it right, telemetry. Well, hopefully you got it right. Next, a network session summary learned from network packets containing the source IP, destination IP protocol, source port and destination port, bytes received and transmitted, and the TCP state which are all Normalized into Interflow records. You guessed it, metadata. Some other important terms are events. An event refers to a specific occurrence or incident that is being monitored or analyzed within the Stellar Cyber platform. It can include various types of activities such as security Alerts, system events, or network events. The term “event,” is often used to describe non-Alert related Interflow records.

**05:00-06:00**

I'll describe what Alerts are coming up soon. Raw event in the context of Stellar Cyber, a raw event refers to the original unprocessed log or data entry that is captured by the sensor. It contains the raw information about a security event or activity, such as network traffic, system logs, or user actions. Raw events are typically collected and stored for further analysis and investigation by the Stellar Cyber platform. Now for Alerts. Alerts are a critical component of Stellar Cyber's ability to discover important events in a sea of data. Using machine learning, or ML, techniques, statistical analysis, heuristic rules, and automated threat hunting playbooks, Stellar Cyber examines raw event data for evidence of potential threats from anomalies or security violations. When threats are detected, Stellar Cyber generates Alerts that are entered into the Alerts index and reported in the Stellar Cyber user interface.

**06:00-07:00**

Later on I'll also cover what an index is. Alerts help drive the action in the home dashboard. The Alert Types page and the Case interface. There are many different types of Alerts available in Stellar Cyber, organized by XDR Kill Chain stage, tactic, and technique with each focused on a specific type of security threat. Looking at the Stellar Cyber platform UI, there's an Alerts page. On the Alerts page you can see there are many different types of Alerts including uncommon application anomaly, external plain text password detected,

etcetera. Throughout the platform, you'll see the word JSON or J-S-O-N, which is JavaScript object notation.

**07:00-08:00**

JSON is a format for structuring data as text in a way that's both easy for humans to read and write and machines to parse and generate. Looking at the right-hand side, you can get an idea of what JSON looks like. With JSON format, objects are collected in key value pairs and closed by curly braces. If we look at the bottom, we see an example of the curly braces. You'll see pfSense firewall or pfSense_FW:, and then a brace and then there are key value pairs there. Data_length, and the value is 76. We have a key of flags and the value of none, key ID, which is a value of 11800. A key is IP version and a value is four. Arrays are ordered lists of values enclosed by square brackets.

**08:00-09:00**

Most of my arrays you'll see are just values of one entry. I have detected_fields, which has an array of source IP_host. Keys are always strings written in double quotes, and values can be strings, numbers, true/ false, null, an object or an array also written in double quotes. You'll be seeing a lot of JSON-formatted objects within the Stellar Cyber platform. One of the most important concepts we'll be talking about within this set of lessons is the Interflow record. In Stellar Cyber, an Interflow record refers to telemetry and metadata collected by Stellar Cyber that is Enriched and Normalized into JSON format by Stellar Cyber to facilitate analysis, correlation, and storage of telemetry and metadata from various sources.

**09:00-10:00**

By standardizing the format of all records, Interflow enables more efficient and effective security and analytics, threat detection, and response activities as data from diverse sources can be easily aggregated, correlated, and analyzed to identify and mitigate security risks. If you take a look at the right, the data that comes into the Stellar Cyber platform after being Normalized into JSON format has the key and value pairs, for example, action is the key. The value is block. In other words, this was a firewall rule from a firewall that took a look at the set of rules and blocked it based on a policy. Here we can see the source IP host, which is 192 with state 1.102 as an array.

**10:00-11:00**

If you look at the bottom you'll see that it was Enriched as well. The desktop_geo are the Enriched portion of the Interflow record that the log source would be gathered. The information that you see in the JSON Interflow record doesn't come with a destination IP geolocation for example, but the record gets Enriched by Stellar Cyber, where we add the geolocation. We add many other fields as well. Interflow records in the UI are also called documents. If you look at the right-hand side in the Stellar Cyber platform user interface,

you will see the word Documents. When you see that, realize this really is a list of Interflow records. The Interflow records may be events or Alerts.

**11:00-12:00**
Another hugely important concept is a Case. We have lots of different modules talking about Cases, but here, just so that we're on the same page, a Case is a set of multiple correlated Alerts and entities, constituting a potential unified security attack, ranked by dynamically updating score, indicating the severity of the attack. In other words, as more Alerts are correlated into the Case, the risk score raises, therefore the Case score raises. Stellar Cyber uses its machine-learning capabilities to generate Cases automatically, grouping related Alerts into a unified Case for improved attack resolution. You can also manually create Cases from a set of Alerts as well. Another term we just used in the explanation of the Case is an entity.

**12:00-13:00**
An entity refers to an asset or user within an organization that is being monitored for security purposes. Entities can include servers, routers, host systems, and individuals. Entities are automatically registered in Stellar Cyber by sensors and can be de-authorized or ignored if needed. Artifact, we don't use this that much within our lessons or within the platform, but it's an important word to know. An artifact generally refers to any piece of digital data or evidence that is created during the occurrence of a security event or incident. Artifacts could be generated by both malicious activities, such as malware infections and hacking attempts, and legitimate operations such as system updates and user activities.

This lesson is complete. Thank you for watching.

# L03 - COLLECTING NETWORK TRAFFIC

## L03A-PRES: Collecting Network Traffic (08m:48s)

**00:00-01:00**
The next lesson is *Collecting Network Traffic*. This lesson, *Collecting Network Traffic*, explains the flows of network traffic and how Modular Sensors obtain network traffic from switch SPAN ports and TAP devices.

In this lesson, I will go through how Modular Sensors collect network traffic and discuss network traffic flows. Modular Sensors collect network traffic traversing a network via SPAN ports on switches or by connecting to TAP devices. On the upper right-hand side I have a compass that basically shows that the internet is towards the North. I have servers towards the East. I have clients towards the West and clients are essentially South as well.

**01:00-02:00**
The way we describe network traffic communication is we have North/ South communication, which is also called vertical communication. North/ South communication is to and from clients and servers that are accessing the internet as that dotted line shows. East/West, or lateral communication, is traffic among clients and servers or even client-to-client or server-to-server. Essentially it's when traffic is not traversing the network to reach the internet. Switch port analyzer or SPAN ports on switches are configured to replicate and mirror traffic from specific switch ports or VLANs to devices connected to the SPAN port.

**02:00-03:00**
Stellar Cyber Modular Sensors receive network traffic from switch SPAN ports for analysis. They perform deep packet inspection, which is called DPI, to identify applications and run IDS signatures to detect threats and look for malware. They also forward network telemetry metadata as Interflow to the Stellar Cyber platform. Looking at this picture, I have the Modular Sensor connected to the switch's SPAN port. The Ethernet zero interface is the interface that is for management. It has an IP address. I can access it via SSH, and that's the interface that also communicates with the Stellar Cyber platform. When communication happens, for example, North/ South communication or vertical communication.

**03:00-04:00**
The SPAN port replicates that data and sends it to the Stellar Cyber Modular Sensor. Likewise, for East/West communication or lateral communication, the Stellar Cyber Modular Sensor also receives that traffic from the SPAN port as well. Here I'll go through how SPAN or mirror ports are configured on switches or give you an idea of how they're configured on switches. Looking at the switch on top, I have port 1/1, which means module one on the switch port one on the switch. That's connected to the Stellar Cyber Modular Sensor on Ethernet zero, which is the management interface.

**04:00-05:00**
The management interface has an IP. That's what is used to communicate with this Stellar Cyber platform.I can also access the management IP locally on the network via SSH. Ethernet one on the Modular Sensor is connected to the switch port 1/2. That will be configured as a SPAN port. I'll talk more about that. Looking at the bottom-left switch, that's connected to port 2/12 of the switch on top, that's module two port twelve. On the right, on the switch on the right is connected to module two, port twenty five, so 2/25. Now looking at the configuration on the right, I can essentially say, look for any traffic coming in on port 2/12 or 2/25, and look at the traffic bidirectionally.

**05:00-06:00**
In other words, it could be coming into the port from the bottom or coming out of the port from the top. In this case I'm saying the destination SPAN port is 1/2. If I see any traffic going in or out of port 2/12, send it to the SPAN port, which will then make it to the Modular Sensor. Or if I see traffic coming in or out of port 2/25, send it to the SPAN port, which will then send it to the Modular Sensor. Here's an example: I have a client on the bottom left communicating with the internet, that's going into port 2/12. That gets mirrored essentially, to the SPAN port, which then makes it to the Stellar Cyber Modular Sensor.

**06:00-07:00**
Likewise, I have traffic coming in 2/12 from the client on the left going to a server on the right. In both cases I have traffic coming in 2/12 but I also have it going out of 2/25. And then the reply traffic comes in 2/25. It's essentially bidirectional. In either case, the mirror port will send it to the Stellar Cyber Modular Sensor, but the Modular Sensor knows to only keep track of essentially one flow. A network TAP device is a device that goes in between switches. Essentially it's a break in the cable. This way I don't have to configure traffic on a switch itself. These devices are usually made so that they will fail open. In other words if that device were to die for any reason it will still maintain connectivity. But like a SPAN port, it will mirror the traffic.

**07:00-07:50**
Traffic coming in from the North/ South communication or in this case really it's South/ North communication, it'll get mirrored and sent to the Stellar Cyber Modular Sensor. Likewise traffic going East/West or lateral will also get mirrored or copied to the Stellar Cyber Modular Sensor. TAPs can connect to switch access ports or trunk ports. TAPs may be configured to forward network traffic to Modular Sensors.

This lesson is complete. Thank you for watching.

# L04 - FUNCTIONALITY OVERVIEW

## L04A-PRES: Functionality Overview (06m:15s)

**00:00-01:00**
The next lesson is *Functionality Overview*. This lesson, *Functionality Overview,* provides an overview of each of the core features that will be explained in detail throughout the *Modular Sensors Concepts and Configuration* training module. This will help you see the big picture of how the features work as a whole before diving into each individual feature.

Hi. In this lesson I will cover Stellar Cyber Modular Sensor functions. We've covered the Stellar Cyber Modular Sensor functions in the big picture as an overview, but we're going to be diving deep into each of these functions and how to configure them. I want to make sure we do a quick recap so it's fresh in your mind. On the left-hand side is the Stellar Cyber Modular Sensor. That talks to the Stellar Cyber Open XDR platform.

**01:00-02:00**

The Stellar Cyber Modular Sensors are deployed within organizations off of network switches or they can be deployed in the cloud, such as in AWS, Microsoft Azure, GCP or Google Cloud Platform, OCI for Oracle Cloud infrastructure, and so forth. The very first function we'll talk about is the log forwarder. Modular Sensors collect parts and Normalize logs sent from security and infrastructure tools and systems into Interflow records in JSON format and forward to the Stellar Cyber platform. Modular Sensors have hundreds of built-in parsers, which we'll talk about, that interpret and Normalize logs into Interflow records. Logs that are not relevant for threat investigation can be filtered at the Modular Sensor to optimize the amount of data that is ingested on the Stellar Cyber platform. That is very important. We'll go through some examples of how to configure all of this.

**02:00-03:00**

Next, I'll talk about network traffic analysis. The Modular Sensors are deployed within an organization off of switches and usually off a SPAN port or TAP device. Being off a SPAN or TAP device, the Modular Sensors will receive network traffic. Modular Sensors capture network traffic and create metadata summaries that are Normalized into interval records in JSON format, and are forwarded to the Stellar Cyber platform. Modular Sensors identify network traffic flows and applications in network traffic using deep packet inspection. That is Normalized into Interflow records in JSON format and forwarded to the Stellar Cyber platform. The Stellar Cyber Modular Sensor also has an IDS engine. Modular Sensors apply IDS rules to network packets to identify threats and create metadata summaries that are Normalized into Interflow records in JSON format and are forwarded to the Stellar Cyber platform.

**03:00-04:00**

Stellar Cyber Modular Sensors also have a malware sandbox that can be configured to identify malware and unencrypted files and create metadata summaries that are Normalized into Interflow records in JSON format and forward to the Stellar Cyber platform. I hope that's becoming a very familiar case for you. Notice that all of it on the sensors gets Normalized into Interflow records in JSON format and forwarded to the Stellar Cyber platform as one of the key functions of the Modular Sensor. The Modular Sensor can also be an Aggregator. Modular Sensors can be configured as an Aggregator for the traffic from

other Modular Sensors, Windows Server sensors, and Linux Server sensors within an organization.

**04:00-05:00**
When sensors are configured to use a Modular Sensor configured as an Aggregator, communication from sensors with the Stellar Cyber platform is from the Aggregator instead of each individual sensor. Stellar Cyber Modular Sensors can also run a Tenable Nessus scanner. With a license from Tenable.io, a security scan can be initiated to run on a Modular Sensor to scan the network for vulnerabilities. We'll go into lots of detail on that as well. Along with performing vulnerability scans, a Tenable.io Connector can be configured to collect telemetry from Tenable that is Normalized into Interflow records. This is really cool and I will show all of that later on in these lessons. Next are Connectors.

**05:00-06:15**
The Modular Sensor can obtain telemetry data from infrastructure identity and security systems and tools via Connectors using APIs that Normalize it into Interflow records in JSON format and then forward them to the Stellar Cyber platform. Using Connectors on Modular Sensors enables the ability to trigger response actions as well to infrastructure identity and security tools from the Stellar Cyber platform. This can be done via automated rules or manual action Another cool thing about the Stellar Cyber Modular Sensors is the fact that it can buffer Interflow records. If there is a connection issue to the Stellar Cyber platform preventing the Modular Sensor from forwarding Interflow records, Modular Sensors can temporarily hold the buffered Interflow records until the connection is resumed to prevent loss of data. It will then trickle the data back so that it doesn't overwhelm the connection once the connection is re-established. That's the end of the functionality overview. Thank you.

This lesson is complete. Thank you for watching.

# L05 - DATA PROCESSING

## L05A-PRES: Data Processing (04m:49s)

**00:00-01:00**
The next lesson is *Stellar Cyber Platform Data Processing Overview.* This lesson, *Stellar Cyber Platform Data Processing Overview,* gives an overview from the perspective of the Stellar Cyber Open XDR platform on how it processes the data received from the Modular Sensors.

In this lesson, I'll give a brief overview of how the Stellar Cyber Open XDR platform receives Interflow records from Modular Sensors, then processes and categorizes Interflow records into indices. That's so the data can be quickly searched based on its category, for example, traffic goes into the traffic indices. Syslogs go into Syslog indices. All Alerts that are generated by the platform go into the Alerts indices.

**01:00-02:00**
Scans by a Tenable Nessus scanner go into the scan indices. Again, by having the indices, I can quickly search through the different categories of data that's been collected by the Stellar Cyber platform. The Stellar Cyber platform then Enriches Interflow records with fields such as threat intel that has the IP reputation. Is it good? Is it malicious? Is it a brute forcer? Is it a scanner? This is good information to have included in the Interflow records that were received by the Stellar Cyber Modular Sensors. The Modular Sensors don't add that information, but once received by the Stellar Cyber platform, this information gets Enriched into the Interflow records. The other information that's important is the geolocation. Where was that source? Where is the destination IP?

**02:00-03:00**
The Interflow records can be also Enriched with the DNS host names. The data source So where did the Interflow record come from? What was the source of the data? Was it a firewall? Was it a Windows Server or Linux Server or was it from the network traffic? Also, the user identity could be added to Interflow records as well. What was the username? That information can be learned from Active Directory as an example. You will see this later on as we go through more examples showing you real records and the Enrichment that occurs. The Stellar Cyber platform then stores categorized and Enriched Interflow records into the Data Lake. Something that we will repeat over and over again:

**03:00-04:00**
The Stellar Cyber platform uses machine learning or ML statistical analysis rules, heuristic rules, and custom ATH playbooks to identify threats and create Alerts. This Stellar Cyber platform correlates Alerts from all data sources using graph machine learning to create Cases. The Stellar Cyber platform provides a streamlined UI and API to enable effective searches and analysis of all data. It also enables containment of threats using automated and analyst-driven response actions using Connectors that integrate with hundreds of security identity and infrastructure tools using APIs. On the Stellar Cyber platform, Interflow records sent from the Modular Sensor are viewable and searchable for threat investigation.

**04:00-04:50**
Interflow are also used by the Stellar Cyber platform with machine learning and rules engines to identify threats and anomalies to create Alerts. On the Stellar Cyber platform,

Interflow records collected from Modular Sensors are among the Alerts used by Graph ML to correlate and create Cases. There are many more features of the Stellar Cyber platform, but I wanted to give a brief overview of some of the features before diving in deep into each of the Modular Sensor functions. Thank you for watching.

This lesson is complete. Thank you for watching.

# L06 - MODULAR SENSOR PROFILES

## L06A-PRES: Modular Sensor Profiles (12m:49s)

**00:00-01:00**
The next lesson is *Modular Sensor Configuration with Modular Sensor Profiles*. This lesson, *Modular Sensor Configuration with Modular Sensor Profiles*, explains the concepts and tasks needed to configure Modular Sensor profiles used to configure Modular Sensors.

Now we'll get right into it. In this lesson, I will cover how to configure Modular Sensors using Modular Sensor profiles. As I just mentioned, Modular Sensors are configured by modifying the settings of sensor profiles and assigning them to Modular Sensors. Modular Sensor profiles are configured by going to **System** on the top right, then selecting **Sensor Profiles**.

**01:00-02:00**
Once you get into **Sensor Profiles**, you can see a list. By default, there are two sensor profiles. The **Default** profile is assigned by default to Windows Server sensors, Linux Server sensors, and some of the older sensors such as network sensors and security sensors. Network sensors and security sensors are no longer available because all that functionality rolls into a Modular Sensor so you can configure which modules you actually need. To start, I will show you how to clone the default Modular Sensor profile so you can change the settings without modifying the existing default Modular Sensor profile. After, you'll be shown how to create a new Modular Sensor profile as well. Check the box next to **Default_modular**.

**02:00-03:00**
Then on the right-hand side, select **Clone**. Note I'm using the word "select." Sometimes people use iPads so I was used to saying click. But now the new terminology we will say "select" instead just in case someone does have a touch-screen device. Once you click **Clone**, or excuse me, once you select **Clone**, you can type in a new clone name. I will call this SC Modular Sensor, of course that stands for Stellar Cyber. I'll finish it off by selecting **Clone**. Okay. The new Modular Sensor profile appears in the list. If you select the pencil icon, next

to the Modular Sensor profile name, on the right-hand side, you can edit the Modular Sensor profile.

**03:00-04:00**
By the way, if you want to delete and start over, you can click, excuse me, you can select the trash can icon. If you want to create a new Modular Sensor profile, you can select **Create** from the top right, then select **Add a Modular Sensor Profile**. Either way is fine. Modular Sensor features are enabled or disabled by configuring and applying Modular Sensor profiles to Modular Sensors. Modular Sensor profile features are enabled or disabled by sliding the lever to the right, which is on and it'll have a little green outline, or keeping the lever off, which will be grayed out. You can select a feature heading to expand and see additional options.

**04:00-05:00**
For example, when I selected **IDS**, now I can see that there's networks, signature, and buffering as IDS options. If I select **Sandbox**, for example, it will close that one and open up the **Sandbox** options. Note that the **Log Forwarder** feature is always on to ingest logs and cannot be turned off. Therefore, there is no lever to slide it on or off. But there are options if I select the **Log Forwarder** to expand the options. To start, I can enter a Profile Name. It is helpful to make the name of the profile something that explains the intended use of the Modular Sensor profile. For example, maybe if I'm doing IDS, I can say this is the Stellar Cyber Modular Sensor IDS profile.

**05:00-06:00**
Or if you have all functions or all features enabled, maybe Stellar Cyber Modular Sensor All, or something that just makes it so that you know what it means. It could also be based on a location, maybe a different location or different customers or different tenants I have. I might have different names to specify those as well. Next, under **Receivers**, you can select in the **Receivers** field to specify the destination for receiving Interflow records. The default Receiver is the Stellar Cyber platform. There are some cases where there's cluster environments or for on-premises or cloud deployments where you might have multiple receivers. If you want to close the selection box, select a point away from the selection box to close it. After completing your configuration settings in the sensor profile

**06:00-07:00**
the **Submit** button must be selected to apply the configuration settings to the Modular Sensor. On the bottom right of the Modular Sensor profile, you select **Submit**. You will then see a Stellar Cyber profile was edited successfully message that appears in green every time you select the Stellar Cyber. You will see a sensor profile was edited successfully message appear after you select **Submit**. As mentioned previously, Modular Sensors are

assigned to the Default-modular profile by default. To assign a new Modular Sensor profile to Modular Sensors, on the top, select **System** then **Sensors**.

**07:00-08:00**
Set number two, select **Manage** and **Assign Profile** on the right-hand side. Step number three in the **ASSIGN PROFILE TO SENSORS** dialog box, under **Sensor Profiles**, select the radio button next to the new Modular Sensor profile that you created, or if you want to leave it by default you can leave it as default. On the right-hand side in step number four, select the Modular Sensor or multiple Modular Sensors. You also have an option to **Select All**. In step number five, select **Submit**. After selecting **Submit** on a Modular Sensor profile or assigning a Modular Sensor to a new Modular Sensor profile, the Modular Sensors using the profile will restart their services using the new configuration.

**08:00-09:00**
Step number one you'll see there that the Stellar Cyber Modular Sensor status will go red, and a message will display that the configuration is out of sync that is normal and is supposed to happen. After a minute you will have to refresh your browser, and then you'll see the **Sensor Status** go green, and the message will be clear. Now I'll show you what happens behind the scenes when you configure a Stellar Cyber Modular Sensor profile, or for the very first time when the Modular Sensor profile gets assigned to the Modular Sensor. This is an example of the communication flow for when the Stellar Cyber Open XDR platform is installed on premises or in a private cloud. One of the very first things that the administrator needs to do on the Modular Sensor is set a CM command to the IP address of the Stellar Cyber platform Data Lake.

**09:00-10:00**
Once that's done, on step number two there's a configuration management connection made to the Data Lake from the Modular Sensor using TCP ports 6640 to 6648, all using TLS 1 .2. The communication from the Modular Sensor to the Data Lake is encrypted. For step number three, either the very first time the Modular Sensor is assigned to the Modular Sensor profile, which, by default, is the default Modular Sensor profile. Or if you're within the profile and you make a configuration change and click **Submit**, or I should say select **Submit**. The Message appears on the user screen that says the Sensor profile was edited successfully, and the Modular Sensor profile is sent to the Modular Sensor.

**10:00-11:00**
The Modular Sensor profile has the Interflow (JSON) receiver name, which is the Stellar Cyber Data Analyzer IP address. When the Modular Sensor receives the Modular Sensor profile, its configuration is updated and it restarts its services with that new configuration. The sensor then starts receiving logs or packets and creates Interflow records to send. On

step number six, Interflow records to the Interflow Receiver on the data analyzer are sent using HTTPS on TCP port 8889 using TLS 1.2. This information is also encrypted.

**11:00-12:00**
The process is different for SaaS deployments. In a SaaS deployment, the administrator on the Stellar Cyber Open XDR platform creates an auth token that contains information on how to reach the Stellar Cyber platform that can be downloaded by the administrator. This is also used to authenticate the sensor. On step two, the administrator goes to the Modular Sensor and copies the auth token from step number one. Next, the Modular Sensor on step three creates the configuration management connection to the Stellar Cyber Open XDR platform using HTTPS port 443 on TLS 1 .2. This contains the auth token information.

**12:00-12:50**
The auth token on step four is then validated and authorized. In step five, the Modular Sensor profile is sent to the Modular Sensor, and it has the JSON Interflow receiver name. In step six, the configuration is updated. In step seven, logs and packets are received and Normalized into Interflow records. Then they are sent on step eight to the adjacent Interflow receiver, which is the Stellar Cyber platform on HTTPS port 8889 using TLS 1.2. That is what happens behind the scenes and it's good to know.

This lesson is complete. Thank you for watching.

# L07 - MODULAR SENSOR CONFIGURATION

## L07A-PRES: Modular Sensor Configuration (03m:31s)

**00:00-01:00**
The next lesson is *Modular Sensor Specific Configuration.* This lesson, *Modular Sensor Specific Configuration,* goes through the parameters that are configured on individual Modular Sensors.

Hi everyone. There are specific parameters that must be configured directly for a Modular Sensor that are accessed by selecting **System**, then **Sensors**. Then select the pencil icon to edit the Modular Sensor parameters next to the Modular Sensors you want to edit. In the Modular Sensor specific configuration you can assign the tenant name. For example, if I have other customers and I want to assign the Modular Sensor to a specific customer, I can do that right from here. For this example I'm using the root tenant.

**01:00-02:00**

You can change the Hostname and DNS Name that was configured during the installation of the Modular Sensor here. That can be handy, especially if you have lots of Modular Sensors and you want to rename them. For step number three, you can validate if the Modular Sensor profile is correct or assign a different Modular Sensor profile. The next thing that's incredibly important is make sure the Timezone is set to the location of where the Modular Sensor is deployed. This is important for the logs to display the proper time. For step number five, if you are sending Syslogs to a Modular Sensor over an unsecured network, it is recommended that you enable Syslog over TLS. This will require that you configure a certificate on the log source and in this Modular Sensor configuration.

**02:00-03:00**
Then select **Submit**. After you submit the settings, you can validate the Modular Sensor status for the configured components. If you go to **System | Sensors**, where actually you should be at this moment, you can select the name of a sensor in the **Sensor Name** column. Then if you look inside, you can see the Service and the Status for each of the services. If they are green, they are running successfully.

**03:00-03:30**
You can also validate that the sensor profile configured is the correct profile. The cool thing about that is that you can also go right to the Modular Sensor profile right from this page. If you select the module profile link, you can actually see the name, and then configure settings as needed. You can then validate the settings are enabled as expected. If you do make some changes, make sure you submit the Modular Sensor profile. Otherwise, if no changes are needed, or you can select the **X** on the upper-right side of the Modular Sensor profile to leave the profile without saving any changes. That's it. I hope you enjoyed this lesson.

This lesson is complete. Thank you for watching.

## L07B-DEMO: Modular Sensor Configuration (16m:18s)

**00:00-01:00**
This lesson continues with demonstrations for *Modular Sensor Configuration with Modular Sensor Profiles and Modular Sensor Specific Configuration*. This lesson is a demonstration of *Modular Sensor Configuration with Modular Sensor Profiles and Configuring Modular Sensor Specific Settings.*

Hi. For this demonstration, I will show you how Modular Sensors are configured using Modular Sensor profiles. I will log in to the Stellar Cyber Open XDR platform with my administrator account, called admin. Okay.

**01:00-02:00**

When you first log in to the Stellar Cyber platform, you start off on the **Visualize | XDR. Kill Chain** page. It's actually located right here: **Visualize | XDR Kill Chain**. The very first thing I'm going to do is go to the Modular Sensor profiles, which are located under **System** then **Sensor Profiles**. In this list, by default, there's the Default profile. You look at the profile type over here, that's a standard profile. If you look at the next one in the list it's the Default_modular. That is the Modular Sensor profile. The default profiles go on to Linux Server sensors and Windows Server sensors, and the default Modular Sensor profile goes on to Modular Sensors.

**02:00-03:00**
There used to be two other sensor types. One was called a network sensor and one was called a security sensor. But now Modular Sensors are all-encompassing, and everything can be configured in a Modular Sensor profile. I'll take a look first at what's inside the default Modular Sensor profile. When I go inside, the Profile Name is Default_modular. The Receiver is the Default_receiver, which is the IP address of my Stellar Cyber Open XDR platform. The Log Forwarder is enabled by default. If I select it, you can see the other options that are available.

**03:00-04:00**
By going further, I can see that I have the Network Traffic and I can expand or collapse that as well. I will stop here just to make sure that everything is working. I can click **Submit** or if I just want to leave without saving any settings, you can click the **X** on the top right. I'll go down and select **Submit**. Here you see the sensor profile was edited successfully. But is this sensor profile in use? Well, what I can do is I can go to **System** and **Sensors**. In my network, let me take one of these columns and I will say autosize all columns. Okay. Now that I've autosized my columns, I can see all the different sensor names.

**04:00-05:00**
The sensor or the Modular Sensor that I'm using for this demonstration is the SC-modular-sensor, which is right here. I can select SC-modular-sensor to get more details. Looking inside, I see that the Default_modular sensor is the sensor profile, and I can actually go right to the Default_modular sensor profile from here as well. Taking a look, the name is the Default_modular. The receiver is Default_receiver, which is the IP address of the Stellar Cyber platform. Log Forwarder is enabled. If I expand it, I can see the options available. Network traffic is enabled. If I scroll down here or expand it, I can also see there are many other network traffic options. We'll be talking about some of these as well.

**05:00-06:00**
But what matters now is that this is the profile that is installed. If I were to click **Submit** right now, **Submit** will save the configuration and apply it to the Modular Sensor. If I want to leave a sensor profile without making any changes, you can select the **X**. Now I will then go back to **System** and **Sensors**. Again, I can autosize all columns. The other option I have here is to edit with the pencil icon. As I was mentioning in the slideware that you can edit sensor parameters directly for some of them. These are crucial parameters.

**06:00-07:00**

For example, I can set or change the hostname of this Modular Sensor. I can assign a Modular Sensor profile, which again right now, mine is set to the Default_modular. Once I have an Aggregator configured, I can select an Aggregator. One of the most important things that you need to set is the timezone. I'm in California, and my sensor is in California, that's what matters the most. Therefore, I have chosen the **America/ Los Angeles** timezone, but all the timezones are available. For example, if I take a look at the list, there are lots of different options to choose from. I'm going to select **America**. I'm going to select **Los Angeles**.

**07:00-08:00**
I can also add a note, which I really like. I can say, "Modular Sensor for Training." Then I will select **Submit**. Again, once you modify the configuration of a Modular Sensor, you will see the status say disconnected and the configuration will be out of sync. After less than a minute, it will come back and sure enough within, it was actually 10 to 15 seconds, my sensor is now green. Now let's take a look at the Default_modular sensor profile.

**08:00-09:00**
If I go here, if I take a look at the Default_modular, if I'm looking at this I might want to ensure that it has a default profile that hasn't been touched, that hasn't been modified. One way I can do that is I can select the Default_modular profile and click the option to **Clone**. When I clone, to begin with it will always add a date. I can say, I'll use SC (for Stellar Cyber) Modular Sensor and I'll say training for this one, and **Clone**. Now, if I modify this Modular Sensor, again I can deselect that one, but if I modify this Modular Sensor profile, again it defaults to the default receiver.

**09:00-10:00**
The network traffic is there and Log Forwarder is configured because it's always on. To begin with, we'll take a look at these options, the network traffic. There's the Application Identification, which is the deep packet inspection I was talking about earlier. There's the Application Session that tracks the state of the applications from the network flows. That's really cool. When I type in `show session` on a Modular Sensor, I can see all the traffic going through. Then there's the Application Metadata. This is how the Modular Sensor is collecting the metadata that it will Normalize into Interflow and send to the platform.

**10:00-11:00**
One of the options we'll be talking about later is excluding certain applications. We also have some other options, which are there by default. You don't need to touch them. You can leave them on, and we'll also talk about buffering a little bit later as well. But right now I'll **Submit**. The sensor profile was edited successfully. Do you remember how to assign this profile to the Modular Sensor? Let's go take a look. If I go to **System** then **Sensors**, the next option for the sensors is to select **Manage**, and then **Assign Profile**. Now I have an extra sensor profile.

**11:00-12:00**
I will select **SC Modular Sensor**. I will assign that to the SC-modular-sensor I have here. If I had a whole bunch of sensors I was assigning this to, I could say **Select All**. I won't do that though. I will select **Submit**. Notice once I select **Submit**, the status has now gone red. That is because the system

is out of sync now. The configuration is out of sync. Now, if I wait a few let's see how long this takes. That was again like 10 seconds or less. Now my Modular Sensor is assigned. Let me just make sure it's the right one in the list. I can select it.

**12:00-13:00**
If I hover over the **Sensor Profile** I see that it's SC Modular Sensor training. If I select it, there's my configuration that I made. Now I can close this out. If you close it out that way, it doesn't cause the sensor to reload its configuration. That's all there is to assigning a sensor profile to a Modular Sensor. Okay. In the last demonstration I showed you how to clone the default Modular Sensor profile to create a new Modular Sensor profile and assign it to a Modular Sensor. Now I'll show you how to create a Modular Sensor profile from scratch. Okay.

**13:00-14:00**
To do that, I will go to **System**, then **Sensor Profiles**. Then I will select **Create** and **Add Modular Sensor Profile**. Note that the standard sensor profile is for Linux Server sensors and Windows Server sensors. They were also for network sensors as well and security sensors. Network sensors and security sensors are no longer available. For new installations, Modular Sensors can do everything that the network sensors and security sensors did, plus as a lot more functionality. You can select the modules that you want to have enabled or disabled. We'll add a new Modular Sensor profile. I will give it the name SC Modular Sensor New

**14:00-15:00**
just so we know it's the brand new Modular Sensor profile. For the Receivers you select in the selection box, select the default_receiver, then you select out of the selection box. By default, the Log Forwarder is always enabled. That's what we'll use to begin with for this example. I'll select **Submit**. Now my new Modular Sensor profile has been created. To assign it, again I go to **System | Sensors**. You don't need to select any other sensor here. Just go to **Manage**, then **Assign Profile**. I can select the new Modular Sensor profile, SC Modular Sensor New, and assign it to the SC Modular Sensor.

**15:00-16:20**
All you need to do is select **Submit**. As always, if we take a look at this line, the Modular Sensor then loads its services again using the new configuration. We'll go red for the status saying it's disconnected. The message will be: Configuration is out of sync. If I refresh it may take about 30 seconds to a minute when assigning a brand new Modular Sensor profile. There you have it. I would say about 40 seconds after assigning a brand new Modular Sensor profile, the status became green again. Just to validate, if I go to SC Modular Sensor and select it, and take a look, it's now SC Modular Sensor New. Alright. This is the end of this lesson. I look forward to seeing you in the next lesson.

This lesson is complete. Thank you for watching.

# L08 - PARSERS, NORMALIZATION, AND INTERFLOW

## L08A-PRES: Parsers, Normalization, and Interflow (15m:06s)

**00:00-01:00**
The next lesson is *Parsers for Normalization into Interflow.* This lesson, *Parsers for Normalization into Interflow,* provides details for how parsers are used on Modular Sensors to Normalize metadata and telemetry into Interflow records in JSON format.

In this lesson, I will discuss more about parsers and how they are used for normalization into Interflow records. Parsers are used to analyze, interpret, and extract relevant information from log files generated by various devices, systems, and applications. Parsers can identify specific fields within the log data, such as timestamps, source and destination IP addresses, event descriptions, and much more.

**01:00-02:00**
Using parsed logs, security analysts can gain insights into potential security threats, detect anomalies, and investigate incidents. Parsers play a crucial role in log analysis and are essential for effective threat detection and response by analysts and the Stellar Cyber Open XDR platform. Note that Stellar Cyber has hundreds of built-in parsers that can interpret logs from generic or vendor-specific sources and more are included in every release. Looking at the right is an excerpt of the Stellar Cyber parser documentation. Here you can see the list of names on the left-hand side for the different parsers. I will cover this next.

**02:00-03:00**
To access the Stellar Cyber documentation for log parts reports, you can go to https://docs.stellarcyber.ai. When the search bar appears, you can search for log parser ports. The entry gets returned called log parser ports that you can select. Also note that there's a help version. By default, if you're on the Stellar Cyber platform, the correct help version will be displayed for the platform you are using. If you go to docs.StellarCyber.ai, you have to select the help version you want to use. If you are not sure, always select the latest help version. When you select log parser ports in the documentation, you can scroll down to see the table containing all the log parsers.

**03:00-04:00**
Looking at the Stellar Cyber parser documentation excerpt on the right-hand side, you'll see that there are several different columns. Each of them is very important The **Device** column specifies the device system or application that is forwarding logs to the Stellar Cyber Modular Sensors for analysis by the Stellar Cyber Open XDR platform The **Port** column specifies the TCP or UDP port to configure on the log source that sends logs to the Modular

Sensor. The port is used by the Modular Sensor to identify the appropriate parser to use to Normalize the logs into Interflow records. The **msg_origin.source** column specifies the vendor's product. You use this field and the Interflow records to find the logs when threat hunting using the index specified in the index column.

**04:00-05:00**
The **msg_origin.category** column specifies the overall category of where the product fits into a solution such as firewall, or pass, which is password as a service, DB logs which is database logs, endpoint, XDR, and many more. The **Index** column indicates the fields that must be present and are not null for the logged data to be entered into the respective index. In some instances, no specific field is required. Just the index name is listed. For many parsers, the remaining data that is not mapped to a specific index is otherwise mapped to the Syslog index.

**05:00-06:00**
For example, for Fortinet data analyzer logs received on port 5542 by the Modular Sensor that are forwarded to the Stellar Cyber platform. Data is added to the ML IDS/Malware index. If the incoming vendor.attack underscore name field is not null. Data is added to the traffic index if the desktop IP is not null. The remaining data is added to the Syslog index. Use the **dev_type** field in the Interflow records to find the logs when thread hunting in the specified index. With help from the documentation you can find the indices used for the specific device type. Here if I look at the pfSense firewall, the index is traffic if the source IP, source port, destination IP, dstport, and protocol fields are within the logs.

**06:00-07:00**
Otherwise, traffic is sent to the Syslog. Find the name of the device under **msg_origin.source**. In this case, it is pfSense_fw for pfSense firewall. For indices, select the box to select all. This is only a test to find out which indices are actually in use by that message origin dot source. In the search box type, dev_type: pfSense_fw. After typing in the search, all the records and charts displayed are reflected by what was typed in the search along with all the indices specified. Here, the legend shows that the indices are Syslog, traffic, and Alerts.

**07:00-08:00**
The documentation just mentioned traffic and Syslog because that's how it gets into the platform. But Stellar Cyber creates the Alerts indices well when it analyzes all the logs that are coming in. Stellar Cyber creates the Alerts from looking at the logs that are in the Syslog and traffic indices. The Alerts are generated by the Stellar Cyber platform. Log event messages received from different sources arrive with different vendor specific formats. On number one on the right-hand side is a Check Point log from a Check Point firewall. Here we

have a time stamp, then the Check Point firewall name, and then there's a FW field, which is the IP address of the firewall itself.

**08:00-09:00**

There's a source, which is the IP address of the traffic coming into the firewall, and a DST is the IP address of where the traffic was going from 192.168.1.10 in this example. There's also a source port and a destination port, and the action of the rule. All these are within the log message sent by the Check Point firewall. Looking at the bottom is an example of a SonicWall firewall. The SonicWall has again a timestamp as the name of the SonicWall itself. There is an IP address for the SonicWall, but this one does not have the FW equals. It's just the IP address itself. Note there's an ID. There's a serial number. There's a whole bunch of different fields that are on two examples.

**09:00-10:00**

Looking at the Check Point firewall, with this SRC equals 192.168.1.10 and SRC_port equals 56034. On the SonicWall, the source is 192.168.1.99: the source port 34098. The format is completely different. If I am trying to analyze and compare log messages from different vendors, it gets very difficult at this point because now I have to understand all the different formats of all the different logs that come in. Well, that's where Stellar Cyber comes in handy. Now here's an example of what happens when the logs are Normalized into Interflow records.

**10:00-11:00**

Modular Sensors use the built-in parsers to Normalize log messages from independently formatted third-party logs collected as Syslog in CEF format, LEEF format. It could be JSON logs, etcetera. They get Normalized into consistent Stellar Cyber Interflow JSON formatted records for all field names and values. Note, the field names are now the same for Check Point and SonicWall Interflow records. Looking at the top right, number one, the Stellar Cyber Normalized Check Point log excerpt shows a key of source IP and a value of 192.168.1.10. I like calling the key the field name, and then the right-hand side is the value. The source port is srcport: 56034. The key is source port. The value is 56034.

**11:00-12:00**

Looking at the Normalized SonicWall log, it is now the same field name, SRCIP and value of course will be what was received by the log itself. That will be different and that's okay. The srcport is the same, again field name, then the value is the source port received. Looking at all the field names and values, they are Normalized into a consistent format. This now allows SOC analysts to use a consistent format when searching for values among all different types of devices and services. It also allows the Stellar Cyber machine learning,

statistical analysis, and heuristic rules to be able to compare and correlate all the data to find threats.

**12:00-13:00**

Additional examples of Normalization from third-party integrations to Stellar Cyber are like the CrowdStrike EDR has a field for the local IP or host name of the device. It's called CrowdStrike.event.local IP. With Stellar Cyber normalization, the CrowdStrike local IP field and similar fields from other third-party integrations are Normalized to host.IP. CrowdStrike EDR has a field for classification of threats. It's called threat info.classification. Again, with Stellar Cyber normalization, the CrowdStrike threat info_classification field and similar fields from other third-party integrations are just Normalized to a field called, "threat." Again, this makes it much easier. Now I don't have to memorize the specific name for the values from all the different integrations that we have. Interflow records in JSON format based on a summary of the sessions including metadata about the session,

**13:00-14:00**

and sends them to the Stellar Cyber platform. Interflow records are created using data that is Normalized from logs, network traffic, and Connectors. Looking at the picture on the right, there's a Modular Sensor and an example of a JSON Interflow record excerpt. The JSON Interflow record contains application and network metadata and telemetry. On the right is an example of the key value of source IP colon, the IP address, which is the value. Notice that this is Normalized because every source IP within Stellar Cyber will be listed as srcip. The destination port as an example is a field name, DST port, and the value is 443.

**14:00-15:06**

The total bytes is metadata as an example of how many bytes have been included in this session so far. The app ID family again is extra metadata. The app path is something called, "ssl.https" to let you know it's encrypted. There's the destination IP and the appid_name. The appid_name actually comes from deep packet inspection. We'll cover more about that later. Modular Sensors send the JSON Interflow records to the Stellar Cyber platform over an HTTPS connection. The Stellar Cyber platform is then able to perform data processing on the received Interflow record. It actually gets stored in the platform where all the machine learning and statistical analysis rules and heuristic rules are applied to look for threats.

This lesson is complete. Thank you for watching.

# L09 - LOG FORWARDER AND LOG FILTER CONFIGURATION

## L09A-PRES: Log Forwarder and Log Filter Configuration (12m:09s)

**00:00-01:00**

The next lesson is *Log Forwarder and Log Filter Configuration.* This lesson, *Log Forwarder and Log Filter Configuration*, explains the process and configuration steps to ingest logs from third-party security, identity, and infrastructure systems and tools as well as how to optimize ingestion by filtering logs that are not essential for security operations.

Now I will discuss one of my favorite topics, which is the Log Forwarder and Log Filter, which is part of the Modular Sensor. This is a key function of the Modular Sensor. Logs are sent to the Modular Sensors from security and network infrastructure systems and tools. Once they are forwarded to the Modular Sensor, Modular Sensors Normalize the logs into Interflow records in JSON format, which are then forwarded to the Stellar Cyber platform.

**01:00-02:00**

The Stellar Cyber platform then uses machine learning, statistical analysis rules, heuristic rules, custom ATH playbooks, and graph machine learning to identify threats, generate Alerts, and correlate Alerts into Cases. Log filters can be configured on Modular Sensors to optimize the data ingested by the Stellar Cyber platform by excluding specific log types that are not required for threat investigation or detection. Examples of security and infrastructure systems include firewalls, EDR, which are endpoint detection and response, NDR, which is network detection and response, vulnerability management systems, email, and many more configured to send logs in the form of Syslog and CEF or LEEF format. JSON and Netflow/IPFix.

**02:00-03:00**

Netflow is Cisco specific. IPFix is more a generic way of forwarding logs. All can be sent to the Stellar Cyber Modular Sensors. Modular Sensors that Normalize the logs from all the different sources into Interflow records in JSON format and forward to the Stellar Cyber platform. Hopefully that is being locked in. I've said this over and over again, and I will keep on repeating this so that It flows nicely off your tongues as well. Modular Sensors Normalize the logs from all the different sources into Interflow records in JSON format, and forward to the Stellar Cyber platform. I know this might sound kind of repetitive, but it is incredibly important to understand the importance of how the logs are parsed, Normalized, and forwarded to the Stellar Cyber platform.

**03:00-04:00**

Note that each log source that can be parsed and Normalized by the Modular Sensors is found in the Stellar Cyber documentation and has a dedicated TCP/UDP port for identification of the log in the Modular Sensor. Now here's an example on the Stellar Cyber platform of how a log forwarder is enabled and configured. Log Forwarder is enabled by default and configured in sensor profiles. To get to Modular Sensor profiles, you select **System** from the top menu and then **Sensor Profiles** on the left-hand side. You find a sensor profile that you want to modify and click the pencil or I should say again, select the pencil icon to edit a Modular Sensor profile.

**04:00-05:00**

The Log Forwarder module in the Modular Sensor profile has additional settings that can be configured by selecting the **Log Forwarder** option. One of the options you'll see is the **Worker's Batch Size**. You should not have to modify this unless directed by our Stellar Cyber Customer Success or Support teams. But you can set the maximum number of records that can accumulate before they are sent to the Data Sink for storage. The default is 100 records, and the range is between one and 10,000. This allows for batching of logs to improve transmission efficiency. One hundred should suffice for most deployments. Data filtering or log filters. One of the other options is to add log filters to filter out logs that are not needed before sending the logs to the Stellar Cyber platform. This is helpful to optimize the amount of data that needs to be sent and ingested by the Stellar Cyber platform. We'll cover more on this later.

**05:00-06:00**

There are no log filters by default. If you do not have log filters, you can leave the sensor profile configuration by selecting **Submit**. Then create log filters as described in the next set of slides. After, come back to the Modular Sensor profile to configure the log filters and assign them to this Modular Sensor profile. As you can see in step number three, to create log filters go to **System | Log Filters**. On the **Log Filters** page, select **Create** as you see in step number four. Specify the following log filter parameters. Enter the log filter name. Enter a value that describes which data logs are being filtered.

**06:00-07:00**

In this environment, there are some QA test devices that use speed tests and some other tests that I do not want to log on the Stellar Cyber Modular Sensor or on the Stellar Cyber platform. Therefore, I will ensure that the log filter name is describing the logs that I am trying to filter, such as exclude QA test services. For the tenant, select **All Tenants**, or if you are only enabling this log filter for a specific tenant, then you select that. Next, for the Log Source, this is optional. You can select a specific product or technology from the drop-down list. In this case, if you are not excluding specific products or technologies, but are instead

excluding specific servers with specific field names within logs, the Log Source does not need to be specified.

**07:00-08:00**

Next, as you see in step number four, select **+Condition**. In step number five, select a field name from the list, such as computer_name. For step number six, for the operator, select an operator, such as “is,” “is not,” or “contains,” or “does not contain,” or the field just actually exists. For step number seven, enter the values. Type in values, such as the prefix of a host name for servers, then press **Enter**. Every time you type in a new value, you can press **Enter** again and that will actually add the value to the Values field.

**08:00-09:00**

For step number eight, optionally add additional conditions that are logically ANDed with the other conditions by selecting **+Condition** again. In this case, I have added another value called dstip, and the operator is “matches.” The IP is 10.20.1.99. With this additional parameter, The log filter will exclude logs that have a computer_name field with a value containing QA1-test or a value containing QA-test and have a destination IP value of 10.20.1.99, which is a test server.

**09:00-10:00**

In other words, if the log received has a field name called computer_name with a value of QA-Test contained in the name of the value, and the destination IP field in the log is 10.20.1.99, do not ingest that log. Likewise, if the log has a computer name field that contains QA-Test and has a dstip value set to 10.20.1.99, then that will also be excluded. When you are done, select **Submit**. Then back on the **Log Filters** page you can see your new log filter was added. If you would like to create additional log filters, go ahead and select **Create**. Now it’s very important to know here that having the log filter does not mean logs are being filtered.

**10:00-11:00**

You actually have to assign the log filter to a Modular Sensor profile under **Log Forwarders** for the log filter to take effect. Also note, if you select the stacked line icon, for a column, you can autosize the column to view the entire column name. As I just mentioned, to apply log filters to a Modular Sensor profile, go to **System | Sensor Profiles**. Edit the Modular Sensor profile. For step number three, select the space for **Exclude by Filters**. When you do that, you’ll have a selection box. In the selection box select your log filters, then select outside of the selection box to close it.

**11:00-12:10**

Then select **Submit** to save your settings. The Modular Sensor will then load the new profile with the new filter settings and will no longer ingest those types of logs. Now for some

troubleshooting tips. If you are on the **Log Filter** page and you notice that there is no trash can icon next to a log filter and you're trying to remove the log filter, the cause for this is that the log filter is in use. You will have to go to your Modular Sensor profile and deselect the log filters from the **Log Forwarding** section. Then after you do that and save the Modular Sensor profile, you will then go back to the **Log Filters** page and the trash icon for deleting an entry will exist.

This lesson is complete. Thank you for watching.

# L10 - LOG FORWARDER EXAMPLE

## L10A-PRES: Log Forwarder Example (05m:35s)

**00:00-01:00**
The next lesson is *Log Forwarder Example*. This lesson, *Log Forwarder Example,* uses a pfSense firewall as an example to describe the entire end-to-end process of configuring log forwarding for the Modular Sensor to ingest pfSense logs in order to parse and normalize into Interflow records and send to the Stellar Cyber Open XDR platform.

Here I'll provide a real example of configuring a Modular Sensor to be a log forwarder for a pfSense firewall. In this example, the goal is for the Stellar Cyber platform to obtain metadata from the pfSense firewall as a data source for use in analysis and threat detection.

**01:00-02:00**
The pfSense firewall can be configured to send its log messages to the IP or hostname of the Modular Sensor along with the specified TCP/UDP destination port that we'll find in the documentation. The Modular Sensor uses the port specified to identify the logs from pfSense to parse and normalize them into Interflow records. The specific TCP or UDP port to configure on the pfSense firewall to send logs to the Modular Sensor is available in the Stellar Cyber documentation at https://docs.stellarcyber.ai. When the **Documentation** page appears, in the search box, you can search for log parser ports. As an example here, I search for log parser ports.

**02:00-03:00**
Then the entry appears called log parser ports, which you can then select. In the documentation use your browser search or command f or control f to find pfSense. When you find pfSense in the table, the first column is the name of the log source, which is pfSense firewall, the second column is the TCP/UDP destination port number, which is 5543. With this information, you can configure the pfSense firewall that will forward the logs to the Modular Sensor IP, which is in this case 10.20.1.15 on port 5543. In the pfSense user interface, from the top menu bar as an example, you can select **Status | System Logs**.

**03:00-04:00**

Then you select **Settings**. For the log message format one of the options is Syslog (RFC 5424, RFC 3339, microsecond-precision timestamps). You can use that log message format, which is selected by default. You scroll down a little bit further you'll see Remote log servers. If you have an IP colon port field available, looking at this example, here's one that's empty. It says IP colon port. If you do not specify a port, port 514 is used by default. Here I did have a free entry, so I typed in 10.20.1.15:5543

**04:00-05:00**
which is the IP address of the Modular Sensor: the port Sensor found in the Stellar Cyber documentation for the pfSense firewall. Modular Sensors identify logs that come in on TCP or UDP port 5543 as pfSense firewall logs so that they can parse them. For remote syslog content, select the logs you want to forward to the Stellar Cyber platform or select **Everything** to send all logs. As this is an example, I am sending everything. In real life if you do not need to have all the logs sent to the Stellar Cyber Modular Sensor, then you can select individual logs. Next, save the configuration. The pfSense firewall is now configured to forward logs to the Modular Sensor IP on 10.20.1.15 with port 5543 as shown in this example.

**05:00-05:35**
The Modular Sensor uses the port to identify the logs as pfSense to parse and normalize into Interflow records. The Modular Sensor forwards the Interflow record in JSON format to the Stellar Cyber platform. That's how it's done. This can be done in similar ways for any product.

This lesson is complete. Thank you for watching.

## L10B-DEMO: Log Forwarder Example (28m:16s)

**00:00-01:00**
This lesson continues with demonstrations for log forwarder example. This lesson is a demonstration of log forwarder configuration and validation using a pfSense firewall as an example.

Hi everyone. In this lesson, I will show you how to configure the Stellar Cyber Modular Sensor as a log forwarder and give an example of using a pfSense firewall that will send logs to the Modular Sensor, which will then be sent to the Stellar Cyber platform. We'll then verify that the logs are making it correctly. Alright. Let's do it.

**01:00-02:00**
First thing I'm going to do is look at the Modular Sensor profile. I'll go to **System | Sensor Profiles**, and remember yesterday or for me it was yesterday, I created a new Modular Sensor profile called SC Modular Sensor new, go take a look at this one. What we see here is I do have the log forwarder enabled. If I take a look, if I expand it, I have Workers, which I don't need to modify, and I have Data Filtering, which we will not touch this until later in this lesson. What data filtering will do again, is make sure that I can filter out applications or, for example, the communications between the sensor and the Stellar Cyber platform that I don't actually need to ingest into the platform.

**02:00-03:00**

Alright. But for now, look–I actually don't have to do anything because by default, the Log Forwarder is enabled. Then if I take a look at the Modular Sensor itself from **System | Sensors** and I take a look at my Modular Sensor. I'll just expand this window a little bit and click the name of the Modular Sensor. Here, it is definitely the right profile. It is SC Modular Sensor New. Again clicking into that I see that the log forwarder is there. Okay. The next step is really what I do on the pfSense, but I do need to figure out what port for Syslog, I need to send to the Modular Sensor.

**03:00-04:00**
One of the ways I can do that is open up the Stellar Cyber documentation. That's under the question mark right from the platform. If you're inside the platform, I can go right to Knowledge Base home or you can go to docs.stellarcyber.ai. What I want to look for now is log parser ports. It doesn't matter the case I type. I can use uppercase or lowercase. It doesn't matter. I can look for log parser ports. If you don't want to see the highlighting, I can select the highlighter right here and that'll remove the highlight from the search.

**04:00-05:00**
I then can scroll down in this help document, which is really, really useful. I can look for the device, which is my pfSense firewall. There are hundreds of devices here, and I can scroll down or I can just search for it but I think it's funner. I know funner's actually not a word, but that's okay. It is more fun to scroll down and look at all the different device sources that are available. If I scroll down further and go right to here, there's some Palo Alto. What I am looking for is pfSense and there it is. So the name is pfSense, the port is 5543. The message origin source is pfSense_FW. We'll need that for later.

**05:00-06:00**
The device category is firewall. The indexes or indices, you can use either index or indices or indexes it's fine. Traffic, if the logs have a source IP, a source port, a destination IP a destination port and a protocol. The log will be sent to the traffic index. Otherwise, all the rest of the pfSense logs will be sent to the Syslog index. On top of that, the Stellar Cyber platform will look for threats and create Alerts from these logs. So you might also see traffic in the Alerts index. Alright. Now that I have 5543 do you think you can remember that? 5543. Okay. I will go to my pfSense firewall.

**06:00-07:00**
Again, the reason why I'm showing you pfSense is because it's free, and you can download pfSense and test pfSense. It's a great firewall to use for testing. If I go to **Status** and **System Logs** you'll see that there are many types of system logs available. There's firewall logs as well and DHCP logs and Authentication logs, all depending on the services I have enabled on the pfSense. But there we go. What I want to go to right now are the **Settings**. In the **Settings**, add the log message format, I'm going to use the syslog message format. If I scroll down there are many different options.

**07:00-08:00**
What I'm looking for in the configuration is where I can send logs remotely. Right now I have logs already being sent to another sensor, which is on the 10.20.1.19, but I also have a sensor with a different IP address. Let's take a look. I have a sensor on my network and I'm going to use the command line of my Mac to access my sensor, which is at 10.20.1.15. And I can log in and type in `show`

`interface`. Here I see yes that it's 10.20.1.15. But again, I can also go ahead and go to my platform and see it here as well.

**08:00-09:00**
Just by going to my platform, go to **Sensors**. If I select my sensor here you see in the pop up here, it's 10.20.1.15. I have all the information I need. Do you remember the port number? I do. It's 5543, but you know what? It's better to double check. Yes. 5543 in the documentation. Excellent. I'm ready to go. I'm going to send it to 10.20.1.15:5543. Usually you have multiple servers or multiple log servers you can send to, just make sure you have an option available. You can keep your other remote log servers as well and add Stellar Cyber as an additional log source.

**09:00-10:00**
In this case I'm going to log everything. Now in real life, I would only log what I need to log. But for now, to get more data, I will just log everything. That's it. I will now select **Save**. There's nothing else to do except verify that it's working. Alright. I will go back to actually I'll go to my sensor first, and I can type in `show log forwarder`. Alright. I do see that pfSense is here. Now I had pfSense running already. I do have some data here.

**10:00-11:00**
But if it's a new installation, you will definitely see this start to increase from zero all the way up. Now I will go to my platform. Do that. I'll go and scroll down here. I can look at **Log Forwarder Stats**, **Log Forwarder Summary**. Again, I can see that I do have log forwarder data coming in. Now the other thing I can do is look at the documentation again. Then the documentation for pfSense, again this is called the message origin source. If I scroll up in the documentation all the way to the top of this table. Here we go.

**11:00-12:00**
Msg_origin.source. This can be used for a search. If I go into the Stellar Cyber platform if you are searching for data in the platform, the best place to go is I**nvestigate | Threat Hunting**. Now here this page defaults to showing you the **Alerts** indices. Again the Stellar Cyber platform takes all the log messages and depending on the type of message categorizes them into one of the indices you see here. This makes it much easier to search for data more efficiently. I know right now if they're by machine learning or by heuristic rules or statistical analysis rules

**12:00-13:00**
that Alerts may be created by the platform, but also, I could have some Syslog, and I may have some traffic as well. Again, I didn't have to memorize that If I look for pfSense. Let's do a search. Doing the search in my browser with the command f or control f, I can then find the line. Here I see the traffic index and syslog index. Again, I don't have to memorize any of this stuff. It's helpful if you memorize it. But again the documentation is always there for you. Alright. Select off this pop up to close the pop up. Now this is the search box. I can type msg_origin.source, or is it the other way around? Did I do it right? Let's take a look.

**13:00-14:00**

I will search for message or how about origin? Yeah. msg_origin.Source. Again, I don't have to memorize this stuff. It's easier if you do though. Use colon and then pfSense_firewall. I'm looking for all messages that have a source of pfSense_FW. Right now again I've had this connected for a while. I disconnected it for this demonstration and I brought it back in. But right now the time is 9:35 in my lab. Right now I have logs from 9:05. Let me see if that's the latest. There we go. Now I just filtered on the time.

**14:00-15:00**
Today is April 5th and now I see 9:35. I am receiving the logs again. Alright. Now they are coming in. If I take a look at one of these logs I see ICMP, which is ping essentially. There's DNS requests. There's tcp3389. tcp3389 is I think that is remote desktop, right? If we go to 3389 port, yes Windows remote desktop. Again, you don't have to remember everything. If I look at these messages here and I scroll through

**15:00-16:00**
I can see all the Interflow records that are coming in. The JSON Interflow. Here's the cool thing. Again, the logs have now been normalized. If I take a look, SRCIP is the Stellar Cyber normalized field name for the source IP address. Take a look here. Stellar Cyber has added the dstip geolocation. We've added the source IP geolocation. Again, this does not come from the sensor itself. It's when the data arrives or the Interflow record arrives at the Stellar Cyber platform. We will, or the platform will, not we, but the platform will add the geolocation. That's really cool. There's some other really cool things that are added as well.

**16:00-17:00**
Let's scroll down through this JSON Interflow record. We also see some other geo information. If you scroll down here, I can see the log information. Scroll down further. Again, yeah the message origin source, which we looked for earlier. We have the message_origin.source. This is, if you take a look, this is the top of it. So you'll message origin. And then underneath it you always do the dot source or message origin dot category or message origin dot processor so I can search for any of these. If I scroll down further, what else do we have? We also have some of the reasons why this log is here. This was a matching firewall rule.

**17:00-18:00**
I matched the rule. If I go down any further, yeah there's is there a there a type? Let me take a look here. Let's say type. Yeah. Source IP type That's something else that Stellar Cyber adds during the enrichment process. We enrich with the geolocation. We enrich with the source IP type. Before I mentioned that there is RFC 1918 addresses. RFC 1918 addresses are, for example, anything in the ten dot zero dot zero dot zero slash eight or the one seventy two sixteen dot zero to zero slash twelve or the one ninety two one sixty eight zero to zero slash sixteen.

**18:00-19:00**
But if it's not that or not what's called the multicast IP address, then it's public. In other words, everything on the internet almost is public. We add that. We also take the IP address for the IP and the destination IP, and we look for the geo, excuse me, the threat intel. Are the IP addresses good IP

addresses? This could be good. It could say Tor node. The reputation is incredibly important. This record gets enriched with that data. Again, the log itself did not have any of this information but Stellar Cyber enriches all the data that comes in. When it creates the Interflow, it adds on all this essential information that is really useful for looking for threats.

**19:00-20:00**
Then Stellar Cyber statistical analysis rules and heuristic rules and machine learning can use all this information as well to look for a threat. Look at what else we added. This definitely was not in the log, but we actually have the username. Who was doing the remote desktop. This is Jerry G. It's actually Jerry Guthmiller. He's my head of enablement. He's performing remote desktop, and his name is in this log. Again, this is all enriched data which is really cool. Alright. I'll close this out, and we know the logs are showing up. That's great. Okay. Now that I've shown you how to use the message_origin.source to look for the pfSense firewall.

**20:00-21:00**
I'll remove this message origin source so you can see what happens if I don't have that selected. Now that I've removed the filter, I can see all the sources coming in. Now in these charts, I have top log sources, I have top sensors. I have documents over time which are Interflow records over time essentially. This shows you the indices, which are Syslog traffic and Alerts. If I scroll in I can actually see that I do have a little bit of light purple there, which is the Alerts. I have the traffic and Syslog. That's the breakdown of each. Right now I'm actually looking for relative time in the last five hours. Okay. Next, if you look at the **Top Log Sources**, one of them is the pfSense firewall.

**21:00-22:00**
If I zoom in a little bit here for you, I can select the plus or the minus. Plus says filter for the pfSense firewall, which is the message origin source. And minus says filter out the pfSense firewall. In other words, on the display I will not see this information. Well, I want to look for just pfSense firewall. I can select it here. If I can scroll back now, all these messages are from what I see from the pfSense firewall in the last five hours. Okay? I will then go to **Filters** and I see the log source was added. By clicking that plus next to pfSense firewall, it added the log source pfSense _FW. I can actually copy this as a query to see what the query looks like.

**22:00-23:00**
I will remove this from the filter right now and close the filters. Now if I paste what I had in the filter and hit Return, I can then see, and I'll zoom in for you as well here, that the message_origin.source is pfsense_fw, or message underscore origin dot source dot keyword is quotes, pfsense_fw. Now the quotes aren't actually needed in this case. There are reasons to use quotes, but pfSense_firewall is one term. Since it's one term, I don't need quotes. Quotes are used to basically specify you have a phrase and we don't really have a phrase. You can use it with or without quotes. You get the same exact results.

**23:00-24:00**
Here, the dot keyword means it must be the exact case. Let me give you an example of that. I'm going to remove this portion and you'll get the same one hundred and thirty thousand, two hundred

and fifty one. But since its message, now underscore origin dot source dot keyword, if I were to say with a capital P, I get nothing because it must be an exact match. That's what dot keyword means. If I get rid of dot keyword and put the colon in here, watch what happens. If I say capital P, there's no change. I could even make this capital FW. No change. When I don't use the dot keyword, the searches are case insensitive.

**24:00-25:00**
The field name is case sensitive, but the value is not case sensitive. That's a good thing to know. We'll cover this over and over again in these examples. Again the other thing is what happens if I don't select the index? So let me leave this page. I will go to the XDR Kill Chain. I'll go back to the XDR Kill Chain. The reason why I'm showing this is as you're analyzing threats or doing some investigation, you may switch between the different pages on the system. Notice though that my filter is preserved. Message origin source is still there. But if I go back to **Investigate | Threat Hunting,** again I see my filters preserved. I'll expand it so you can see more.

**25:00-26:00**
But now I only have 13 Interflow records showing. When I was looking at it before, I had over a hundred thousand Interflow records. I'm looking here and I go oh maybe it's the time frame but no I was using five hours before. What's the difference? And then if? you take a look and I'll scroll down here you can see the uh in this case there's now 13 records. Because it is a live system. As I move around more will increase. As I'm going through now and I go wait a minute, I am seeing only Alerts. That's because when I go back to the **Investigate | Threat Hunting** page, the indices section defaults to **Alerts**. I can go back down and select **Syslog** and **Traffic**.

**26:00-27:00**
Now I'm back over a hundred thousand again. I thought I did something wrong actually. Yeah. I really didn't. But you know it's really important to understand what happens. Just so it's clear now, let me go back to Visualize. I'll go to another page, maybe **Analyst View**. I'll go back to **Investigate | Threat Hunting**. Again every time you come to this page, it defaults to Alerts. You got that. I hope so. Alright. This is actually another important thing to understand about this page as well. Let's say for instance, I have Alerts but I'm looking for the past week. That could be a lot of data in a production system. But look what happens if I select multiple indices: **Syslog** and **Traffic**.

**27:00-28:16**
Now the time interval is the last 24 hours. The last 24 hours is the maximum time you can search when multiple indices are selected. If I were to bring this back to just for example, Traffic. Now I can go back and set this for a longer period of time. Note that as well. On a single indice you can search pretty far back. Let's go over here. I can search for the last week, for the last month. Whereas once I put multiple indices in, now I have Syslog and Traffic, the max I can go to is 24 hours. Okay. You got that? Awesome. Okay. I will clear this filter out now and I hope you enjoyed this lesson. Thank you very much.

This lesson is complete. Thank you for watching.

# L11 - LOG SOURCE CONFIGURATION FOR SINGLE TCP OR UDP PORT

## L11A-PRES: Log Source Configuration for Single TCP or UDP Port (05m:55s)

**00:00-01:00**
The next lesson is *Log Source Configuration for Single TCP or UDP Port.* This lesson, *Log Source Configuration for Single TCP or UDP Port,* shows how to configure the Stellar Cyber platform to enable the Modular Sensor to receive logs from systems using a single port, such as the default Syslog port 514.

In this lesson I will cover what happens if you do not have the option to configure a log port on a log source, and you have to configure the default Syslog port, TCP or UDP 514.

**01:00-02:00**
As we demonstrated previously for the log parsing to function, the Modular Sensor needs to identify the log and the product that it belongs to so it can assign a parser. If you send logs to the Modular Sensor on port 514, which is the default, the Modular Sensor then needs to identify the log from a specific vendor. Stellar Cyber has a way of accomplishing this. Stellar Cyber has a log source configuration that provides you the ability to map log sources to parsers, which gets configured on Modular Sensors. To configure log sources, go to **System** on step number one, **Log Sources** on the left-hand side. Now I will configure the log source by selecting **Create**. This will open the **Add Log Source Configuration** dialog.

**02:00-03:00**
The first option I will configure is the name. In this case, I am using only port 514 on the Check Point firewall to send logs to the Stellar Cyber Modular Sensor. This will be a perfect device to do that with. The name will be “Check Point Firewall.” Next, I will specify a tenant name. If I have multiple tenants, this will be the tenant that has the Check Point firewall inside. In my case, it's the root tenant. Next, I'll specify a log source address. Here I type in the IP address for each log source device and enter **Return**. In this case, the log source devices must be Check Point firewalls. You can enter multiple after every IP address, you hit **Return** or **Enter**. Type in another IP address.

**03:00-04:00**
Hit **Return** or **Enter** and it creates a bubble box for each IP address you enter. Note that the addresses from the log source to the Stellar Cyber Modular Sensor cannot be NATed or Proxied. Otherwise, the IP addresses will not work. Next, I will specify the ingestion port. This is the port for incoming log messages from log source devices. The default Syslog port for UDP and TCP is 514. If you are using Syslog with TLS, you can set it to 6514. The other reason to set the ingestion port is if your Check Point firewall or any other type of device is using a different ingestion port, you can enter that here as well. When you're done, select **Submit**.

**04:00-05:00**
The settings are automatically applied to Modular Sensors and the root tenant in this example. Modular Sensors will now be able to parse and normalize Syslogs from Check Point firewalls coming in on port 514, as long as they match the IP addresses specified. From the Modular Sensor command-line interface, you can view the log sources with the following command: `show logforwarder port-ingestion`. Here I see the rule that's added to the Modular Sensor. For example, on lines seven, eight, nine, and ten, I have the different IP addresses 10.20.1.241 and 10.20.1.240. On TCP and UDP port 514 will get mapped to the parser port 5519

**05:00-05:55**
which is the port the Stellar Cyber Modular Sensor can use to identify the Check Point firewall parser. Now as an example, here's the Check Point firewall in the diagram. That is 10.20.1.241. The Check Point firewall is configured to forward logs to the Modular Sensor IP 10.20.1.15 with port 514, which is the default Syslog port. The Modular Sensor uses the port ingestion table to identify the logs from Check Point to parse and normalize it into Interflow records. Then the normalized logs as Interflow records in JSON format are sent to the Stellar Cyber platform.

This lesson is complete. Thank you for watching.

## L11B-DEMO: Log Source Configuration for Single TCP or UDP Port (05m:28s)

**00:00-01:00**
This lesson continues with demonstrations for log source configuration for single TCP or UDP port. This lesson gives a demonstration of how to configure the Stellar Cyber platform to enable the Modular Sensor to receive logs from systems using a single port, such as the default Syslog port 514.

Okay. In the slide presentation I explained about the log forwarder, but what happens when the port cannot be set?

**01:00-02:00**
In other words, I learned the port of the pfSense firewall, but maybe I have some devices that I can't change the port or I'm working with a a company that I am managing as a managed service provider and they don't have the ability to change the port that's being that's being used to forward traffic to a Modular Sensor to identify the parser for log ingestion. Well, Stellar Cyber also has a method to solve this, which is our log sources capability. If I go now to **System** and go to **Log Sources**, from the log source configuration, I can now make it so if a log comes to a Modular Sensor on the default TCP UDP port of 514, or maybe a different port that's being used by the organization, I can map that to the correct parser.

**02:00-03:00**
Let's give an example here. I'll go to **Create** and there's a list of parsers here already, which are essentially log sources that Stellar Cyber has parsers available for. As an example here, I'll just choose the Check Point firewall like I used in the slides. Now I can say alright. For the root tenant or maybe a specific tenant, depending on your organization, or if I'm managing an organization and that one organization cannot change their port number I can do it just for the one tenant. Now I'm going to specify a specific IP address for the log source in the root tenant: 10.20.1.240.

**03:00-04:00**
Maybe I have another Check Point at 10.20.1.241. Now, and I click away from the selection box to clear it, now I say okay. If in the root tenant in this case, I have a log source of 10.20.1.240 or 10.20.1.241, sending logs on the default port of 514 using UDP or TCP, that is Check Point firewall log traffic. Alright. Then this Modular Sensor can apply the parser for the Check Point firewall. Now I can select **Submit**. That's all I need to do. I don't need to do anything else. Modular Sensors will automatically be updated.

**04:00-05:00**
If I go to **System**, I go to **Sensors**. There's nothing red here. I don't need to modify anything. But if I now go to my sensor and I type in `show logforwarder` and I can do a question mark here and go oh what was the command? `show logforwarder port-ingestion`. It's not here yet. There we go. It looks like it takes a moment or two. There it is. I now see that these rules got added. One for each UDP and TCP from 240 to 241. That's cool.

**05:00-05:30**
Now I can parse traffic and I don't have to make the specific port for a parser. I think it's easier just to use the parser port and I think it's a recommendation from Stellar Cyber to use the parser port. But in cases where you can't, Stellar Cyber has you covered. Okay.

This lesson is complete. Thank you for watching.

# L12 - VALIDATE LOG FORWARDER

## L12A-PRES: Validate Log Forwarder (07m:46s)

**00:00-01:00**
The next lesson is *Validate Log Forwarder*. This lesson, *Validate Log Forwarder,* provides the steps to validate the parsed and normalized logs are being ingested by the Modular Sensor as Interflow records in the Stellar Cyber Open XDR platform.

You might be asking yourself, “How to validate that the log forwarder is working?” Or you might be asking me, “Hey Paul. How do I know that the log forwarder is working?” Well, there are several ways to validate the log forwarder’s working and that the Stellar Cyber platform is ingesting log data. From the Stellar Cyber platform, if you select **System | Sensors** and select the name of your Modular Sensor in the list, you will see a page with a bunch of details.

**01:00-02:00**
You can then scroll down and select **Log Forwarder Summary**. You also want to make sure that the time frame is relevant. In this case, I will say the time type is **Relative** and the time interval is in the **last 24 hours**. Looking at the bottom now under the log forwarder summary, I see that I have plenty of logs coming in. Another option to validate log forwarder data is being ingested by the Stellar Cyber platform is from the **Operational View**. If you go to **Visualize | Operational View**, then in **Event Origins** look for pfSense. Here's the **Event Origins** chart.

**02:00-03:00**
There I see Linux agent Modular Sensor, and pfSense_firewall. Perfect. For the pfSense firewall, I can select the plus right next to it, which will filter for all the events that are related to the pfSense firewall. On the top left, on the **Filters**, you'll see a little bubble pop up. That lets you know a filter has been applied. Next, if you select **Filters**, the filters bar will appear. On step five you'll see the pfSense filter was added, but you yourself may not know what that actually means. What does it mean that the pfSense firewall is added as a log source?

**03:00-04:00**
You can actually figure out what query that actually is. One of the options is to as step number five, **Copy as Query**. On step number six, after selecting **Copy as Query**, select the **x** to remove the pfSense firewall query. This will make sure it doesn't interfere with some of the tasks we're doing in a few minutes. Next, on step seven, I can paste the query. There you see again the key value pairs. We have message_origin.source of pfsense_firewall, or message_origin.source.keyword pfSense_firewall. Now keyword, we'll talk about much, much later. Keyword is just saying that it must be an exact match. That's all the dot keyword means in this case.

**04:00-05:00**
Next, if you scroll down after entering the search, on the **Operational View** page, you'll see all the Alerts associated with the log source specified in the search. Note that Alerts have an **Alert Type** and **Alert Score** fields, which can also be column headers. This can help you determine how critical the Alert is. You can also view the Interflow events and Alerts from the log source by going to **Investigate | Threat Hunting**. Once you're in **Investigate |**

**Threat Hunting**, next to **Indices**, you can select the **Alerts** button. It looks like an **Alerts** button by default because that's the default indice that's specified.

**05:00-06:00**
But if you select **Alerts**, you'll get a drop-down list with many other indices. In this case, we know that the pfSense firewall from our documentation will have logs that are categorized as Syslog or Traffic and also Alerts that are generated by the Stellar Cyber platform. In this case, I've selected all three Alerts, Syslog, and Traffic. To close the pop-up box, select outside the pop up box to close it. Next, on the top search bar to the right of **Filters**, select the **L**, which is for the Lucene query. This will expand the search box. Here you can see that the same query that we copied previously is still applied.

**06:00-07:00**
If you select **L** also it will bring up a little pop up letting you know what Lucene actually is. Here, you see the legend shows data is received for Syslog, Traffic, and Alerts. If you zoom in, you can see that all three colors are there. If you look at the middle of the chart you will see that there are some Alerts there that have the lighter purple color. You can also select a single indice from the indices selection box to view events for specific indices. For example, I can look for just Alerts, or just Syslog, or just Traffic from the pfSense firewall. You can also validate log forwarder by viewing details on the Modular Sensor by going to **System | Sensors**.

**07:00-07:45**
If you select and A**dd a Checkbox** next to the Modular Sensor of your choice, you can then select **CLI access** from the CLI pop-up window, or from an SSH to your Modular Sensor, you can type `show logforwarder`. Logforwarder is one word. You wait about 10 to 20 seconds, and you'll get the result back. Here, I can see a number of logs associated with the pfSense_firewall. I know that it is working.

This lesson is complete. Thank you for watching.

# L13 - LOG FORWARDER PRESERVING RAW LOGS

## L13A-PRES: Log Forwarder Preserving Raw Logs (05m:17s)

**00:00-01:00**
The next lesson is *Log Forwarder Preserving Raw Logs.* This lesson, *Log Forwarder Preserving Raw Logs*, shows how to configure Modular Sensors to include the original raw log message in the Interflow record that is sent to the Stellar Cyber Open XDR platform.

You might be asking yourself, can the Stellar Cyber Modular Sensor log forwarder function preserve the raw log that it received from various security and infrastructure tools? The answer is yes. Raw logs can be preserved by the log forwarder by configuring the log forwarder settings in the Modular Sensor profile.

**01:00-02:00**

If you go on the top right on step number one to **System**, then select **Sensor Profiles**. You can then select the pencil icon to edit a Modular Sensor profile. As described in the log forwarder lessons, Stellar Cyber normalizes all log messages into Interflow records in JSON format. However, the original raw log is not preserved by default because metadata from raw log was already collected and normalized into Interflow records, there may not be a need to preserve the raw logs, and saving raw logs might actually cause a higher level of ingestion on the Stellar Cyber platform. However, sometimes organizations have a need to preserve the original raw log, and we can configure that. If there's a need to preserve the original raw log within the Interflow record, you can enable **Raw Log Capture** in the log forwarder configuration.

**02:00-03:00**

Under **Data Filtering**, you then go down to **Raw Log Capture** and slide the lever to the right to enable raw log capture. After completing your configuration settings in the sensor profile, you must select the **Submit** button to apply the settings to the Modular Sensor. If all was successful, you'll see, "Sensor profile was edited successfully." After this, it will take a minute for the configuration to be reloaded on the Modular Sensor. Next, in the Stellar Cyber platform, to view the raw log within an Interflow record, go to **Investigate | Threat Hunting**.

**03:00-04:00**

Next to **Indices**, in this example I will select **Syslog**, **Traffic**, and **Alerts**. You can select **All** until you know which indices contain the data you need. Next, in the search toolbar type `parser_raw_message` as the field name `:*`, which is a wildcard. I'm saying look for all the Interflow records that contain the field name parser raw message. Next, if I scroll down to the event list called documents here and select **i** in the **Action** column for an event, I can view the Interflow record. In the Interflow record, select **Details**. Now I can scroll down to find parser_raw_message.

**04:00-05:16**

There I'll see the raw log message in its entirety as it would appear without normalization and enrichment. Note that the name of the IP fields in the raw log for example is source IP has SRC_IP. Destination IP has DST_IP. Here I can validate that normalization happens. Here the source IP now in the Stellar Cyber normalized portion is SRC IP as a field name colon, the IP address. The destination IP is DSTIP colon, the destination IP address. This is a perfect

screenshot for showing normalization. If we did not have normalization, you would get what you see in the raw logs from every single device that you have integrated into the platform. Normalization is essential for being able to find information quickly and for machine learning, statistical analysis rules, and heuristic rules to be applied.

This lesson is complete. Thank you for watching.

## L13B-DEMO: Log Forwarder Preserving Raw Logs (19m:26s)

**00:00-01:00**
This lesson continues with demonstrations for log forwarder preserving raw logs. This lesson is a demonstration to show how to configure Modular Sensors to include the original raw log message in the Interflow record that is sent to the Stellar Cyber Open XDR platform.

Hi everyone. For this lesson, I want to show you how to preserve the original raw message or the original raw log that gets sent to the Modular Sensor from a log source. Now this might be needed by some organizations for some compliance reasons.

**01:00-02:00**
It might be needed because SOC analysts within an organization are used to the original message, and they know they understand the fields of the original message. Whatever the reason why you want to preserve the original raw message before it gets normalized and before it gets enriched, I can show you how to do that. Now what we'll do here is go to **System | Sensor Profiles**. Again I'll modify SC Modular Sensor New. We'll go to **Log Forwarder**. Here I can slide the option to enable **Raw Log Capture**. I'll go ahead and submit that. If I go back to my sensors list, I will now see that–

**02:00-03:00**
–as we always see now, this sensor will be restarted using the new configuration. While that's happening though, I'll go to **Investigate | Threat Hunting** Okay. So now I want to search for the parser_raw_message:*, and I see that I don't have any messages yet coming in with the parser raw message field. Actually, maybe I do. Does anyone see what could be happening here? Every time I go to the **Investigate | Threat Hunting** page on the Stellar Cyber platform, we default to the **Alerts** indice. What we want to be looking at right now is either the **Traffic** or the **Syslog** indices. Keep that in mind.

**03:00-04:00**
Now if I were to specify **Traffic** for the indice, take a look now. I see messages that contain the parser raw_message field. If I take a look at one of these now, I'll take a look at this one. One of the best ways to see this is from the **Details**. I can look for the parser raw message field. Let me scroll down and there it is. I see the log type, this is the firewall, I see the firewall

rule. I see exactly how it comes to the Modular Sensor. You'll notice for example the desk IP is dst_IP. We see the dst port as DST_port.

**04:00-05:00**

Let's see if we can find the source as well. There must be a source port in here. My vision isn't as good as it used to be. The source IP is 192.168.1.103. Okay. This is not normalized. Right? Don't forget if it's normalized, what's it going to be? If you take a look at the source IP here, and look at the source IP here, the source IP field after normalization is SRC IP. Before normalization for every product it could be totally different. That's why normalization is incredibly important. Notice there's no geolocation. There's nothing letting us know that the destination IP reputation is good or bad. Well, in this case, the destination IP is a broadcast. It's a local network message. Let me find a better one to use.

**05:00-06:00**

Let's try DNS or let's try this one right here. That's another broadcast. If I want to find some other messages here, let's take a look. I want something that's not a broadcast. 3389. Do you guys know what 3389 is? 3389 is a remote desktop. If I take a look here, wow! Here I see the reputation of the source is malicious. I see that the source country is Panama. If I go to **Details** now, and see if we can find the parser raw message. In the parser raw message, I just have that it's the following. Right?

**06:00-07:00**

It's TCP. It has an IP address in it. It's actually really, really hard to read or understand. But the fact that we normalized the message and to then readable fields in JSON format, now I can get a better idea. I can look at the destination IP, the region, which is my device. It's in my lab. It's a private address. Then if I look at the source itself, the country is Panama, the reputation is malicious. It learned that through ETPro in DHS. Right? These are some threat intel's we are using. At this point I should take a look and start to block some IPs in my firewall.

**07:00-08:00**

I can also look right here as well. If I go to **Key Fields**–**Key Fields** is, when we are not analyzing the threats, we can make it so that you can look up some of these IPs. For example, I go to VirusTotal and it takes me to a different site. The VirusTotal is saying, "Hey, by the way, this IP address is malicious." That's pretty cool. Right? I can do that right from the platform itself. What port is being used? It is 3389. I want to make sure that I block this IP or make sure that no one has access to my devices internally using port 3389.

**08:00-09:00**

At this point I would check my pfSense firewall. Let me go back in here again. Okay. I can go even further. If I had a Connector to my firewall, I could actually block the IP address right here. I could run a script, I can actually add a log filter. If it wasn't important, I could actually just create a log filter and filter out that type of message from even appearing. I can trigger an email if I ever see something like this again. A lot of things you can do at this point though I know that I will need to block some IP addresses or actually only permit some IP addresses to access my network on Port 3389.

**09:00-10:00**

Again, this is a test lab environment, so that's okay. But you always want to make sure you are safe and using the Stellar Cyber platform you can see that. But again, the raw message itself, it's great to see that you can do it. But you notice there was not much in the raw message. It's the Stellar Cyber platform on top of the raw message that's doing the normalization and enrichment that gives you the value. Now I think it's a good time to do a little bit of digging into the threats I just noticed when looking at the parser raw messages. Again, getting the extra value out of the platform. One of the things I'll do is I don't really care much for the parser raw message anymore. I'll clear that search out. Again, I see all my messages coming into the traffic log in the last 24 hours.

**10:00-11:00**

What I was looking at before was the remote desktop, which is port 3389. If I do a search now for 3389, again I don't need to actually specify a field name, we do 3389, and I see here, here's one right here. If I go into the more info, I see that there is a source reputation of the scanner from this one, which is from the Netherlands. So this source host if I go to VirusTotal for this one, let's see if this is a bad host. Well, this one actually shows as clean, which is interesting but who knows? Maybe it's not clean now. Alright.

**11:00-12:00**

But they're actually doing some scanning on my port 3389. Okay. What I want to do though is see if I could find any other source reputations that are bad. I'll go to the JSON, or, actually I'll use JSON. I can look for the reputation right here. There's a destination IP reputation, but I want the source reputation, let's find that. There it is right here. Search down further. Here we are. Source reputation of scanner. Okay? I go, "Wow! I wish I knew what other reputations there were." I can use this as a copy right now source reputation scanner.

**12:00-13:00**

If I were to copy this, and do a source reputation of scanner, I can now see that I have 927 with a source reputation of scanner. What I can do as well is actually select the **Column**. I can go here and say, reputation. I can make that a column that I have now. There we go, scanner. If I were to clear this out, I have good, and then you had scanner before. But what if

I want to see more? What are the other fields that I've collected? Do I have to search through everything to find the other types of reputations? The answer is no. We have something built in called the Interflow dictionary. I can open the Interflow dictionary and do a command f and search for a reputation.

**13:00-14:00**
There it is right there. I want the source reputation though, so I'll scroll down further. It's also going to hit Reply, but source reputation. If I select this, I see that I have good, or scanner, or malicious. I have a lot more good than everything else. That's what the chart is showing. But if I want to see all the malicious, I can actually use this Interflow dictionary, select the plus here, and then I can close the Interflow dictionary, and every single record I see here is malicious. One of the other options you can do as well, if I find that inside the log here, let's look for the reputation again.

**14:00-15:00**
I'll scroll down here. That's DST IP reputation, Source reputation. There we go. From here you can create a query. If I go up here to the search, and command V, we now will have a query for you to use. The keyword is saying that this malicious has to be the exact case. I don't care about the case, so I'm going to get rid of this OR. But I can say, for example, AND dstport and 3389. All of these are malicious for this destination port. There are a lot of cool things I can do here. But should I be concerned?

**15:00-16:00**
I am seeing a lot of activity for remote desktop that's a source of malicious. Well, let's take a look even further. If I go to the **Details**, this is from my firewall that's reporting it. But one of the most important things here is that there was an action of block. Right? At least I know that the traffic didn't make it. I'm actually logging blocked attempts. Many times you don't log blocked attempts, but I like to do it because I want to know what's happening. But I can go a little bit further on this as well. I know this field name is **action** and **block**. So if I were to say AND–excuse me. Let's try it over here. I should have the same exact results: 411. But again, let me close this out because what I want to know next–I'll get rid of the AND action block–

**16:00-17:00**
–is what other actions there are for that firewall. Again, I can go to the open Interflow dictionary, look at the **action**, which is the first one. I see that I only have an action of **block**. Maybe because I have a filter. Let me go here. That's malicious. Let me get rid of malicious. I'll clear all here. Here we go. Clearing all filters. I see that I have some records with pass, some with block, and some with deny. If I were to say pass and just filter on these, alright

now I can say, okay–let me close all the filters, and I'll close the Interflow dictionary. Again, my filter stays there, it's pass, or I can copy this as a query.

**17:00-18:00**
Close it out, go to **Filters**, paste that in here and it does the same thing. Again, we always give you the action keyword, which is the exact case. I don't care about the exact case. Now I can start combining these things. Okay. What's been good is 53 DNS. Well, DNS has been permitted. Okay. But do I have any action pass for the dst port 3389? I don't. Right? That's good news. We just make sure that dst port is actually correct by itself. Yes. The dst port 3389. But in all these cases, the 6,524 records, if I say action AND action block.

**18:00-19:00**
Yes most of them right there OR action and we'll say a deny. I'll put this in a grouping that says the destination port AND one of these two–either action block OR action deny. Okay. That's the case here. All these records will have action block or action deny. But if I were to say AND action pass, nothing. Okay. That's good news that no one was able to get in or was permitted into, viewing those records.

**19:00-19:25**
Again, I will make sure one more time for the **action** field. Yes. Pass, block, or deny. Okay. That is good to know. Thank you for watching. I hope that you found this lesson valuable.

This lesson is complete. Thank you for watching.

# L14 - NETWORK TRAFFIC ANALYSIS

## L14A-PRES: Network Traffic Analysis (10m:22s)

**00:00-01:00**
The next lesson is *Network Traffic Analysis Details and Configuration.* This lesson, *Network Traffic Analysis Details and Configuration,* explains how to configure to collect network traffic off SPAN ports or TAP devices and provides details of how the Modular Sensor analyzes network traffic to create sessions in metadata and forward it as Interflow records to the Stellar Cyber Open XDR platform.

Here I will describe network traffic analysis. Network traffic analysis is essential for network detection and response, which is done by the Stellar Cyber Open XDR platform. Network traffic from switches is sent to the Stellar Cyber Modular Sensor via SPAN ports or TAP devices.

**01:00-02:00**
Modular Sensors use deep packet inspection, DPI, to identify applications, create a session state, correlate app processes, and add metadata such as the number of bytes used by the application during a session. A summary of the traffic and metadata is normalized into Interflow records and forwarded to the Stellar Cyber platform. Now I will describe how network traffic analysis works on the Modular Sensors. To begin though, I'll describe this network diagram. On the bottom left is Laptop A. It has an IP address of 192.168.1.190. Does anyone remember what type of IP address that is?

**02:00-03:00**
You guessed it. It is an RFC 1918 address, and it is a private IP address. Laptop A connects wirelessly to the WiFi access point, which is then connected to the bottom switch on the left. That is an access layer switch. That access switch connects to a distribution switch on port 2/12. The Modular Sensor's management interface is connected to 1/12 of the switch. The management interface is what's used to connect to the Stellar Cyber platform. Port 1/14 on the switch is connected to Ethernet 1 on the sensor. If we look at the configuration, we've basically said any traffic coming in on port 2/12 of the distribution layer switch, either from or to that port, mirror that traffic to the Modular Sensor on port 1/14.

**03:00-04:00**
Or basically all the switch knows is it's mirroring the traffic to port 1/14. The fact that we have a Modular Sensor on it means that the Modular Sensor can collect that traffic. Next, the distribution layer switch is connected to a firewall. The firewall is connected to a router, which then takes you to the internet. On the internet is where the AWS management console exists. AWS again is a cloud infrastructure company–Amazon Web Services. The IP address of the console used to manage AWS, or my AWS environment. This example was actually the IP 3.3.10.1.

**04:00-05:00**
This is a public IP address. In this example now Laptop A, again with IP address 192.168.1.190 connects to the AWS management console with an IP of 3.3.10.1 using HTTPS, which is TCP port 443. That's a mouthful. Now when that connection is made, the traffic is mirrored. The SPAN port mirrors the traffic to and from the switch on port 2/12 and sends to the Modular Sensor on port 1/14. Traffic from Laptop A, in this case to AWS is captured by the Modular Sensor. Now my laptop will have a screen showing the AWS console.

**05:00-06:00**
Looking at the Modular Sensor now, the Stellar Cyber Modular Sensor performs deep packet inspection, or DPI, on traffic to identify applications. There are over thirty eight hundred applications detected by Stellar Cyber's deep packet inspection engine. On the

sensor, if I were to have CLI access, I can type `show dpi`. Looking at `show dpi`, I can scroll down to look for AWS console. There I see AWS_console has an app ID of 3574, and it is a web category. Modular Sensors create a session table based on IP flows learned from network packets using stateful firewall technology, which includes the following metadata.

**06:00-07:00**
It has the 5-tuple. You'll hear this a lot and see this a lot when talking about firewall routers and switches. The 5-tuple is the source IP address within a packet, the destination IP address in a packet, the protocol, which is either TCP/UDP/ICMP, the source port within a packet, and the destination port within the packet. Together, there's five entries and that's called the 5-tuple. Next, the application is determined from deep packet inspection. Do you remember what that number was? Well, let's take a look. What you're seeing here on the Modular Sensor is the output of the command `show session`. If I look at the `show session`, I will actually see the entry. You'll see the IP address 192.168.1.190, the source port.

**07:00-08:00**
You'll then see the destination IP, which was 3.3.10.1/443, which is the destination port, which is HTTPS, and 6 is the protocol. There's the 5-tuple. The protocol 6, by the way, is TCP 17 UDP ICMP, which is Ping, if you've ever heard of that, is one. Next, you'll see some of the metadata associated with it as well. You'll see the number of forwarded packets and the number of forwarded bytes. You'll also see the backward packets, BWD packet, and the backward BWD bytes, which essentially is the return traffic for the session. But remember we talked about it earlier, there was an app ID and the app ID is 3574. If we go back and look at that table, 3574 was the AWS console.

**08:00-09:00**
Pretty cool. By doing this, the Stellar Cyber Modular Sensor doesn't need to send every single packet to the Stellar Cyber platform. It can determine all the packets are related to this session and start recording the number of bytes and packets that are sent and received, and then that information as metadata can be put into the Interflow to send to the Stellar Cyber platform. When Modular Sensors collect metadata for each network session flow and normalize it into Interflow records in JSON format, it looks something like this. Looking at the JSON Interflow record excerpt, you can see that there's a source IP, which is normalized SRC IP. There's a destination port 443. There's the number of bytes, which is metadata. There's the App Family, which is metadata.

**09:00-10:00**
There's the app ID family, which is metadata, which is web. There's the app path, which is letting you know that it's SSL or HTTPS communication. There's the destination IP and the app ID name. There's a lot more information that goes along with this as well, but this is

what gets sent to the Stellar Cyber platform. To configure network traffic analysis, in the Modular Sensor profile assigned to the Modular Sensor, turning on network traffic enables the Modular Sensor to form network traffic analysis, which again is essential for network detection and response. After completing your configuration settings in the sensor profile, the **Submit** button must be selected to apply the configuration settings to the Modular Sensor. Note there are a lot of options available under **Network Traffic**.

**10:00-10:25**
These are usually left as default. Go ahead and click **Submit**, and then you should see, "Sensor profile was edited successfully."

This lesson is complete. Thank you for watching.

## L14B-DEMO: Modular Sensor Network Traffic Analysis (19m:01s)

**00:00-01:00**
This lesson continues with demonstrations for Network Traffic Analysis Details and Configuration. This lesson is a demonstration that shows how to configure Modular Sensors to collect network traffic off SPAN ports or TAP devices and provides details of how the Modular Sensor analyzes network traffic to create sessions and metadata and forward it as Interflow records to the Stellar Cyber Open XDR platform.

Now I will show you some in-depth analysis that's being performed by the Stellar Cyber Modular Sensors. As an administrator of the Stellar Cyber Open XDR platform, I have SSH and console access to the Modular Sensor.

**01:00-02:00**
SOC analysts never need to go to the Modular Sensor itself to look at the data because it's all in the Stellar Cyber platform that you see here. But I think for learning right now and getting a better idea of how the sensor works, I will SSH to the Modular Sensor to get a command-line interface so you can see exactly what's happening on the Modular Sensor itself. I have a terminal window on my MacBook. I will type in `ifconfig`, which is interface config, and look for en0. In the terminal window now I can see that my IP address of my MacBook is 10.20.1.112. Again, going back to my Modular Sensor that's 10.20.1.15.

**02:00-03:00**
We're on the same subnet. I will now SSH to my Modular Sensor. To do that, you have to have the username, which is Aella in this case. That is the default username, and I go to 10.20.1.15. I'll enter my secret password and I am in. I can type in `show interface` as an example. Here I can see that the IP address of my Modular Sensor's management interface is 10.20.1.15. Ethernet0 connects to my switch for management purposes. Ethernet1 on the

Modular Sensor connects to the switch SPAN port. Alright. Now I will type in `show question mark` as an example.

**03:00-04:00**
This gives you an idea of the commands that you can type in. We have some other commands we'll be using: `show session`. We'll also be using `show dpi`, which is the deep packet inspection information. This shows you all the applications. Later on we'll do some or I think but previously we showed `show logforwarder`. That was a very useful command. So `show dpi` gives you an idea of the applications that the Stellar Cyber and Modular Sensor can identify. Stellar Cyber can identify thousands of applications; I think over 3,800 or 4,000 applications that we can identify on the Modular Sensor. Some examples I can show you. Here's one like on my phone if I was running Apple News, I could see that.

**04:00-05:00**
I can look for oops, let me try that again. Command f. Look for Zoom. If I was running a Zoom call, I could see 2928 is the application ID for Zoom. Another thing that we can look at is I'll be using AWS console. AWS console has an application ID of 3574. Once we identify the application, we'll correlate it with an application ID. That'll be used so that we can track what's going on. But if I scroll to the top of this list here Let me put it this way. We have this. Let me go all the way down to the top of this list here. You'll see that there are lots and lots and lots of applications.

**05:00-06:00**
Okay. Now we see the application ID, the app name, and the app category. For example, you can see Bittorrent would be 15. There is VMware itself. I just saw that appear. VMware is 220. Alright. There's Yahoo. There's Google, Google Maps, everything. Most of the common applications in the world you will see here. Again, AWS console is 3574. What does that mean and why do I care? Well, let's take a look. I'll go to AWS console.

**06:00-07:00**
I'll look around. Now these are all TCP sessions for a web app, so they're not persistent. You will have to be quick to view them in the Stellar Cyber Modular Sensor. I have another terminal window open. I'm doing this so that I can show you the sessions on one tab and then show you the DPI on the other tab. Let's take a look here. I will also open up a connection. There we go. To the same Modular Sensor. Alright. On this one I will type in `show session`. Okay. But what I'll do now, as I told you these sessions are short lived. Here we go. Let me go back in. The sessions are short lived, so I need to make sure I capture it right away.

**07:00-08:00**

Then if I do a `show session` let's see if we can find it. There it is. There's the 5-tuple and lots of other information as well. Here I see application ID 3574 from my Mac 10.20.1.112. My source port going to 13.248.199.77, which is the AWS console on 443. This is just HTTPS with TCP, right? But we're able to identify that it's not just HTTPS. It is HTTPS that is AWS console 3574 and that could mean different things when correlating and looking for Alerts or identifying Alerts. We also see the number of packets being sent, the number of bytes being sent

**08:00-09:00**

the number of packets received, and the number of bytes received. Alright. Again, we can identify lots of things in applications. I had multiple connections to AWS console, but also AWS does some round robining and it takes me to other IP addresses as well. You'll see that there are multiple connections out from my Mac to AWS. Next, another thing I could look at is Zoom. Let me end. There's been no communication on this, so let me see if I can start a new Zoom session. Here we go. I'm starting a meeting right now. Let's do a `show session`.

**09:00-10:00**

Okay. I see 2928. I'm not sure if that's it. We'll find out. If I do it 2928. There it is, right there. Zoom: 2928. I see that yes, I had a connection going out from my Mac source port 52467. Again, source ports are what are called ephemeral. They are made up. They're above 1024 and less than 65535. Then this is going to Zoom 170.114.52.6. Again, on 443. AWS was on 443. This is on port 443. Again, because it's all HTTPS communication, and being able to identify the application is key.

**10:00-11:00**

Then again I see the packets out. The packet or the bytes set, the packets received, and the bytes received so far. Some pretty cool stuff happening on the Modular Sensors. Let's go ahead and take a look here. We see this connection is 170.114.52.6. On the command line of the sensor itself, you can do a shell command. In other words, I can run some commands that would ordinarily be in Linux. For example, there's a command called Dig. Dig let's me find that it does a host name to IP address lookup, a DNS lookup essentially. So dig: `shell dig`, and I'll say `zoom.com`.

**11:00-12:00**

Here we see zoom is 170.114.0.12. But again, they round robin as well. If I take a look here you'll see it's a different IP address, but it starts off the same. It looks like they have this subnet here 170.114 or at least some of that subnet. Alright. It gives you an idea of what's going on but the shell dig command can be very useful. Okay. Now let's go back on to the Modular Sensor, but see it from the perspective of network traffic analysis. I can go to

**Investigate | Threat Hunting**. I'm not going to look for an Alert right now. Let's look just for the traffic itself. For the traffic, let's see what we can find. There we go.

**12:00-13:00**
Already, right off the bat there's some of my AWS console. 10.20.1.112 to 3.136.230.111. Right? If I were to look through that list one of the IP addresses would be 10 or 3.136. I go do I have any other AWS console? I can type an AWS console from the top search bar. Sure enough, here's the AWS console. You notice that it goes to a lot of different destinations from my source IP to many destination IPs because it turns out Amazon is quite big. They have a lot of IP addresses. A lot of the world runs on AWS. Alright. Some of the other things–remember I had Zoom before so let's look at Zoom.

**13:00-14:00**
See all the Zoom sessions that I had. Here I can see that my source IP 10.20.1.112 went to this destination IP and port 443, which is the app Zoom. But there's been other connections as well because the Zoom app is doing other other things as well. Okay. There's the AI and a whole bunch of other modules within Zoom. Remember that on the Modular Sensor itself. Again, Zoom is 2928. Then if I look at the session table for 2928 as an example, we see the source IP, the source port, and the destination IP, and the destination port in this example. The number of bytes transmitted.

**14:00-15:00**
Again, the fact that we are recording the number of bytes I don't need to send every single packet to the Stellar Cyber platform. I only send a summary of the metadata, which contains the 5-tuple and the application ID, but the packet sent, and the out bytes sent, and the packets received and so forth. Alright? That's really cool. But again that's not a lot of information. Let's take a look now at one of these on the platform and open up an Interflow record. Now looking at the Interflow record, and I click on **Details** or click **JSON**–don't forget everything's in JSON formatted Interflow records–you'll see there's a lot more information in here. For example, the dstip_geo. This is called enrichment.

**15:00-16:00**
We are enriching the Interflow record that came from the Modular Sensor to the **Traffic** index. Now we can see the dst geo. This one doesn't actually show the city, but I know it's in the United States. Right? It also gives a latitude line and a longitude line so I can pinpoint where it is on the map. The other thing I have is the source geo (srcip_geo). Sunnyvale. This is where the lab exists where my Modular Sensor exists and where the communication is starting from. Alright. Again, we have the app ID name Zoom. The field name is app ID underscore name, and the application is Zoom. I see the number of out bytes that were

sent. Another really cool thing is dst IP reputation. We're using what's called threat intel and we are identifying whether or not the destination IP is from a good source.

**16:00-17:00**
Is it Bittorrent? Is it a brute forcer? Do they have a good reputation? In this case yes, it turns out that Zoom does have a good reputation. Okay. That's all called enrichment. We see a lot of data coming in. I actually see the origin, so we actually add that as well. When you're looking around you could see where that came from. We are adding the host as well for the communication. There's a lot of really detailed information that could be used especially for the correlation with other records within the system. I could do the same thing for the AWS console. Right? Now I know that by the way, I could type in `appid_name:AWS_console`, right?

**17:00-18:00**
Because I found the field name when looking at the Interflow record. Searches are field name and field value. Take a look right here. Again, these are the AWS consoles from my source IP. If I take a look here at a record, I'll go to the JSON and there again I see the source IP, which is my MacBook. The destination IP geo is the geolocation of that IP address for the destination. If I scroll down further, I see that it is Interflow traffic, the destination IP geo, the destination IP reputation is good. That's definitely what you want to see. If I scroll further down, we have a lot of information that gets added

**18:00-19:01**
to make it easier to identify and correlate this information with other as well. The source IP geo, which is my lab environment, it's also good so that's also a good sign. Again, what comes from the Modular Sensor is the 5-tuple. Source IP, destination IP, source port, destination port, and protocol. Then Stellar Cyber is able to use threat intel, DNS lookups, geolocation, and a lot of other information that you'll see here within the record to fill it in to give it more context. When you're using it in identifying threats, you have a lot more context to use. Pretty cool stuff. Alright. I will stop there.

This lesson is complete. Thank you for watching.

# L15 - NETWORK DETECTION AND RESPONSE (NDR)

## L15A-PRES: Network Detection and Response (NDR) in the Stellar Cyber Platform (06m:45s)

**00:00-01:00**

The next lesson is *Network Detection and Response (NDR) in the Stellar Cyber Platform*. This lesson, *Network Detection and Response (NDR) in the Stellar Cyber Platform,* explains some key concepts of how data collected from Modular Sensors in the Stellar Cyber platform is analyzed and utilized.

Hi everyone. In this lesson, I will cover some of the capabilities of network detection and response, also known as NDR, in the Stellar Cyber platform. On the Stellar Cyber platform traffic metadata from Modular Sensors arrives as Interflow records and is stored in the **Traffic** index within the Stellar Cyber Data Lake.

**01:00-02:00**
Remember from previous lessons, that Modular Sensors can perform deep packet inspection to determine the application, create network session flows, and report that as Interflow to the Stellar Cyber platform. The traffic Interflow records are visible on the Stellar Cyber platform by selecting **Investigate** then **Threat Hunting**. You then select the **Traffic** indices where it says indices as step number two. For step number three, you can filter on, for example, aws_console. That was the application found with the deep packet inspection. You can then scroll down on the page to view the Interflow records, called documents, by expanding an individual entry in the list.

**02:00-03:00**
To view the details of the Interflow record, for the traffic to the AWS console, select an Interflow record entry, also again known as a document, from the list. Under the **Actions** column on the right-hand side, select the little **i** for more info. More info gives the ability to view key fields, details, Interflow in JSON format, and perform actions. As shown in step number three, select **Details** to view Interflow record field names and values, including the enrichment performed by the Stellar Cyber platforms such as the geolocation, host names, threat intel, reputation, desktop IP, type, duration, and much more. Again, this enrichment is really helpful for identifying threats.

**03:00-04:00**
For step number four, the Interflow record can be viewed in JSON format by selecting JSON, or J-S-O-N. JSON, or JavaScript object notation, is a lightweight data interchange format that is easy for humans to read and write and easy for machines to parse and generate. It's a perfect format for analyzing data and using data with the machine learning and rule-based algorithms on the Stellar Cyber platform. In step number five, **Key Fields** provide essential details that are crucial for understanding and analyzing the data that are all visible by selecting **Key Fields**.

**04:00-05:00**

Number six, **Actions** provide the ability for a SOC analyst to mitigate threats by initiating response actions with integrations to security infrastructure and identity tools by selecting **Actions**. The list of actions available, as you see in number seven, is based on services and Connectors configured within the Stellar Cyber platform, such as block an IP on a firewall or a security switch, disable a user in Active Directory, or contain a host with an EDR, which is an endpoint detection response platform, and there's so much more. You can run scripts, send Slack messages, and trigger an email. Again, the response actions are incredibly valuable, especially when trying to mitigate threats and inform people of threats as well.

**05:00-06:00**
Another key component for network detection and response on the Stellar Cyber platform is the automation. On step number one, if you go to the **Respond** menu on the top menu bar, then select **Automation**, you can create an automated threat hunting, or ATH, playbook that can be configured to automate response actions based on conditions learned from network traffic analysis. The response actions, or ATH playbooks, can also be run manually by SOC analysts. When a condition is met, for example, an NDR threat is detected, you can trigger an email, send a Slack, create a webhook responder. You can block an IP on a firewall or security switch, you can create an Alert.

**06:00-06:45**
You can disable a user, such as with an IDP, as Active Directory or Azure. You can run a script of your choice. Again, these are all things that can be scheduled automatically when a triggered condition is met or you can have a SOC analyst run one of these ATH playbooks if they see, for example, the Alert get created. Thank you for watching.

This lesson is complete. Thank you for watching.

# L16 - OPTIMIZING NETWORK TRAFFIC INGESTION

## L16A-PRES: Optimizing Network Traffic Ingestion (04m:30s)

**00:00-01:00**
The next lesson is *Optimizing Network Traffic Ingestion*. This lesson, *Optimizing Network Traffic Ingestion,* provides explanations and configuration examples to optimize ingestion of network traffic by excluding network applications and services that are not essential to ingest on the Stellar Cyber platform.

In this lesson, I will cover optimizing network traffic ingestion. Why is optimizing network traffic ingestion important? The following requests from Stellar Cyber partners and customers give good examples of why optimizing network traffic congestion is important.

One customer wrote, “I need to bring down the ingestion to be in compliance with the license limit of Stellar Cyber.”

**01:00-02:00**
Others wrote, “We want to reduce data ingestion to have room for a longer period of log retention.” This could be for compliance reasons. “We are only interested in a small portion of the tenant's network traffic. The rest can be discarded.” “A certain forwarded log event is not relevant to our security posture, but is a high portion of our overall log volume. For example, firewall deny actions. If the firewall is actually doing the job of protecting the network, why do I need to send that to the Stellar Cyber platform for analysis?” in their case. There are some organizations that might want to see the firewall deny actions, but you have a choice on what you can ingest.

**02:00-03:00**
In the Modular Sensor profile, network traffic application metadata settings have been set by default to optimize the amount of network traffic ingested. You can optimize data ingestion further by setting the **Collection Scope** to exclude certain applications or only collect certain applications. For example, select **Exclude certain applications** and an option to exclude the following applications or application groups appears. Select inside the box to get a search box with a list of applications and application groups. In the search box, by typing the name of an application or application group, the list narrows. In this case, I've typed in YouTube.

**03:00-04:00**
Here we see some of the applications: YouTube, YouTube_HD, YouTube Kids, and YouTube music. You can select applications and application groups by selecting the check box next to the application or application group name. Select outside the pop-up box to close it. By excluding applications or application groups, these selections are added as traffic filters to the Modular Sensor. In this case, I am excluding YouTube traffic and speed test traffic. Packets that match the applications or application groups within the traffic filter are discarded. In this case, the Modular Sensor will discard all YouTube traffic and all speed test traffic. Modular Sensors will not create or report Interflow records for the excluded applications to the Stellar Cyber platform.

**04:00-04:30**
Once you have selected the applications or application groups, select **Submit** at the bottom of the Stellar Cyber Modular Sensor profile to save and apply the settings to the Modular Sensors assigned to this Modular Sensor profile.

This lesson is complete. Thank you for watching.

# L17 - EXAMPLE OF OPTIMIZING NETWORK TRAFFIC INGESTION

## L17A-PRES: Example of Optimizing Network Traffic Ingestion (09m:08s)

**00:00-01:00**

The next lesson is *Example for Optimizing Network Traffic Congestion*. This lesson, *Example for Optimizing Network Traffic Ingestion,* gives an end-to-end example of configuring traffic filters to exclude Modular Sensor configuration management and ingestion communication to the Stellar Cyber Open XDR platform from being ingested by the platform.

In this lesson, I will give an example of optimizing network traffic ingestion. One, we can filter sensor communication to the Stellar Cyber platform. In most cases, this data is irrelevant and should be filtered.

**01:00-02:00**

This not only applies to Modular Sensors, but it also applies to Windows Server sensors and Linux Server sensors as well. For example, you can filter the Modular Sensor IP that opens a configuration and management connection to the Stellar Cyber platform Data Lake. This connection uses TCP port 6640-6648. It also opens up a connection to the Stellar Cyber platform Data Lake on port 8443 for downloading software updates. The Modular Sensor IP also communicates to the Stellar Cyber platform for data ingestion on port 8889 to the Stellar Cyber platform Data Analyzer. You can also filter log sources to the sensor, which is the traffic mirrored to the Modular Sensor.

**02:00-03:00**

For example, Fortinet sends logs to the sensor port 5517. The sensor log forwarder processes the logs and the packets to 5517 are also mirrored to the sensor. The mirror traffic in this case can be filtered. Here's an example: When configuring the traffic filtering, it is helpful to have a network diagram. In this case, I have Stellar Cyber sensors on the 10.20.10 subnet. I have Modular Sensors on that subnet, Windows Server sensors on that subnet and Linux Server sensors. Each of these opens up a configuration management connection to the Stellar Cyber Open XDR platform using TCP ports 6640 to 6648 over TLS to the Data Lake.

**03:00-04:00**

Again, this is for on-premises or cloud deployments. For SaaS, this would only be port 443. Stellar Cyber sensors also open a connection for Interflow ingestion using TCP port 8889 over TLS to the Data Analyzer IP on the Stellar Cyber Open XDR platform. To optimize the traffic, for on-premises or cloud Stellar Cyber platform deployments, you can create a custom application for Stellar Cyber platform communication from sensors to the Data Lake for configuration management, and also create a custom application for Stellar Cyber platform communication from sensors to the Data Analyzer for Interflow ingestion.

**04:00-05:00**
Using the last diagram as a reference, for this example, I will create a custom application in traffic filters for the Modular Sensor communication to the Stellar Cyber platform Data Analyzer for data ingestion. For example, the Data Analyzer I am using has an IP address of 192.168.1.150/32. What is slash 32? That is the netmask. Slash 32 says that I care about every single bit in this IP address. In other words, the complete IP address. Sometimes you'll see /24. What that means is the 192.168.1.0. We want to make sure that this is /32 to ensure that we are identically matching this IP address. The ports that are used are TCP 6640 to TCP 6648.

**05:00-06:00**
Here on step number one I will go to **System | Traffic Filters**. At the top there are two tabs, **Custom Applications** and **Application Groups**. In this example, separate traffic filters will be created for each IP address and port combination to be filtered. For step number one, I'll select **Custom Applications**. For step number two, I'll put an application name in. Now I like to make sure that this is easy to understand and easy to recognize because if you make a mistake or type in a different name, it might be harder to realize what you are filtering in the traffic. In this case, I'll use the tenant name. I'll use the word "sensor" to say this is sensor communication.

**06:00-07:00**
DA is saying that traffic is going to the Data Analyzer, and the port number is the port that I'm actually filtering. In this case, I'm using Customer_one, which is the tenant name dash sensor dash DA, dash 6640. Next, since I have a customer called Customer_One, for the tenant, I will select Customer_One because that's where these sensors are deployed. Now if the sensors are for my own network, I would just say the root tenant unless I have a tenant specified. Next, I will say Matched on Protocol. I will select TCP. For number five, I'll enable Match on Source, and I'll enter the IP address or subnet that contains the sensors.

**07:00-08:00**
In this example I could use a single IP of 10.20.1.15/32, which is my Modular Sensor. But I have other sensors as well on my Linux Servers and on my Windows Servers that are on the

same subnet. I can say 10.20.1.0/24, which covers the entire subnet. Next, I will enable **Matched on Destination**. For on-premises and cloud deployments, I could enter the IP address of the Stellar Cyber platform Data Analyzer. As an example, in my lab I have 192.168.1.150, and I'm entering /32 to say that specific IP address. If I used /24, it would say the entire 192.160.1.0 subnet, which I don't want in this case.

**08:00-09:07**
Here I'll enter the TCP port example 6640. Now I can select **Submit**. I will do this for each of the ports and combinations of subnets. I select **Create** to create additional traffic filters. Note traffic filters are not applied until they are assigned to the sensor profile in the application metadata section. When done, as an example, I have created these 12 traffic filters. If we take a look, I have Customer_One-Sensor-DA-6640, same as 6641, all the way to 6648. Then also filtering out 8443, which is the software downloaded to the Data Lake, and 8889 as well for the configuration and management connection.

This lesson is complete. Thank you for watching.

## L17B-DEMO: Example of Optimizing Network Traffic Ingestion (27m:07s)

**00:00-01:00**
This lesson continues with demonstrations for optimizing network traffic congestion. This lesson is a demonstration that provides explanations and configuration examples to optimize ingestion of network traffic by excluding network applications and services that are not essential to ingest on the Stellar Cyber platform.

Hi everyone. In this example, I'm going to provide a demonstration of how to create traffic filters so that I can optimize the ingestion of traffic into the Stellar Cyber Modular Sensors, which then would get forwarded to the Stellar Cyber platform.

**01:00-02:00**
If it's traffic that I don't need to analyze on the Stellar Cyber platform, then I can filter it out so that I don't cause unwanted ingestion. To begin, I will take a look at the sensor profile I am actually using to see what I have configured. I can go to **System | Sensor Profiles**. The Modular Sensor profile I'm using is the SC Modular Sensor IDS. I can click the pencil to edit. If I scroll down to create a traffic filter I have the ability to select the field here. I can select **Network Traffic** and scroll down to **Application Metadata**. If I expand **Application Metadata**, I see that I have a **Collection Scope** of **All applications**.

**02:00-03:00**

If I want to exclude certain applications, I click the radio button there. Then from here I can choose an application. Now, by default, Stellar Cyber has a lot of applications and application groups built in. For example, there's YouTube. If I type in YouTube, I can see that YouTube is here. If I don't want the Modular Sensors to look at YouTube traffic or capture YouTube traffic, and send that information as Interflow records to this Stellar Cyber platform, I can create a filter, as I did right here, for YouTube. Now the way this selection box works I click into the box. I get a selection window, and then I click out of the box, and then whatever I have selected goes here.

**03:00-04:00**
I don't have any custom applications right now. I'm going to leave this Modular Sensor profile and create a traffic filter of my own. From **System**, I go to **Traffic Filters**. I have no traffic filters to find. But if I look at **Application Groups**, there are lots of applications available. What I'm seeing here are the application group categories. If I wanted to find YouTube, for example, I can go to mm–I think it's mm_streaming, yes multimedia streaming. If I select the hamburger icon here I can say auto-size all columns, and in this list of multimedia streaming apps is YouTube, as an example.

**04:00-05:00**
If I block the whole category from being ingested, or if I filter based on this category not blocked again, we're not blocking any traffic. I'm just filtering traffic that I'm going to ingest and send to the Stellar Cyber platform. But in this case, I'm going to remove the filter, go back to **Custom Applications**, and create a traffic filter or a set of traffic filters to filter the ingestion of the Stellar Cyber communication for the Modular Sensors to the Stellar Cyber platform. Again, this is information that does not need to be analyzed by the Stellar Cyber platform. To begin, I can create an application. The application name–again what I want to do is make sure that I can recognize the application.

**05:00-06:00**
I'm going to apply this application to all my tenants. I will actually call it All_Tenants. This is communication from my sensor to the Stellar Cyber platform Data Lake. For the configuration management connection, this is TCP port 6640. I'm going to match based on TCP. Now here's where I could decide to be more specific or less specific. If I say **Matched on Source**, which I'll do in this example, I will match on the subnet that contains my Modular Sensor, my Windows Server sensor, and my Linux Server sensors.

**06:00-07:00**
I'll type in 10.20.1.0/24. I am saying any communication from the 10.20.1.0/24 subnet matches this rule. But then it also has to match the destination. As I mentioned before, this is the communication from the sensors to the Stellar Cyber platform Data Lake because this is an

on-premises deployment. If this is a SaaS deployment, then I would just use the IP address of the Stellar Cyber platform. I will use 192.168.1.175, which is the IP address of my Data Lake. Now here's where people can make mistakes. This is /32.

**07:00-08:00**

Each of these–these are called octets that are separated by the dotted notation–is eight bits. This is eight bits. This is eight bits, so that's sixteen. This is eight bits, so that's 24. If I say another eight bits, that's 32. If I were to put /24, then I would essentially be saying–and I'll scroll this up a little bit so you can see it better–essentially I'd be saying I will apply this rule from anything in this subnet 10.20.1.0 to anything in the 192.168.1., essentially 1.0/24. I want to actually ensure that it's only to my Data Lake. I need to use /32.

**08:00-09:00**

Next, I'll put in the port, which is 6640. Click off anywhere other than the selection box, and now the **Submit** button appears. I can now auto-size all the columns. Now I see a clearly defined rule. It's for all tenants. It's for the sensor talking to the DL, on port 6640 from a source subnet of 10.20.1.0 to a destination IP of 192.168.1.175 on port 6640 with TCP and all tenants. Alright. I need to do this a few more times. To do it again, I will add another [selects the **Create** button] and do essentially the same thing.

**09:00-10:00**

The application name is going to be All_ Tenants_sensor to Data Lake on port 6641. Now, I will then do the same thing 4642, 43, 44, 45, 46, 47, 48. I'll apply this to all tenants. If at a specific customer environment and they're in their own tenant, I could create a filter just for their specific tenant. For example, here I have Customer_One and Customer_Two. I'm going to Matched on Protocol TCP. I will match on source 10.20.1.0/24. I'll Matched on Destination 192.168.1.175 again /32.

**10:00-11:00**

The port will be 6641. Now, if I were to turn off **Matched on Source**, I could make my life a little easier. I can essentially say any communication to the Stellar Cyber Data Lake that's using 6640, 6641, 6642 will match this rule. In reality, that's probably what I'd do in most situations. But what I want to show is that it is being more exact. For this, I will use the 10.20.1.0/24 because that's where all my sensors are anyway. Alright. I'll submit that.

**11:00-12:00**

I will continue on. I'll do just one more, so it's really locked in. I'll go back to **Application Name**. Again, this is All_Tenants-Sensor-Data Lake-6642 and make that TCP, and Matched on Source 10.20.1.0/24. The Data Lake IP is 192.168.1.175. The port, in this case is 6642. So six six four zero, six six four one, and six six four two.

**12:00-13:00**
I will stop for a minute and put the rest in, but I will make it so that you don't have to watch me doing that. I'll be right back. Voila, I am finished creating the traffic filters, or am I? If you look at the custom applications I have now, there's one of these entries that doesn't look like the others. Which one is it? Maybe you saw me make the mistake on the last configuration of a traffic filter I did before taking a break. Now if we take a look at the destination IP, one of them doesn't have a /32. That's why it's really good also to filter here or take a look here and see the traffic filters and see if they're all aligned with each other.

**13:00-14:00**
What's really great now is I can just go back and modify this one. Scroll down to the bottom and use /32. Click **Submit**. Now they're all in line with each other. Next, I will apply these traffic filters to my Modular Sensor profile. I can go to **System | Sensor Profiles**. I'm using the SC Modular Sensor IDS. I will edit this row. Go to **Network Traffic**. Select **Application Metadata**. Now for **Collection Scope** I will exclude certain applications.

**14:00-15:00**
Here now, if I were to start typing in the name, I had all tenants. I have all my tenants DL 6640. Again then type in all again, do the next one. Type in all again and do the next one. I'll do that until they're all selected. If it wasn't a custom application, I could actually do this in one swoop here. I'll scroll down. Type in all again until I have them all selected. That's it. Click off.

**15:00-16:00**
Now I can see that they're all there and I'll go ahead and click **Submit**. Actually, select **Submit**. Alright. Now if I go to my Modular Sensors, we'll go to **System | Sensors** and we'll see here that the, what does it say? If I go to here, and auto-size all columns, the configuration is out of sync. It's out of sync because I just modified the profile and actually modified my Linux Server sensor profile as well. This will take a minute for it to come back. I'll refresh my screen and there it is. It's back. I will also readjust the columns. Now my traffic filter is applied.

**16:00-17:00**
I will no longer see the traffic from my sensors to the Stellar Cyber platform. Okay. Now that my Modular Sensor profile has been updated and assigned to the Stellar Cyber sensor well, first of all let me just make sure. I modified the SC Modular Sensor IDS profile. Let me go to my Modular Sensor just to make sure that that is the profile in use. I am pretty sure it is because it was red a minute ago. Alright. If I were to look at my SC Modular Sensor profile. I look right here, yes the SC Modular Sensor IDS is assigned to my Modular Sensor.

**17:00-18:00**

If I select into it, go to **Network Traffic**, go to **Application Metadata**, I see these applied. Now, I'll close that out. Next, if I go to **Investigate | Threat Hunting** I can actually see where, or how much traffic is coming from the 6640 to 6648 ports. To begin with, I will select all indices because I don't know which indices contain the communication from the Stellar Cyber Modular Sensors to the Stellar Cyber platform. Now that I've selected all indices, I will search for a destination port. In Lucene queries syntax is dstport:, and I'm going to use a range.

**18:00-19:00**

For a range I use a square bracket. I have a space or no space. It's your choice. I'll put a square bracket and I will say 6640. Capital T-O is a range to 6648. Close the square bracket, hit **Return**. Now I see that I have a legend called **interflow_traffic**. Well who would have thought? If I look at Interflow traffic here and I filter for that, and look down here, I can see that, yes, I had my Linux Server sensors and scroll up here and my Modular Sensor all using those ports. The application that it did see.

**19:00-20:00**

Let me click on this, and go to **Details**. Here I can look through and find, first of all DST port. DST port is 6640. The destination IP host is 192.168.1.175, which is my Data Lake IP address. The source IP, if I scroll down these are in alphabetical order, is 10.20.1.15. If I look at the time this came in, which is at the top here, this came in on April 3rd at 4:17 PM, which is 16:17. I should not receive any more of these messages. Look over here and see the same thing: 16:17.

**20:00-21:00**

The rest are my Linux Server. The Linux Server is using a different policy. Therefore, the messages are still coming in, at 16:57 is the last one. Okay. Now that we have all the configuration management connections being filtered, the next thing I will do is create two more traffic filters. I will create the traffic filter for the ingestion on the Stellar Cyber platform Data Analyzer, or if it's SaaS, the Stellar Cyber platform itself. I'll create a new custom application. Again, I'll give a descriptive name of All_Tenants.

**21:00-22:00**

This is for sensors talking to the DA (the Data Analyzer) using port 8889. This will be TCP. I will Matched on Source. The source is 10.20.1.0/24. I will match on the destination, which is the IP address of the on-premises platform Data Analyzer. The 192.168.1.150/32 to match the IP address, and the port is 8889. I can go ahead and submit that. I will create one more to the Data Lake.

**22:00-23:00**
This one will be the All_Tenants again. From the sensor to the Data Lake on port 8443. This is for software that the Modular Sensor downloads or even sensors download to get updates. This will be TCP **Matched on Source** 10.20.1.0/24. See? I wasn't paying attention. This has to 10.20.1.0/24. The port number is 8443.

**23:00-24:00**
Again, it's okay to make mistakes. Just make sure you correct them. Of course I did that on purpose. Now I'll go back to my Sensor Profiles. Go to **Network Traffic | Application Metadata**. Click in the selection box here or select in the selection box. I'll type in "all" again to see if I have anything new. I do. I have my DL 8443. Try that again. All and I had all tenants to the DA. Again, I will apply.

**24:00-25:00**
As soon as I apply, if I go to my sensors now, **System | Sensors**. Yeah. I see that this time, it's only the Modular Sensor. Now, last time I had multiple administrators on the platform. They were both red. That was because actually somebody else was modifying the Linux Server sensor. Always be aware of what you're doing. Right now the configuration is out of sync, but it will come back shortly. There we go. After a quick refresh, it is now all working correctly. If I go to **Investigate | Threat Hunting**, let's just take a look. If I select **all** here and look for just let's say, dst port 8443.

**25:00-26:00**
Of course this is in the last 24 hours. Nothing was there. If I were to choose a dst port of 8889, let's take a look. There's a lot more records there that will be filtered out. What do these look like? This is your TCP 8889. These are from the Modular Sensor to the Data Analyzer, just as we had configured. This is exactly what we want to filter out. Now if I scroll down, it's all the traffic flows and the message class is Interflow traffic. Seeing this message class Interflow traffic, I can actually do this as well. I can go over here. Okay. Now let's see if I see the message_class Interflow traffic.

**26:00-27:10**
I'll go to MSG_class: Interflow traffic. Here I see 1,285,000 records. If I look over here, I have this in **Alerts** and in the **Traffic** index. If I go to the **Traffic** index, and only select that here. Again, I see one 1,285,000 records. Now let's see what it's been like for the last week. Seven million, eight-hundred thousand and seventy nine records. Okay? Filtering out this traffic isn't a bad thing to do. It's actually quite good for optimization.

This lesson is complete. Thank you for watching.

# L18 - INTRUSION DETECTION SYSTEM (IDS) CONFIGURATION

## L18A-PRES: Intrusion Detection System (IDS) Configuration (07m:53s)

**00:00-01:00**
The next lesson is *Intrusion Detection System (IDS) Details and Configuration.* This lesson, *Intrusion Detection System (IDS) Details and Configuration,* provides the explanation and configuration of the IDS module on the Stellar Cyber Modular Sensor.

In this lesson, I will cover the Modular Sensor Intrusion Detection System, also known as IDS. Looking at the picture on the left, IDS is built into the Stellar Cyber Modular Sensor. It works in conjunction with the network traffic analysis that we enabled earlier.

**01:00-02:00**
After receiving network traffic from SPAN ports or TAP devices, Modular Sensors use a Suricata engine with over 60,000 commercial IDS rules to detect threats within network traffic, which also includes traffic from OT devices, which is operational technology. The Modular Sensors create metadata summaries from IDS data that are normalized into Interflow records in JSON format. The Modular Sensor forwards IDS metadata as Interflow records to the Stellar Cyber platform. After receiving the IDS metadata or Interflow records, the Stellar Cyber platform uses machine learning to reduce the number of events generated by the IDS engine and performs complex analysis and correlation with other data sources to generate Alerts.

**02:00-03:00**
To authorize the IDS engine and optionally the APT, which is the Malware Sandbox licenses for the Modular Sensor, you go to **System | Sensors**, then step two go to **Manage** and then **Authorization**. To authorize the IDS, you ensure that the modules are selected from the left-hand side. In step number one, there's the **Authorize** menu. If you look to the right of that is the **Unauthorize**. In this case, we want to authorize. It looks like during the installation process of the Modular Sensor, the authorization of the Modular Sensor was done with the IDS. Well, that's why it's in the right-hand table, but by default, it may not be checked.

**03:00-04:00**
As you see here for SC Modular Sensor under **IDS License**, I now have it checked. APT license is also checked, which is the Malware Sandbox. Log Forwarder is always checked because it is always enabled by default. Step number three we click **Submit**. After submitting authorization, the sensor status will display the message disconnected and configuration is out of sync. The status will be red. After a few moments, if you refresh the browser, the message will clear and the sensor status will be green. The Modular Sensor

Intrusion Detection System is enabled by editing the Modular Sensor profile and system sensor profiles. In the Modular Sensor profile, enable **Network Traffic** in step number one

**04:00-05:00**

which is required for capturing network traffic used to apply IDS rules. Step number two, enable IDS. Inside that, it is used for configuring the home and external networks to apply IDS rules and for determining the signatures to apply. Configure Networks as a policy to determine when IDS rules will be applied. On the right is a picture with an example with IDS networks with a home of private and an external network of public. In step number three, you can select **Private**, **Public**, **Any**, or **Specific** for the home network. In this case, I will select **Private**. For the external network, you can also select **Private**, **Public**, **Any**, or **Specific**. In this case I will use **Public**.

**05:00-06:00**

In other words, the home network is my private network within my organization. The external network is public, which is the internet. Home and external networks can be set to **Private**, which is RFC 1918 IP addresses, which are 10.0.0.0/8. In other words 10 dot anything or 172.16.0.0/12, which is actually 172.16.0.0 to 172.31.255.255 and 192.168.0.0/16, which is 192.168, anything. Public is all but private RFC 1918 addressing and not multicast IP addresses, which are in the range of 224.0.0.0 to 239.255.255.255. **Any** of course, is all networks.

**06:00-07:00**

**Specific** is used to define custom IP ranges, if there are specific networks inside a format, for which to apply IDS rules within your organization. For example, there are some universities that have public IP addresses. But they are really considered internal network addresses for their organization. You could define specific IP address subnets to use for your own private networks. An example here, I have IDS networks with a home of **Private** with an external network of **Specific**. The **Specific** IP in this case is 3.3.0.0/16 and 2.2.1.0/24. You can have multiple entries.

**07:00-07:55**

For the IDS configuration you also have the signature. The **Signature** section lets you configure a list of IDS rule sets. These rules are integrated by Stellar Cyber from third-party threat intelligence. You can choose which rules to use, but you cannot add your own rule set. Again, you select into the box, and you'll get a set of rules to choose from. By default, the rules are there. But if there are additional rules that get added in a new release for an example, you will see them appear in this list. After completing your configuration settings in the sensor profile, the **Submit** button must be selected to apply the configuration settings to the Modular Sensor.

This lesson is complete. Thank you for watching.

# L19 - IDS IN THE STELLAR CYBER PLATFORM

## L19A-PRES: IDS in the Stellar Cyber Platform (07m:05s)

**00:00-01:00**
The next lesson is *Intrusion Detection System (IDS) Stellar Cyber Platform Operations*. This lesson, *Intrusion Detection System (IDS) Stellar Cyber Platform Operations*, explains how to view the IDS event data received from Modular Sensors and how the Stellar Cyber Open XDR platform uses machine learning to create actionable Alerts from the IDS events.

In this lesson, I will cover the Intrusion Detection System as performed with operations in the Stellar Cyber platform.

**01:00-02:00**
Now that the Modular Sensor profile has been applied to the Modular Sensor with the IDS functionality enabled, the Modular Sensor will now apply IDS rules to all the traffic it receives based on the home network and destination network specified to generate IDS events. To view IDS events that have been received by the Stellar Cyber platform, go to **Visualize | ML IDS**, which is a little bit further than midway down the list. Ensure you do not have any filters set that are not intended. You can expand the **Filters** sidebar and just make sure you don't have any filters. Otherwise, you may not get the results you're expecting to see. Set the **Time Interval** such as the last twenty four hours. You could make it longer if desired. View events and make sure that you scroll down to see them all.

**02:00-03:00**
Here we do see that there are ML IDS events. Stellar Cyber Machine Learning is applied to the ideas events from the Modular Sensors to identify threats and create Alerts. In other words, the Modular Sensors themselves use the Suricata engine to identify threats from the Suricata rules. Those get sent to the Stellar Cyber platform. The Stellar Cyber platform then applies machine learning, statistical analysis rules, and heuristic rules to identify threats along with using graph machine learning to correlate Alerts into what are called Cases. In this view we can see that some Alerts have actually been created.

**03:00-04:00**
Here we're looking for private-to-private exploit attempt anomalies, private-to-public exploit attempt anomalies, public-to-private exploit attempt anomalies, and public-to-public exploit attempt anomalies. In this situation, I have nine private-to-private exploit attempt anomalies, one private-to-public exploit attempt anomaly, and one public-to-private exploit attempt anomaly. Each of these can be investigated. Under the Alert that I want to investigate or under the Alert type, I'll select **Investigate**. This will take you to the **Alerts** page on the Stellar Cyber platform. In the **Alerts**

page, all the charts and Alert records are displayed from the Alert that we are investigating right now, which is the private-to-private exploit attempt anomalies.

**04:00-05:00**
If you scroll down, you'll see the actual Interflow records that are related to the Alert. The **ML-IDS Events** and related Alerts can also be viewed by going to **Investigate | Threat Hunting**. Next to **Indices**, select **Alerts**, which is the indices or index displayed by default. Clear the checkbox next to **Alerts**. Select the checkbox for **ML-IDS/Malware Sandbox Events**. Step number five, with the **ML-IDS/Malware Sandbox Events** indices selected, all ML-IDS-related charts and events are displayed. Here are a few troubleshooting tips for the Intrusion Detection System on the Modular Sensors.

**05:00-06:00**
Let's say you're on the Stellar Cyber platform and there's no IDS data being received. The potential solutions for that are number one, ensure the Modular Sensor is authorized and has the IDS license set to **Authorized**. That's the little check box for the IDS license. Number two, validate the sensor profile used by the Modular Sensor has IDS enabled. Don't forget that the Modular Sensor could be assigned to a different Modular Sensor profile. Number three, validate the IDS configuration in the Modular Sensor profile has the proper network selected and signature selected. The other thing is validate the maps are appropriate for the networks that you have if you created your own specific networks. Number four, ensure that the Modular Sensor is receiving traffic from SPAN ports or TAP devices.

**06:00-07:05**
This is important because the Modular Sensor needs to receive the network traffic to perform the IDS functionality. Next, on the left-hand side, again another symptom, limited IDS data is being displayed in the Stellar Cyber platform. I'm going to say this one over and over again. Ensure the correct filters and time ranges are set on the page you are on. If you're on the **Alerts** or **Visualize | ML IDS** pages, ensure that the time range is set in the upper right. On the left-hand filter bar, make sure that you are using the filters you are expecting to see. If you have not set a filter, make sure you just clear the filters out. There's a little option to do that. For step number two, from **Investigate | Threat Hunting**, ensure the **ML-IDS** index is set. Okay. That's it for now. Thank you for watching this lesson.

This lesson is complete. Thank you for watching.

## L19B-DEMO: IDS in the Stellar Cyber Platform (33m:35s)

**00:00-01:00**
This lesson continues with demonstrations for *Intrusion Detection System (IDS) Stellar Cyber Platform Operations*. This demonstration explains the concepts and configuration of the IDS module on Stellar Cyber Modular Sensors and provides details on investigating IDS events and Alerts in the Stellar Cyber Open XDR platform.

Hi everyone. In this lesson, I will show you how to enable IDS in Modular Sensors. The first thing you need to do is go to **System | Sensors**.

**01:00-02:00**
The IDS module is a licensed feature and that's available by going to the **Manage | Authorization**. Here in the **Authorize Sensors** page, I can go and take a look at my Modular Sensor, and I see that I have an IDS license. All I have to do is check the box for **IDS License** to enable it, then go ahead and select **Submit**. Though I've authorized the Modular Sensor to use the IDS module, it's still not configured yet. How do I configure a Modular Sensor to use the IDS module? You guessed it. I have to go and configure the Modular Sensor profile. To do that I go to **System | Sensor Profiles**. The Modular Sensor profile I am using for these examples is SC Modular Sensor New.

**02:00-03:00**
I will go ahead and select that now and select the pencil to edit. To use IDS, one of the first things you need to do is ensure that you have network traffic being collected as well. The IDS module is applying signatures and rules to look for threats in network traffic. If you are not forwarding network traffic to the Modular Sensor, the IDS module isn't going to do you much good. Here, I do have **Network Traffic** enabled. I do have my Modular Sensor configured off a SPAN port on my switch. The next thing I need to do is enable **IDS**. Now, **IDS** is enabled, but for what traffic?

**03:00-04:00**
Let's take a look. To configure the traffic, we can look at the networks. The IDS module uses a home network and an external network. What does this mean? Well, in every rule that's used for IDS, there's something that will go from either home to home, or home to external, or external to home. It needs to know which networks you consider home, and which networks are considered external. Now, typically a home network is a private network. It is RFC 1918, which is my 10.0.0.0/8 CIDR block, which is 10.0.0.0 to 10.255.255.255.

**04:00-05:00**
It's also 172.16.0.0/12 to 172.31.255.255. There's also the 192.168.0.0/16, which is the 192.168.0.0 to 192.168.255.255. Next, what is the external network? The external network is using a public IP address. Public addresses are, number one, not private. Everything that's not RFC 1918 and not multicast. Multicast is the 224.0.0.0 to 239.255.255. With that in mind now, the rules will say, look for a threat in the home network to the external network.

**05:00-06:00**
Or look for a threat in home to home or look for a threat in external to home. Now private to public does not apply to everyone. In some organizations or even universities the private network actually might be a public IP address. They're assigned a certain range of public IP addresses. They're trusted home network contains public IPs. If that's the case, you can actually change what a home network applies to. You could say the home network applies to **Any** but then you'd be looking at all traffic. You can specify **Specific**. Now for **Specific**, you can enter IP range in the form of CIDR notation. I can select **plus**. Now let's say my organization uses the 2.2.0.0/16 network.

**06:00-07:00**
Now I could say anything that is from 2.2.0.0 to **2.2.255.255** is essentially my home-to-home network. Or I can say my home to public is the 2.2.0.0/16 to public, which is everything else. We give you the ability to specify what your home network is. Okay. I also have the ability to specify additional IP ranges. Again, in the form of CIDR notation. I might have a 2.3.0.0/16. I might have a 2.4.1.0/24.

**07:00-08:00**
I might even have a 3.1.1.0/24. Alright. You can have multiple IP CIDR blocks that are used to identify your home network. But in most cases, I'll clear all these. It's usually going to be **Public**. So excuse me. In most cases, it's going to be a **Private** as my home network to an external of **Public**. Now, if you really want to see how this works, I think one of the best ways is to actually look at a Suricata engine rule. I have one here. One of the best ways to understand this concept is to actually take a look at a Suricata engine rule.

**08:00-09:00**
I have a few lined up here. In my trusty text editor, I have copied a few Suricata rules for us to take a look at. if we take a look at the first one, you'll see that the rule has a variable saying look for home network to external network. In this case, it would be a private IP address to a public IP address. This one I can say if a packet has an external IP address, which is a public IP address to the internal IP address, which is my private network, then this is an external to home. If I look over here again, I have a rule that shows a home network to home network. That's a private to private. This will make more sense even when I take a look at the ML-IDS in the Stellar Cyber user interface.

**09:00-10:00**
In the configuration of the IDS, you can also take a look at the signatures that are being used. These are the Suricata rules. Go to **Signature**. I can look at the selected rules. By default, we have selected most of the rules that apply to most organizations. If I scroll through here I can see the rules. If you want to check and see if there are other rules available, you can select outside of these bubbles here. If you do that, you'll get a selection box and you can scroll through this list as well. One of the questions that pops up a bunch is, “Hey I have operational technology in my organization. Do you have any rules that cover that?” The answer for that is absolutely we do.

**10:00-11:00**
If we scroll over here you'll see that we have scada rules and that is selected. By default, the IDS in Stellar Cyber does take into account operational technology as well to look for threats. Again, to close the selection box you can click outside the selection box. Now what's important is that these aren't individual rules. These are groupings of rules. The scada rules, there's lots of scada rules, there's lots of netbios rules by default, Stellar Cyber is looking for approximately 60,000 different rules to detect threats in network traffic. Also, if we look here there's some buffering involved. If the Modular Sensor loses connectivity to the Stellar Cyber platform, maybe there's an internet outage at the remote site, we can buffer the Interflow records.

**11:00-12:00**

Alright. I will go ahead and select **Submit**, and this will enable the IDS module. To view the status, I can go to **System | Sensors**. You'll see the configuration is out of sync. That is because it is now loading the new configuration. Okay. After about ten to twenty seconds, the Modular Sensor status became green, which is success. That means that it has loaded the new configuration. It may take a couple extra seconds for the modules themselves to load. If I select the Modular Sensor name from this page, I can get the sensor details. Looking at the sensor details on the right, I do see that the modules have loaded. I have the IDS Analyzer, I have the IDS Rules and the Traffic Monitoring. Everything is green, so that looks good.

**12:00-13:00**
If I scroll down, I could also get additional statistics as well. One of the tabs here is the **ML-IDS Malware Sandbox** stats. I can see right now I have 76,000 ML IDS Interflow records that have been collected. If I look over here as well I see that this is from March 24. Right? So it is important to also know that everything in this **Sensor Details** page is related to the time I have specified up here as well. Turns out that I was using the IDS module previously. But now what I'll do is go ahead and select a more reasonable time frame since we just enabled the IDS module. Now the stats below will be more current.

**13:00-14:00**
Within the last hour, I have received some ML-IDS Interflow records. If I scroll down I can actually see that yes I am receiving traffic. Okay. Everything is now looking good, and I will close out of this page. Now that the IDS module is enabled, there are some places you can go to view ML-IDS-specific information. ML, again it's machine learning, IDS. Even though we've enabled IDS on the Modular Sensor, the IDS engine is looking for threats in network traffic using the Suricata rules. When it identifies threats, it creates Interflow records and sends those back to the Stellar Cyber platform.

**14:00-15:00**
The Stellar Cyber platform then uses machine learning and statistical analysis rules and heuristic rules to look for threats. If it identifies a threat, that will get created as an Alert. Alerts themselves can be correlated into Cases using graph machine learning. For now, let's take a look at some of the IDS-based Alerts. We go to **Visualize | ML-IDS**. Here are the main Alert types associated with IDS. There are private-to-private exploit attempt anomalies. There are private-to-public exploit attempt anomalies. There are public-to-private exploit attempt anomalies. And public-to-public exploit attempt anomalies.

**15:00-16:00**
In the last hour there have been no threats on my internal lab network. Let me extend the range. Again, I have been using the IDS engine previously. I had turned it off just so I could turn it on again. The first thing I'm going to do is take a look at, let's say, last week. Okay. Now looking at this chart, I can actually take a look at the signatures that have the most activity. If I take a look at this one right now, the ID 2027267, the signature is for an attack response possible lateral movement. Again that's the East/West Movement. There's a file creation request and remote system 32 directory. The category is attempted user privilege, gain, and severity is critical.

**16:00-17:00**
There's been 22,000 entries that have matched this threat. Looking over here as well for this signature, I have the 2027757 is a DNS query for a top level domain as TOs that's the country of Tonga. Or I guess the Republic of Tonga. That's potentially bad traffic there. The severity is high. Once the IDS module on the Modular Sensor detects a threat, it creates the Interflow record and sends that to the Stellar Cyber platform. That is the raw event. But what's cool about that is the other 22,000 entries here and there's 10,000 entries here and there's 1.6k here and 251 entries here. That's a lot of records to sort through.

**17:00-18:00**
Well, the good thing is the Stellar Cyber platform is applying machine learning on top of this and heuristic rules and statistical analysis rules. Based on that, the Stellar Cyber platform has detected six Alerts. Six is a much more reasonable number to deal with. Here to investigate these six, I will select **Investigate**. Now you can see I've been taken to the **Alerts** page. The main difference between going here directly is that by using the **Investigate** for those specific Alerts, the correct filter has already been applied. From here I can see the private-to-private exploit attempt anomalies. I can see the Kill Chain stage for MITRE as Propagation. The tactic is lateral movement. The technique is exploration of remote services. There's even an explanation right below.

**18:00-19:00**
For example, a private IP address initiated a large number of exploit attempts identified by a given signature or a rarely seen exploit attempt to another private IP address, investigate that signature. Okay. If I look over here now, I can scroll down and look at those actual threats. If I take a look here, I have an Alert score of 82. That means if you see an Alert score, the Alert score is what Stellar Cyber has identified as a higher Alert. There's also something called Score, which you might see in a IDS raw event message, and that score actually comes from the IDS engine itself. But the Alert score is generated by the Stellar Cyber machine learning.

**19:00-20:00**
I'll click on the Interflow record for an Alert, which is called More info. In the more info, I see again at the top that this is a private-to-private exploit anomaly. I see the stage is Propagation. The tactic is lateral movement. If I want to know what this actually means, I can select lateral movement here and that takes me to the MITRE ATT&CK framework page. If I scroll down in here, I can get lots of details on what this means. I'll go back to my Alert. I can also select the technique here that also links to the MITRE ATT&CK framework page. Over here now I can see what exploitation of remote services means. Again, you don't have to remember everything. There is a lot of information out there. What's important, more than anything else, is to know how to get that information when you need to get to it.

**20:00-21:00**
Luckily, Stellar Cyber links this information right from the platform itself. Now if I scroll down I can look at the **Key Fields**. Looking at the **Key Fields** here, I see the **Tenant**, which is the root tenant. I see the **Exploit Signature**, which is the attack response, possible lateral movement. There's the **Exploit Severity**, which is critical in this case. There's the **Actual**. **Actual** is a weighted number. If I wanted to

determine what that weighted number means, I can use a little **i**. Here, it's the actual number of times this signature was found in the period. With critical ideas signatures counted as two, high counted as one, low counted as point five, and others counted as one. Again, this is a weighted number.

**21:00-22:00**
I can see the source host and the destination host. The source host is 10.20.1.15. Does anyone remember what that IP address is? It is the IP address of my Modular Sensor, and the destination host is winad-San Jose, Training DC two, so that is my Windows server. The cool thing about looking at this is that I now know where this came from. My Modular Sensor is also running a vulnerability scanner. I will explain more about that later, but the cool thing about that now is it did a vulnerability scan, which looks for threats. The vulnerability scanner actually made a threat that was identified by the Modular Sensor as well in the IDS engine, and therefore, it was reported.

**22:00-23:00**
I can filter out this communication, and most likely I would. I would say, "Look I don't want to look for threats from my Modular Sensor." But for this lab environment it's a great way to see a whole bunch of Alerts. Now I can also see if this Alert is related to a Case. A Case is created by graph machine learning on the Stellar Cyber platform. It correlates related Alerts. Here I can see that the correlated Alerts have a score of 99.6. In other words, the Case score. The platform links you to the Case as well. Looking over here, I can see the Case is a private-to-private exploit anomaly and 30 others.

**23:00-24:00**
In other words, there have been 30 Alerts that have been correlated together. I can see if I scroll down, all the Alerts that are related. Now the cool thing about a Case is that I can analyze it. I can see the external hosts that were involved. I can see the internal hosts that were involved. I can see the URLs that were attempting to be accessed. Over here as well I can look at a timeline and see what happens. I can move this out of the way. Here I see the IP addresses involved. The ten dot twenty dot one to fifteen is my Modular Sensor. These are the URLs that were being accessed on the 10.20.1.12 Windows server. I also see some remote hosts as well.

**24:00-25:00**
If I select the number here, it filters the Alerts in the timeline on the left-hand side. If I want to see what this one is about, it's an external firewall denial anomaly. What does this one say? It is a host with 80.66.88.215 has actions blocked by the firewall too many times trying to access the 10.20.1.12 server. I could investigate more. We cover how to process Cases in the SOC analyst modules, but just so that you can get an idea of what you can do, if I do detect a threat I can disable the user in an identity platform such as Active Directory. I can run a script. Depending on the Connectors that I have connected to the Stellar Cyber platform or to the Modular Sensor, I can have a lot more different actions.

**25:00-26:00**
I will go back to the Alert. If I go further down, I can get the details of the Alert. The Details are all the field names and field values collected There's also the JSON format version of this as well that you can also export here or copy to a clipboard. What's also really cool is that I can take an action.

Depending on the Connectors I have configured either on the Modular Sensor or on the Stellar Cyber platform, I have a set of external actions that are available. With an Active Directory Connector I can disable the user, I can run a script that goes on a server.

**26:00-27:00**
If I have a Connector to an EDR platform, I could contain the endpoint, and there's many more actions depending on the Connectors I have. Here I would be able to disable the user. In this environment, I have Connectors to Active Directory. I could locate the user associated with this Alert and disable the user. I will demonstrate that in the Connector lessons. Another very important concept to understand is that the IDS module on the Modular Sensor is looking at these 60,000 Suricata rules to identify threats. When network traffic matches one of the rules, the IDS module creates the Interflow record and sends that to the Stellar Cyber platform as a raw event.

**27:00-28:00**
It is a raw event as an Interflow record, but it's still a raw event. Now if I had to look through 32,300 events by myself, that can take a long time. That's where machine learning comes in and the statistical analysis rules and the heuristic rules. Letting the Stellar Cyber platform do the work gets rid of the noise, so you can focus on the real threats. Now if I scroll down and look at the threats that Stellar Cyber has determined out of all those 32,000 events, these are the six that are determined to be real threats, and they have been given a score. The first one I see here is a score of 82. I can then go ahead and analyze that as well. By the way, the score with an 82 had 23,580 raw events associated with it.

**28:00-29:00**
Now another important place to go to investigate IDS-related Alerts and events is the **Investigate | Threat Hunting** page. From the **Investigate | Threat Hunting** page, the default indices is **Alerts**. Again, Alerts are the statistical analysis rules, the heuristic rules, and the machine learning Alerts that are created from the raw events that are received. If I go over here, I'm looking at Alerts for the past week. What if I just want to see Alerts related to IDS?

**29:00-30:00**
Well, you can do that quite simply from this interface. If I scroll down to here where it says **Top Log Source Classes**, if I look at the legend, one of the legend entries is **ids_event**. Let me zoom in so you can see it better. Over here I will create a filter that includes ids_event. Now this whole page is for everything that is related to an IDS event. You'll also see here that there's a filter created. Scrolling down I see ids_event is a message class. What if I want to know what that actually means and what it looks like? The cool thing is you can copy as query. I'll also remove this filter now because I'll do the same query up top.

**30:00-31:00**
I'll close these filters. If I expand this content search here, I'll paste in that query. I have message_class is the field name. The value is ids_event. This query also has or message class.keyword colon quote IDS event end quote. Now keyword just says that the IDS event must be in this exact case. I really don't care right now. All I really care about is that the message class is an IDS

event. If I select **Return** again. Let me minimize this. If I select **Return**, I go from 283 back to eight But if I want to filter more now, I can take a look at one of these events.

**31:00-32:00**
Let me take a look at ones by sorting on the **Alert Score**. This Alert with an 82 is the Alert we were looking at in the **Visualize | ML-IDS** page. But what if I want to get more information? If I expand this Alert and maybe I just want to search for similar to the same destination IP. I'll look over here. Let me expand this more. So the destination IP 10.20.1.12, I can also copy as query. I'll minimize that. You'll have a green notification box that the query was copied. I'll go back over here. I'll zoom in one more time and I will type in AND. AND has to be uppercase.

**32:00-33:00**
These operators OR and AND must be in uppercase, and I can paste. This query is saying show me all IDS events that have a destination IP of 10.20.1.12 in the **Alerts** indices. In other words, it has to be an Alert. Now minimize this again, zoom out, and I'll press **Return**. Now I have three events that are going to the same IP. To validate that, if I expand one of these, I can look for the destination IP is 10.20.1.12. If I were to scroll down further, I can look for the message class. There it is right there: Message class is IDS event.

**33:00-33:35**
It's really cool to be able to do the copy as query and to focus and filter on what you really want to find, to find related events. I can go ahead and say **Clear all** if I have anything here just to make sure. You do want to make sure that if you are done filtering or searching for an item, that you do clear your filters.

This lesson is complete. Thank you for watching.

# L20 - MALWARE SANDBOX

## L20A-PRES: Malware Sandbox Details and Configuration (06m:32s)

**00:00-01:00**
The next lesson is *Malware Sandbox Details and Configuration*. This lesson, *Malware Sandbox Details and Configuration,* provides the concepts and details on how to configure the Malware Sandbox functionality on a Modular Sensor.

Hi everyone. Here I will describe the malware sandbox built into the Modular Sensor. A Malware Sandbox is a security mechanism that allows for the analysis and execution of potentially malicious files in a controlled environment. It is used to detect and analyze the behavior of malware without risking the security of the host system.

**01:00-02:00**
The sandbox isolates the malware from the rest of the system, allowing security analysts to observe its actions and gather information about its behavior, such as file modifications, network connections,

and system changes. By running the malware in a sandbox, security professionals can gain insights into its capabilities and develop appropriate countermeasures. Within the Modular Sensor, on Modular Sensor feature number four, I will describe the Malware Sandbox. The Malware Sandbox is enabled in the license called APT for advanced persistent threat. After receiving and identifying files in the network traffic from SPAN ports or TAP devices, Modular Sensors identify, extract, and reconstruct malware traveling through the network within unencrypted protocols

**02:00-03:00**

such as HTTP, FTP, which is the file transfer protocol, SMB, which is Windows file sharing, and SMTP, which is simple mail transfer protocol. Modular Sensors forward files identified as malicious to an external HTTPS server for storage. Modular Sensors normalize malware metadata into Interflow records and forward to the Stellar Cyber platform. From the Stellar Cyber platform, you can view and analyze observations of malware traveling through the network and arriving at specific hosts as well as view summaries of the events related to malware. Enabling the Malware Sandbox requires authorization of the APT Malware Sandbox license for Modular Sensors.

**03:00-04:00**

To authorize, you can go to **System | Sensors**. Then on the **Sensors** page, select **Manage**, then **Authorization**. If you look at the radio buttons on top, there is an **Authorize** and **Unauthorize** radio button. You're going to ensure the **Authorize** radio button is selected. If The Modular Sensor itself was not authorized. it would be on the left-hand side. As it has been authorized by the administrator already, I see the SC Modular Sensor on the right as a selected sensor. Underneath, I have an IDS license and an APT license. I have checked both of them.

**04:00-05:00**

You just need to make sure if it's not checked, to check the APT license. After you're done, you can select **Submit**. As a side note, APT stands for advanced persistent threat. It refers to a prolonged and targeted cyber attack in which an intruder gains access to a network and remains undetected for an extended period. APT attacks are initiated to steal data rather than cause damage to the target organization's network. After submitting authorization, the **Sensor Status** will display as red and the message will show that it is disconnected and the configuration is out of sync. But you will see the authorization as authorized for the sensor.

**05:00-06:00**

After a minute if you refresh the browser, you will then see the message go away, and the status will be green, and the sensor of course is authorized. The Modular Sensor sandbox is enabled by editing the Modular Sensor profile in **System | Sensor Profiles**. In the Modular Sensor profile you must enable **Network Traffic** because to find malware, you actually have to capture network packets. You then ensure you enable **Sandbox**. One of the options under **Sandbox** is that you can set the max file size in megabytes, exclude MIME types, and a region close to the sensor to minimize latency in sending files.

**06:00-06:31**

You can leave the region as **Automatic** and it will pick up the region from the sensor, or you can select a specific region. After completing your configuration settings in the sensor profile, the **Submit** button must be selected to apply the configuration settings to the Modular Sensor. Hopefully by now this is a very simple task for you.

This lesson is complete. Thank you for watching.

# L21 - MALWARE VISIBILITY IN THE STELLAR CYBER PLATFORM

## L21A-PRES: Malware Visibility in the Stellar Cyber Platform (06m:33s)

**00:00-01:00**
The next lesson is *Malware Visibility in the Stellar Cyber Platform*. This lesson, *Malware Visibility in the Stellar Cyber Platform,* explains how to view details about files collected and analyzed by the Malware Sandbox module in the Stellar Cyber Open XDR platform.

Hi! In this lesson, I will show you how to validate that the malware module is enabled in the Stellar Cyber Modular Sensor. I'll also show you how to view the occurrences of malware within the network traffic received. I'll also show you how to mitigate threats by blocking an IP using a Connector to a firewall.

**01:00-02:00**
To begin, for step number one, let's go ahead and select **System** and **Sensors**. That takes you to this **Sensor** page. On the **Sensor** page, you will select the name of the Modular Sensor that has the Malware Sandbox enabled. For step number three, if you look on the right-hand side, you have a **Service** and **Status** for Malware Sandbox. Right now we see that the Malware Sandbox is green for **Status**. That's a very good sign. You can also view the ML IDS Malware Sandbox output chart and see the number of events. Here we do see that the events are increasing.

**02:00-03:00**
Therefore, we know it's working. Also the chart below gives you details on this bits per second for example, for the malware and IDS. To view the Alerts related to malware and take an action to mitigate threats, you can go to **Alerts** on the top menu bar, then search in the search bar: type malware. By doing that and making sure you hit **Enter** or **Return** after typing "malware," you will see, for example, uncommon process anomaly or external other malware. Let's take a look at the external other malware. On the right-hand side for step number three, you can select **View**. When you see the chart appear, you can scroll down to view the malware or other malware documents.

**03:00-04:00**
You can locate an Alert record and select the **i** for more information. This will bring up the Interflow record. In the Interflow record, you can select the **Key Fields** and **Details** to look at the IP addresses and usernames and other important information associated with this malware event. If you want to

mitigate, for example, there's a host out there that has been infected, and you want to block the IP address or prevent that host from sending any other information. You can go to **Actions**.

**04:00-05:00**
One of the options available, depending on your Connectors, if I have a Connector to a firewall, I have Block IP. If I have a Connector to an EDR platform such as CrowdStrike or BlackBerry Cylance, or any of the EDRs that exist, you can have a Contain Endpoint option as well, which is actually probably one of the better solutions to do, but Block IP will be a good example for this demonstration. In the action configuration after selecting block IP, for step number nine, I can select the **Connector**. I have a Connector to a FortiGate firewall. I can select the IP source host, which actually comes from the Interflow record. I can specify a **Direction** for the network traffic for the firewall policy.

**05:00-06:00**
In this case, I will select **Any**. Then you can specify a duration. Maybe I just want to block an IP for five hours while I'm doing some research investigation or just even a few minutes. I can say 30, then put minutes for the duration. It's your choice. You can even put "forever," but I think that will be drastic. I don't think we need to do it forever. At that point, when you're done, click **Submit**. After submitting, if you go to the top menu bar and select **Investigate | Actions**, you can see the list of actions that have been performed. In this case, I see that the firewall action did succeed.

**06:00-06:35**
If, for example, the username was wrong for my API or something of the sort, you might get it failed. I do have some tests below that do show failed, as an example. But once it succeeded, that IP now will not be allowed to go through the firewall. There are many other actions that can be performed. This just gives you an example of where to check your actions are working.

This lesson is complete. Thank you for watching.

## L21B-DEMO: Malware Visibility in the Stellar Cyber Platform (24m:12s)

**00:00-01:00**
This lesson continues with demonstrations for malware visibility in the Stellar Cyber platform. This demonstration explains how to configure the Malware Sandbox on Modular Sensors and shows how to view details about files collected and analyzed by the Malware Sandbox module in the Stellar Cyber Open XDR platform.

Next, I will show you how to enable the Malware Sandbox. Malware Sandbox is used in conjunction with IDS. To do that you go to **System | Sensors**. On the **System Sensors** page, the Malware Sandbox is another authorization.

**01:00-02:00**
If I go to **Manage | Authorization** you'll see here that there's something called APT license. That's advanced persistent threat, which is the Malware Sandbox. I'll select that to enable that option and then submit. Once the authorization is made, the sensor will reload the configuration so that the

module will function. It only takes a second. Now I will go and configure the **System | Sensor Profiles**. Again, the Modular Sensor profile that I am using for this demonstration is the SC Modular Sensor New. I will select the pencil to edit this profile.

**02:00-03:00**
To enable the Malware Sandbox feature, there are three options that need to be enabled. The first is **Network Traffic**. The Malware Sandbox analyzes network traffic to find files. Therefore, **Network Traffic** must be enabled. Second, you must have **Malware Sandbox** enabled. That one is pretty obvious I hope. The maximum file size for a file that will be analyzed is ten megabytes. This is the default and the maximum size possible. Next, you can exclude MIME types. By default we've excluded types that are not really considered a threat. However, if we go in here, and select next to one of these bubbles, there are many other options as well that you could exclude if you find they're not important to exclude in your environment.

**03:00-04:00**
Select outside of the box to close the selection. Next, the region. There are regions all around the world where the Malware Sandbox can store files that are uploaded from the Stellar Cyber Modular Sensor. When malware is found, the malware will be stored in a location of your choice, and it should be the closest region to where your Modular Sensor is deployed. The Malware Sandbox requires the IDS module as well for a couple of purposes. One is for the reconstruction or reassembly of files from packets that are being forwarded to the Modular Sensor. Two, it's for the definition of what files should be checked.

**04:00-05:00**
This is based on the IDS networks that were specified. From the home network to external network or external network to home network and home network to home network, as an example. With these three things: Network Traffic, the sandbox feature, and IDS, I do not need to configure anything else. The sandbox will work appropriately. I'll submit. To check on the status, I go to **System | Sensors**. I see that my Modular Sensor is already a status of success. I will expand that out so we can see it. There we go. I'll select my Modular Sensor. If I go over here, I see that Malware Sandbox is running successfully.

**05:00-06:00**
If I go over here as well and look at the ML-IDS stats, I do see that I do have traffic coming in and being processed for ML-IDS, which is also being processed for files. Next, I will go to **Visualize** and **Malware Sandbox Overview**. In this Malware Sandbox view, I will start with the **Overview**. In the **Overview**, I can see that there are lots of good files found, but there was also a malicious file. It's called eicar_com.zip. I can select the **(i)** to get more info. Looking at this, I see the file name here. One of the most important things to understand about analyzing files is that the file name could be anything.

**06:00-07:00**
It might even be the name of a file that you use on a regular basis. You cannot rely on file names whatsoever. No one's going to create a malware file and call it, “malware file” because then you won't

click it and open it and get exposed. One of the most important things to understand is that every file has a hash. A hash algorithm looks at the entire file and creates a 128-bit output representing that file only. If I change a file in any way, then this MD5 hash will also change. There's also SHA-1, which is a hash algorithm as well that uses 160-bits and there are other different versions as well. But the most important thing is the name of the file can be changed, but the contents of the file will always produce the same hash.

**07:00-08:00**
Now let me give you an example. Here, I will look at a file that I have on my Mac. This file is called "testfile1.text." I know, it's a very complex file name. I can say `vi`, which is an editor, and I can look at this file. This file only has, "Stellar Cyber training, example of hash function." Now Macs come built in with an MD5 hash function, and I can hash. I can run the hash on this file. Here I see the 128-bit output starting with "b," and ending in "6." Now let me rename the file. I'll say, "CP testfile1.text to testfile2.text."

**08:00-09:00**
If I do the MD5 hash again, on testfile2.text, you'll see it's the exact same hash. If I had malware in one of these files and I copied the name to something that looks right, it's still malware, and it will be detected. It does not matter what the file name is. But if I were to modify testfile2.text in any way, I'll just add a one inside the file name. I run the MD5 hash again, you'll see that the hash has changed. That's the concept of looking for malware. It's looking for essentially a signature or a hash of the file to see if known bad malware is found. Alright. I hope that was helpful. Okay. Now I will take a look at this MD5.

**09:00-10:00**
One of the options you can do is do a VirusTotal lookup on that hash. Sure enough, I see that VirusTotal has found that this is a bad file. The other thing I can do is copy this to clipboard. Over here I can see if there's any other file with this hash. I can do a search. I can zoom in a little bit for you. [Types] md5: the hash. Again, it only finds the one file. That's what we're actually expecting because I only had one occurrence of it. Okay. Now I will close out this filter. I'll select **Clear all**, minimize the filter box, and we'll go back to the main screen.

**10:00-11:00**
Now I will go through the tabs. Looking at the tabs on the **Visualize** | **Malware Sandbox**, if I had ransomware I would see accounts here or trojans, spyware, PUS or potentially unwanted applications, or other malware, but I also have good files. Looking at the good files, hopefully that you have a higher number of good files than the rest. If I scroll down and take a look, I can see that I have a lot of events related to good files and hopefully that is the case. I can also take a look at these good files. Just like I did for the malware, if I'm investigating a threat or I want to see if someone else had received the same file as somebody else, I could take a look at the MD5 hash of the file.

**11:00-12:02**

I can copy that to clipboard or add as an including filter. If I go over here, from the top. Again, I can zoom in here, do a paste, and search for MD5:, and I'll zoom back out. Here I see there are three events related. This destination host here, this host here, and this host here all have the same file. Maybe this file was part of an upgrade that was done by each of these three servers. They are all Windows servers. That is potentially what happened. But either case it's good to be able to have access to the file hashes so that you can use that in your threat investigations.

This lesson is complete. Thank you for watching.

# L22 - MODULAR SENSOR Aggregator

## L22A-PRES: Modular Sensor Aggregator (09m:32s)

**00:00-01:00**
The next lesson is *Aggregator Details and Configuration.* This lesson, *Aggregator Details and Configuration,* explains the concepts and configuration of Modular Sensors to be an Aggregator as well as how to configure other Stellar Cyber sensors to use the Modular Sensor as an Aggregator.

In this lesson I will cover the Modular Sensor Aggregator, and its uses and how to configure it. Looking at the picture on the left-hand side, the Aggregator is listed as one of the core features of the Modular Sensor.

**01:00-02:00**
When configured as an Aggregator, the Modular Sensor acts as a proxy to forward traffic from other Modular Sensors, Windows Server sensors, and Linux Server sensors to the Stellar Cyber platform. Sensors send Interflow records to the Modular Sensor Aggregator using TCP port 8080. It's therefore important that you don't have any type of firewall or any type of access control blocking TCP port 8080 between the sensors and the Modular Sensor Aggregator. When the Interflow records get sent by these sensors to the Aggregator, it will then forward them all from a single IP of the Modular Sensor to the Stellar Cyber platform. The Aggregator feature reduces the number of connections that are made from the Modular Sensors and server sensors within an organization to the Stellar Cyber platform over the internet.

**02:00-03:00**
For some organizations that's important. For a lot of other organizations, it doesn't matter that each Windows Server sensor or Linux Server sends directly to this Stellar Cyber platform. This is a preference based on your organizational needs or your customer's organizational needs. You can configure multiple Aggregators because the sensors can have a primary and backup as well. When the Aggregator feature is not in use, all sensors for an Interflow records directly to the Stellar Cyber platform. Just so it's clear, when the Aggregator feature is not in use, all sensors–Linux Server sensors, Windows Server sensors, Modular Sensors, forward Interflow records directly to the Stellar Cyber platform.

**03:00-04:00**

With the Aggregator feature enabled, sensors can be configured to forward Interflow records to the Stellar Cyber platform through the Aggregator. To configure the Modular Sensor to be an Aggregator, modify the Modular Sensor profile in **System | Sensor Profiles**. In the Modular Sensor profile, enable Aggregator as in step number one. For step number two, select **Submit**. That's pretty simple. That's all there is on the Modular Sensor. The rest of the work is to tell the other sensors within the organization to use this Modular Sensor Aggregator. You can configure other Modular Sensors as Aggregators for scale. You can also configure other Aggregators for redundancy as well.

**04:00-05:00**
When you configure a sensor, you have the ability to configure a primary Aggregator and secondary Aggregator. Okay. Next, configure Modular Sensors, Windows sensors, and Linux Server sensors, within a network to use the Modular Sensor as their primary Aggregator. As a reminder, TCP 8080 must be permitted between sensors and the Modular Sensor Aggregator within the network. Now to configure Modular Sensors, Windows Server sensors, and Linux Server sensors to use the Modular Sensor as their Aggregator, on step number one, select **System**, then go to **Sensors**. This will bring up a list of sensors you have deployed.

**05:00-06:00**
For each sensor on step number two, that will use a configured Modular Sensor as its Aggregator sensor, select the pencil icon to edit the sensor parameters. This will bring up the list of parameters for a sensor. For step number three, select the primary Aggregator, which is the host name of the Modular Sensor configured as an Aggregator. Once you select a primary Aggregator, a secondary Aggregator is available to be selected. If you do not have a secondary Aggregator, that is okay. This is optional. Next, select **Submit**. After selecting **Submit**, the sensor status will display the following: The messages column will show as disconnected configuration out of sync, the Sensor Status will be red, and the authorization will be authorized.

**06:00-07:00**
After a minute if you refresh the browser the messages will show empty. The Sensor Status will show green and the Modular Sensor will still be authorized. After the Modular Sensor status is green, you can check the status of the Modular Sensor Aggregator. Now that the Modular Sensor has been configured as an Aggregator, and another sensor has been configured to use the Aggregator, you can check the status of both using CLI commands. Select the checkbox next to the Stellar Cyber Modular Sensor that is the Aggregator, and then you'll see an option for CLI access. Select **CLI Access**. In the **CLI Access** window you can type `show Aggregator`.

**07:00-08:00**
After typing `show Aggregator` in about twenty seconds or so, you will see a message that shows DS, which is data sensor, is not routed through an Aggregator. That is true because it is in itself the Aggregator. This device acts as an Aggregator, and it will show you that it has received traffic from sensors, and has sent traffic to sensors, and it is connected to a sensor. In this case, it's ten dot twenty dot one dot twenty three and the connection is established. That's a very good sign that everything is working. Next, similar to what you just did with the Modular Sensor Aggregator, you can also check the status of the sensor that is using the Modular Sensor as its Aggregator.

**08:00-09:00**
For step number one, from the **System | Sensors** list, check the box next to the name of the sensor that is using the Modular Sensor as its Aggregator. For step number two, select **CLI Access** from the **Sensor CLI Access** window. For step number one, type `show Aggregator`. For step number two, you'll see after about twenty seconds, that the DS, or data sensor, is routed through the Aggregator, which shows the correct IP address of the Modular Sensor, which is 10.20.1.15. You can also see that the connection is established to the configuration and management connection of the Stellar Cyber platform.

**09:00-09:31**
At this point everything looks great. The connection is working and data is being sent from this sensor to the Modular Sensor, which is then forwarded to the Stellar Cyber platform. All done there. That's all we need to do for this.

This lesson is complete. Thank you for watching.

## L22B-DEMO: Modular Sensor Aggregator (07m:51s)

**00:00-01:00**
This lesson continues with demonstrations for Aggregator details and configuration. This demonstration explains the concepts and configuration of Modular Sensors to be an Aggregator, as well as how to configure other Stellar Cyber sensors to use the Modular Sensor as an Aggregator.

Hi everyone. In this lesson, I will show you how to configure a Modular Sensor as an Aggregator. From the **System | Sensors** page, I can see that I have three sensors. The first one here is a Modular Sensor, the second one is a Linux Server sensor, and this one is a Windows Server sensor.

**01:00-02:00**
By default, every sensor communicates directly with the Stellar Cyber Open XDR platform. When the Windows Server sensor creates an Interflow record from events, it will then forward that to the Stellar Cyber platform. When the Linux server sensor generates Interflow records from events, it will then forward that directly to the Stellar Cyber Open XDR platform. Of course, the Modular Sensor does the same. In some organizations, you may want to limit the connectivity of servers talking directly out to the internet or to the Stellar Cyber Open XDR platform. If you want to consolidate the communication, you can configure the Stellar Cyber Modular Sensor as an Aggregator.

**02:00-03:00**
To do that, I go to **System | Sensor Profiles.** The profile that I'm configuring right now for my Modular Sensor is the SC Modular Sensor New. Here, I can enable the Modular Sensor as an Aggregator. It's pretty simple. That's all I have to do now. You'll see here, by the way, it says Aggregator: Act as a proxy to forward traffic from sensors to DP. When you see the letters DP in capital letters, that is the Stellar Cyber Data Processor. The Data Processor is another for the Stellar

Cyber platform. I will now select **Submit**. The interesting thing about this is that this doesn't change anything yet.

**03:00-04:00**
Even though the Stellar Cyber Modular Sensor has the aggregated capabilities, other sensors now need to point to that Modular Sensor. I can select the pencil of the sensor that I want to edit. Here, there's an option for **Primary Aggregator**. I can go ahead and select SC Modular Sensor. Note that if I had multiple Modular Sensors in the organization, which might be either for redundancy or for scale. In other words, I'm sending everything to one Modular Sensor, I might want to have multiple. In this case I only have one Modular Sensor so I don't have a secondary and that's okay. I will go ahead now and select **Submit**. And that's all I need to do.

**04:00-05:00**
The Linux Server sensor I see here will now reload its services to use the new configuration, using the Modular Sensor as its Aggregator. If I take a look now, I hit refresh. I see that now that it's active. To validate the status of the Aggregator what I can do is go to the CLI shell. One way is from this screen right here where I can select the Modular Sensor and select **CLI Access**. From the CLI window, I could type in `show aggregator`. This will take about 10 to 20 seconds.

**05:00-06:00**
Here I can see that this device is an Aggregator that it has received from the sensor and sent to the sensors. In other words, it has received from sensors and it is sending traffic to sensors back. There's another sensor that's connected, which is the 10.20.1.23. To see this a little clearer I'll use my SSH. An SSH to aella@10.20.1.15. I type in my secret password. Here I could type in `show aggregator`. As you saw on the previous screen, I can see the communication is to the Stellar Cyber Open XDR platform Data Lake, which is 192.168.1.175. I see the port, which is 6644.

**06:00-07:00**
I see that this device is an Aggregator. It is receiving traffic from sensors and it is sending traffic back to sensors. It is connected to the sensor 10.20.1.23. If you look over here 10.20.1.23 is the Linux Server sensor. I can do the same thing again for the Windows Server sensor as well. That's all there is to configuring an Aggregator. I could configure an Aggregator per tenant for example, or for each site that my sensors are in so I can have one set of communication from that site back to the Stellar Cyber Open XDR platform. Those are some of the main reasons why I might do this. It's really just to aggregate all the communication from each site.

**07:00-07:51**
This is an optional feature. Now you also want to be careful here as well though. If I do have a Modular Sensor that's receiving all the Interflow records from all the other sensors, you have to make sure that there's enough bandwidth, interface speed, and processing power on the Modular Sensor that is an Aggregator. In this case, if you are finding that you have too many connections being made to the Modular Sensor, you can configure multiple Modular Sensors as Aggregators. Then you have a choice of which servers point to each Modular Sensor. Okay. Well that's it for this lesson. Thank you very much.

This lesson is complete. Thank you for watching.

# L23 - MODULAR SENSOR TENABLE NESSUS SCANNER

## L23A-PRES: Modular Sensor Tenable Nessus Scanner (15m:27s)

**00:00-01:00**
The next lesson is *Tenable Nessus Scanner Details and Configuration.* This lesson, *Tenable Nessus Scanner Details and Configuration,* provides the concepts and configuration of how to enable the ability to initiate scans from the Tenable cloud website to run on the Nessus vulnerability scanner module that can be enabled on Modular Sensors.

Another cool feature of the Stellar Cyber Modular Sensor is that it contains a Tenable Nessus scanner. A Tenable Nessus vulnerability scanner is used to scan servers either by IP address or by network or by IP ranges

**01:00-02:00**
to look for vulnerabilities within the operating system or those devices themselves. You can even configure the Nessus scans to log into servers–Linux Servers and Windows Servers–for example, to, again look for threats and then report those back. A lot of organizations use Tenable Nessus vulnerability scanning capabilities, but they usually have to install a separate server within a network and run the Tenable Nessus scanner from that. Stellar Cyber Modular Sensors actually have a Tenable Nessus vulnerability scanner built in. You still need to have an account on Tenable.io. Once you get an account on Tenable.io, then you can then use that license to link the vulnerability scanner that's built into Modular Sensor to run a security scan.

**02:00-03:00**
From the Tenable website, which is cloud.tenable.com as an example, I can actually go to an option called **Scans** and run a scan that will then be triggered from the Modular Sensor, and then reported back to the Tenable cloud site. Once the Nessus scan is complete, the Tenable site creates and makes a report available with the results. Here's an example of the results you will see when running a vulnerability scan. An important note, for configuration of Tenable Nessus's vulnerability scanner on the Modular Sensor virtual appliances

**03:00-04:00**
you will want to add an additional 1.5 gigabytes of RAM and add an additional vCPU. Alright? You can do this after the fact. Again once you determine you want to run the Tenable Nessus vulnerability scanner, you can go back to your virtual environment and increase the memory and vCPU and then essentially restart the sensor. It will then have the correct amount of resources to run properly. Nessus's scan functionality does require purchasing a vulnerability management service from Tenable.io. Once you have that purchased from the Tenable cloud website, you go to **Sensors**, then **Nessus Scanners**.

**04:00-05:00**
The next step is you add a Nessus scanner, which is on the right-hand side, and you'll be provided a linking key. You can copy the linking key and that will be used in the configuration on the Stellar Cyber Modular Sensor. Back in the Stellar Cyber platform, Tenable Nessus is enabled by editing the Modular Sensor profile in **System | Sensor** Profiles. In this Modular Sensor profile, you enable Tenable Nessus. For step number two, you paste the linking key from Tenable.io copied from the **Add Necessary Scanner** section. Note that the linking key I have here is not valid. For you I've used it in my test lab, but then disabled it.

**05:00-06:00**
For step number three, you enter the host, which in our case is cloud.tenable.com. Step number four, enter port 443. That's not there by default so please make sure you do enter it. After completing your configuration settings in the sensor profile, The **Submit** button must be selected to apply the configuration settings to the Modular Sensor. After you select **Submit** on the Modular Sensor profile, the Modular Sensors using the profile will restart their services using the new configuration. Again, we've gone through this several times but again it's good to see. With repetition we remember. Alright. Step number one the **Sensor Status** will go red, and the message will display the configuration is out of sync.

**06:00-07:00**
After a minute, you will have to refresh your browser, and you will see the **Sensor Status** go green, and the message will be clear. To validate the status of the Tenable Nessus module, go to **System | Sensors**. Then select the name of the Modular Sensor with the Tenable Nessus scanner enabled. You then verify the **SERVICE STATUS** for Tenable Nessus is green. Here the one I have is green so we are good to go. You can get more details of the status of the Tenable Nessus client from the Modular Sensor CLI.

**07:00-08:00**
Again from **System | Sensors**, select the checkbox next to the Modular Sensor that has the vulnerability scanner enabled. For step number two, select **CLI Access**. You will then have a **CLI Access** window that appears. From that window you can type `show tenable`. By the way, if you are on the network, and you are the IT administrator, for example, you can SSH to the Modular Sensor and run the same set of commands. Step number two, about twenty seconds after typing in `show tenable` from the **CLI Access** window you will see a working result hopefully that says, "Linked." It'll show you linked to the cloud.tenable.com:443, which is the port number.

**08:00-09:00**
Back on Tenable, the name of the Modular Sensor will show up in **Settings | Sensors | Nessus Scanners**. Then select **Linked Scanners**. If you have a lot of scanners available, you can type the start of the host name and the list of linked scanners will be filtered. Under **Linked Scanner**, the Modular Sensor will be listed. Then select the name of the Modular Sensor. From the **Overview** tab, you can see the status of the Nessus scanner within the Modular Sensor. Here we see that the status is "Online." Next, you can create a vulnerability scan on Tenable.

**09:00-10:00**

If you select the stacked line, often known as the hamburger icon, you can go to **Scans**. Then on the right-hand side for step number three, **Create Scan**. Now there are many different types of scans. The Advanced scan gives you a lot more flexibility. For this case, I will just do a Basic Network scan, which is an in-depth scan. It just means that it's less for you to do. Enter a name for the vulnerability scan. Mine is called, "SC Labs Training Basic." For the scanner type, choose **Internal Scanner**. You then select the Stellar Cyber host name as the scanner from the scanner list. By default it's set to auto. Please make sure you select the scanner to ensure that it is the scanner being used.

**10:00-11:00**

Again, I have a note to verify that you have selected the Stellar Cyber Modular Sensor as the scanner. For step number eight, there are targets. Targets can be IP ranges. For example, 10.20.1.21-10.20.1.23. That will do 21, 22, and 23. Comma, an individual host such as 10.20.1.11. You could even do a CIDR notation and say 10.20.1.0/24 and that will target the entire subnet. But again, every scan is very in-depth and does take time. For the very first time, you might want to consider using a single IP address, then after you know everything is working, then increase the range.

**11:00-12:00**

The next thing you can do is specify email recipients. I could put in, for example, one of our heads of training in Australia. His name is David Shirley, and myself as well are the email recipients. I can then also give the correct permissions. The permissions may only need to be **Can View**, but mine is **Can Edit** just in case. Next, you could save and not launch, but I will save and launch so that I can see the test actually happen. Next, back on the Stellar Cyber platform, to validate the scan is started on the Modular Sensor Tenable Nessus's client, SSH to the Modular Sensor

**12:00-13:00**

or use the CLI access in the Stellar Cyber platform by going to **System | Sensors**, and select the checkbox next to the Modular Sensor. Then select **CLI Access**. From the **Sensor CLI Access** window, you can type `show tenable`. Then wait about 20 seconds. When the results come back you will see that the scanner is linked to cloud.tenable.com:443, which is the port number. In this case, I actually have a scan started, so there's one job pending. Back on the Tenable cloud, the status of the scan will be displayed as follows. First you will see initializing, then pending, then running, then publishing results, then finally, the scan will be completed, and you can then view the results.

**13:00-14:00**

It may take a few minutes to over an hour for a scan to complete depending on the number of targets and the type of scan you perform. To see the scan, select the scan name and select **See All Details** in the upper right. The resulting report provides details for vulnerabilities for target hosts specified in the scan. The email recipients configured in the scan are also sent a copy of the vulnerability report. In case you run into any problems on the Tenable Nessus scanner, if you see for example, that it does not appear on Tenable.io, on the Modular Sensor you can type `show tenable`.

**14:00-15:27**

If you see that the scanner is not linked to a manager, you must ensure that the linking key that you copied from the **Add Network Scanner** is copied and pasted into the Tenable Nessus configuration in the Modular Sensor profile assigned to the Modular Sensor. For example, if you added the linking key to a different Modular Sensor profile, you will see the message on the left–that it's not linked. Or if you had a typo in the linking key, the same thing will happen. Also, when configuring the Tenable scan on cloud.tenable.com for example, you might get an error message that appears: The following targets were not routable. In this case, please ensure the Modular Sensor is specified as the Modular Sensor in the configuration in the Nessus scan. Remember previously when I talked about something you must double check? By default, the scan will be set to using a scanner of auto. You want to make sure that the scanner is selected as the Modular Sensor. Okay. These are just some tips for solving some problems I ran into. Hope you enjoyed the lesson.

This lesson is complete. Thank you for watching

## L23B-DEMO: Modular Sensor Tenable Nessus Scanner (12m:47s)

**00:00-01:00**
This lesson continues with demonstrations for *Tenable Nessus Scanner Details and Configuration.* This demonstration provides step-by-step details for configuring a Tenable Nessus vulnerability scanner on a Modular Sensor, adding the Modular Sensor vulnerability scanner to Tenable.io, and creating a scan on Tenable.io that runs on the Modular Sensor Tenable Nessus vulnerability scanner.

Hi everyone. In this lesson I will show you how to configure the Tenable Nessus scanner module on the Modular Sensor. With this module from Tenable.io. or cloud.Tenable.com, you can actually launch a vulnerability scan right on the Modular Sensor to scan the local network.

**01:00-02:00**
This is a really cool feature because if you don't use a Modular Sensor, oftentimes organizations have to install a separate server to run a Tenable Nessus scanner. Having the ability to run the Tenable Nessus scan right from the Modular Sensor definitely saves a lot of time and resources as well as makes it really easy to run the scans. First, let me show you how to configure a Tenable Nessus scanner. As we've done many times before, to configure a Modular Sensor, we have to modify the Modular Sensor profile. I can select the SC Modular Sensor New and edit by selecting the pencil. Here we've enabled the Log Forwarder, the Network Traffic, Sandbox, and IDS. None of these are actually needed for being a Tenable Nessus scanner.

**02:00-03:00**
Once you slide the lever to enable the Tenable Nessus vulnerability scanner, there are a few parameters that you need to configure. One is the linking key, two is the host, and three is the port. The port is 443, but the linking key I need to obtain from Tenable. Let me show you how that's done. Let me close this out. I have an account on cloud.Tenable.com. I'll go there now. From cloud.Tenable.com, I have the option to go to **Settings**. In **Settings**, the option I'm looking for is **Sensors**. Under **Scanning**, there are **Sensors**.

**03:00-04:00**
Here I see that there are quite a few sensors within my organization. What I want to look for are Linked Scanners. If I take a look at the scanner name that I have, it's called SC-Modular Sensor. You'll see that I don't have it in this list. To enable the Tenable Nessus scanner, I can select the option to **Add Nessus Scanner**. I have a linking key. I will copy this linking key. Now that I had the linking key copied, I will go back to the Stellar Cyber platform and I will go to SC Modular Sensor New and edit. Next, I'll enable the Tenable Nessus's vulnerability scanner and I will copy the linking key. The port will be 443. That's just HTTPS. Then I select **Submit**. Enabling the Tenable Nessus's vulnerability scanner on the Modular Sensor is a very simple task, but there's also something you should be aware of. It does take a little bit more CPU and memory. The best way to see what you should do here is go to the documentation. In the documentation of the Modular Sensor profile towards the bottom is the **Tenable Nessus** section. In the **Tenable Nessus** section you can see this note: "This feature is supported on Modular Sensors running version 434 or later. Stellar Cyber recommends that you provision the Modular Sensor with an additional 1.5 gigabytes of RAM and one vCPU when you enable this option."

**04:00-05:00**
You can go–usually before you enable the Tenable scanner to enable this option. That will definitely give you better performance. Okay. Now I will go to **System | Sensors**. I will check my Modular Sensor. I see that Tenable Nessus is enabled. Great. Now I will go back to cloud.Tenable.com. Okay. Back in the **Linked Scanners**, I will look for my Modular Sensor. There it is! It is now connected and functioning. That is perfect. The next step that I need to do is configure a scan that uses this Modular Sensor vulnerability scanner.

**05:00-06:00**
To do that, you select the three-line stacked icon. I go over here and go to **Scans**. Then from **Scans**, I can go ahead and create a new scan. Here, there are many different scans. There's an Advanced Network scan, which I could use and make it even simpler than a Basic Network scan. A lot of the time I will use the Advanced Network scan just to do a really quick task. The Basic Network scan is a full system scan and is suitable for any hosts. It does take a little bit more time though. I will actually do that one; I'll do the Basic Network scan. I'll give it a name. I'll call it, Training SC Modular Sensor.

**07:00-08:00**
Okay. Since this is a Basic scan, I'll put that as well. Next, here's something that got me. This is an internal scanner. But the scanner by default is auto select. I went here and I saved my scan and started it. It couldn't find my host. It said that my hosts that I was trying to scan weren't on the same network. Then I found out because you do need to select the scanner. Now for the targets, you can use a range of IP addresses with a dash or you can use CIDR notation or even host names.

**08:00-09:00**
I will start off with a simple IP address because I want this to be as quick as possible. I will use 10.20.1.12, which is one of my Windows servers. Next, there's the email recipients for notifications. I'll use my email. Then, when the scan is done, I will get an email letting me know that it's done. If I want to give other people access to use my scan, I can give them **View** access. Now there's also something

else that's really cool: I can create credentials. I'm going to allow my Tenable Nessus scanner to log in to my Windows server to look for vulnerabilities. I'll go over here and type in `Windows`.

**09:00-10:00**
Here this is the host. I'll select **Windows**. I will type in my administrator. I'll put in the administrator's password. I really should create a separate account, maybe as an administrator called Tenable, but for now, for ease of use I'm just going to use my administrator account. The domain is SC labs.ai and I will save this credential. I'll go back to the settings. It looks good. I have my target. I have my Email recipient, and now I can save and launch. This will go from pending and then initializing and a whole bunch of other stages. This will take up to an hour.

**10:00-11:00**
I will leave this for now and then come back later. After a few minutes on cloud.Tenable.com, I see that my scan is running. It's running with one scan target. The other thing I could do either from the Stellar Cyber Open XDR platform, or from a command shell and SSH, I can see that there is a job pending on the Modular Sensor. In this example, I will just use a terminal window and I will open up a connection to my Modular Sensor. There we go. I'll type in my password. Then all I need to do is type in `show tenable`.

**11:00-12:00**
Here I can see, as I showed you before, that the scanner is linked to cloud.Tenable.com using 443. We also see a scanner has one job pending and that's the scan that's running. Great! This is exactly what we're looking for. I will continue to wait for the scan to complete. Okay. Checking back on the Tenable website. I see that my scan is publishing now, so that's great to see. It looks like it'll be a little bit longer. I will check back again. Okay. I did receive an email showing me that the vulnerability scan was completed. I have my report from Tenable. This shows me that there are some critical severity vulnerabilities that I need to fix on my servers, and I will do that.

**12:00-12:47**
That's really cool. I was able to perform the vulnerability scan from my Modular Sensor to servers on the network and then that information gets reported to Tenable. Then Tenable was able to send me a report showing me the results of the vulnerability scan. Then, with a Connector, which is configured in another module, you can actually get the info from the scans into the Stellar Cyber platform. Thank you very much. I hope to see you in another lesson.

This lesson is complete. Thank you for watching.

# L24 - MODULAR SENSOR PLATFORM Connector WITH TENABLE

## L24A-PRES: Modular Sensor Platform Connector with Tenable (30m:26s)

**00:00-01:00**

The next lesson is *Stellar Cyber Platform Connectors Example with Tenable.io (Cloud)*. This lesson, *Stellar Cyber Platform Connectors Example with Tenable.io,* explains Stellar Cyber platform Connectors and provides an example of how to configure a Tenable.io. Connector on the Stellar Cyber platform to collect vulnerability scan data and reports as well as collect assets learned by the Tenable vulnerability scan.

Hi everyone. In this lesson, I will cover Connectors. I will also provide an example of using Stellar Cyber Connectors that connect with the Tenable.io service, which is in the cloud.

**01:00-02:00**

Stellar Cyber, being an Open XDR platform, has hundreds of built-in Connectors for integrating with security, identity, and infrastructure systems and tools. Looking at the picture on the right, you'll see I have the Stellar Cyber platform, and at the bottom of the platform are Connectors. The Connectors use APIs to connect with cloud security services and apps, infrastructure tools, identity tools. As an example here, the APIs can connect to identity-management tools such as identity platforms Azure, OKTA, Google. They can connect to email services such as Microsoft 365 or Google Mail. They can connect to EDR platforms such as SentinelOne, CrowdStrike, BlackBerry Cylance.

**02:00-03:00**

They can connect to vulnerability-scanning services, which is what this example will be, such as with Tenable.io or Qualys and even other tools such as cloud-based, next-generation firewalls or CASB or WAFs. They could also provide response actions to these as well to block IP addresses and much, much more. The Connectors can also be used to connect with, for instance, Salesforce as an app. Let's go a little bit further now. Connectors are an essential component of the Stellar Cyber Open XDR architecture that enable these third-party integrations with APIs as I just mentioned. Connectors run on Modular Sensors or on the Stellar Cyber platform, depending on which has the best reachability to the security, identity, and infrastructure systems and tools in place.

**03:00-04:00**

Connectors to cloud-based systems run on the Stellar Cyber platform, as this provides the most direct reachability to the internet or these cloud systems. Connectors enable automated or manual response actions using third-party integrations. For example, an automated threat hunting, or ATH, playbook can be created to block an IP obtained from an Interflow record on a firewall if an Alert is detected. Another example, when analyzing an Alert, an analyst can trigger a response action to contain an endpoint using an EDR solution. That's pretty cool. They don't actually have to log in to that third-party solution. They can use the APIs right from the Stellar Cyber platform.

**04:00-05:00**
Connectors to on-premises or private cloud systems typically run on Modular Sensors deployed within the same network for direct reachability to these local systems. They can run queries, perform response actions, all directly to the local services or systems. In this example, I have local services, including Active Directory, a firewall, and a vulnerability scanning service as well, which can be deployed internally within an organization. Stellar Cyber online documentation provides details on whether the Connectors run on Modular Sensors or the Stellar Cyber platform. As an example, I will show you how to connect with Tenable.io. We've already had examples on using the vulnerability scanner on the Modular Sensor, which was from Tenable.io.

**05:00-06:00**
Now I'll show you how to build a Connector as well. Therefore, you can see the results of the security scans. Right now if you want to understand what type of Connector and where it goes and how it's installed, you can look at the documentation. You can search for Connector types and functions by going to HTTPS docs dot stellarcyber dot ai. At the very top there's a drop-down box for the specific version. As per the **Collect** column, if we take a look at the columns, there's the **Connector** column on the left, the **Collect** column next then the **Respond** column, **Alert Integration**, **Indices**, **Runs On**, **Interval**, **External Actions**, and **HTTP Proxy supported**.

**06:00-07:00**
Looking at this diagram, I will use Tenable.io Connector to collect data from Tenable.io. If I go to the right, this data will go into the **Scans** indices. It will run on the DP. The DP is another name for the Stellar Cyber platform and the interval for collection is configurable. The default is 60 minutes. If I do happen to have an HTTP proxy in my network to get out to the internet, this is supported when configuring this Connector. The Stellar Cyber online help documentation also provides the steps needed to configure the Tenable.io Connector.

**07:00-08:00**
In this example, a Connector will be created for the Stellar Cyber platform to access Tenable.io via APIs that will collect scanning data that was performed by the Nessus client on the Modular Sensor. Pretty cool. Okay. Again, in this example, the Tenable access is going to be cloud at Tenable.com where I have a valid account that I have paid for. Next, after logging into the Tenable cloud platform, select the profile name in the upper right-hand corner of the page. This will enable you to generate API keys that will enable the Stellar Cyber Tenable Connector to access Tenable using APIs. As an example, from the profile of my account, I can go down on the bottom left and select **API KEYS**. Then I can select **Generate** to generate API keys.

**08:00-09:00**
Note that if I have other API keys in use, once I select **Generate**, those will no longer be available. Please make sure if you do this that you will not hurt any other types of API integrations that you have. If you have your API key stored somewhere else, then you can use those. But if you need to generate new keys, for example, for a brand new account, then select **Generate**. Again the warning will appear. This will replace any existing keys and unauthorize all applications currently using them.

Are you sure you want to continue? Again, just be very careful with this. If you know it's a new system or your new account, because your account will affect your keys. Alright. I'll continue because this is new for my deployment.

**09:00-10:00**
Once this is done, the API keys are generated. Note these keys will not work for you. They've already been erased. You will need both the **Access Key** and the **Secret Key** to paste into the configuration of the Tenable Connector when configuring it on Stellar Cyber. Next, to create a Tenable Connector that will be run on the Stellar Cyber platform, go to **System | Connectors**. On the **Connector Overview** page, which is step number two, select **Create**. You can specify the Connector parameters for Tenable.io. First, Tenable.io is a vulnerability scanner. Next, specify the following Connector parameters for Tenable.io.

**10:00-11:00**
For step number three, the category select **Vulnerability Scanner**. Note that there are other categories as well for the hundreds of Connectors available on Stellar Cyber. Please ensure you select **Vulnerability Scanner**. For step number four, you have the type. Select **Tenable.io**. Note that you'll also see there's a Nessus Connector and there's also a Tenable.SC Connector. Even though this is performing a Nessus scan, the Connector itself is going to cloud.Tenable, which is the Tenable.io Connector. Please make sure you select that. For step number five, ensure **Collect** is selected. Actually, you don't really have a choice; the Tenable.io Connector is collect only.

**11:00-12:00**
You are only provided a single option, which is defaulted to be checked, which is **Collect**. Next, for step number six, specify the name. The name I'm using here is, "SCLABS-Tenable- Connector." Number seven, select the tenant. Again, if I have a customer that has their own tenant, I could have the Connector only be available for that tenant, but since this is a deployment for my corporation, the root tenant is all I need to specify. Next, as the Tenable Connector is connecting to a cloud service, the **Run On** is for the Data Processor. The Data Processor is another word that's used to say essentially the Stellar Cyber platform.

**12:00-13:00**
When you see Data Processor, it is synonymous with this Stellar Cyber platform. For step number nine, select **Next**. For step number 10, please note that you have the API keys on Tenable. Again, as I mentioned before on the Tenable.io site cloud.tenable.com, when I log in I have a profile in the upper right-hand corner. When I select my profile, I can scroll down to API keys and select that. If I generate new keys, those keys are displayed, and you must take those keys and put them in a safe place because once you leave that **Keys** page or the **API Keys** page, for example you log out and log back in again, those keys will not be visible again.

**13:00-14:00**
In that case, you would have to generate new keys if you did not have them saved. Please note though when you generate new keys, those keys do make the other ones no longer function. Therefore, be very careful when you create or generate new keys. Please make sure you store your

keys somewhere safe. In step number 10, we'll copy the access key and place that in the **Access Key** field. It's also very important that you make sure you copy all the numbers. What happened to me one of my copies is, for example, on the number six, on the access key at the start, I actually started from number five. I pasted it and nothing was working, and you're getting frustrated saying why isn't it working? Then you go back and check and you realize, "Oh, I didn't copy the full access key."

**14:00-15:00**
It can definitely happen if you have a smaller screen you're working with as well. Alright. Make sure you have the full key from the start to the end and paste it into **Access Key**. Next, for the **Secret Key**, do the same thing. Copy the secret key from the first number to the last and paste it into **Secret Key**. I'll actually go back a step that's not shown here on the **Interval**, which is below, the **Interval** defaults to 60 minutes. That's how often the Connector will query Tenable.io to find if new scans or new assets are created. That should be fine by default. If you have a test environment and you want to see results happen sooner, you can make that a smaller number like, for example, 10 minutes.

**15:00-16:00**
In my case though, I will leave this as the default of 60 minutes. Next, is the **Asset Synchronization**. When targets are scanned inTenable, assets get created, which are essentially a representation of the system being scanned. That information can be sent to Stellar Cyber. Stellar Cyber can send Tenable information as well about assets. Select the checkbox for **Asset Synchronization** on step 13. For step 14, if you select the field next to **Content Type**, you'll get a selection box. In that selection box, choose **Scans and Vulnerabilities**. Then for step number 15, select **Next**. Alright.

**16:00-17:00**
You are now done with your configuration. For step number 16, select **Submit**. This is important for on-premises or cloud deployments. This does not apply to SaaS deployments of Stellar Cyber. But when prompted by the Stellar Cyber platform to add the Connector to the default Data Analyzer profile, please ensure you select **OK**. Step seventeen, select **OK**. If the Connector is not assigned to a Data Analyzer profile, it will not run. Let me reiterate this again. If you select **Cancel**, the status of the Tenable.io Connector will show a gray circle that will inform you that you need to add the Connector in a Data Analyzer profile.

**17:00-18:00**
You can find the Data Analyzer profile in **System | Data Analyzer Profiles**. Data Analyzer profiles are only used for on-premises or private cloud deployments of Stellar Cyber. For SaaS, Connectors are automatically added to the Stellar Cyber platform, so you will not see this pop up. Just for the on-premises or cloud deployments, optionally, you can review the Connectors assigned to a Data Analyzer profile by going to **System | Data Analyzer Profiles**. Then edit the profile. If you look here under **Added**, you'll see the SCLABS Tenable Connector. You'll also see right above that there's something called **Available**. If you had other Connectors that you didn't add,

**18:00-19:00**
you can select it from the drop down and to the right of the drop down select **Add**. Then when you're done select **Submit**. But if you are now done reviewing this page and no changes are needed, go to

the upper right-hand corner and click the little **x** to close it out. Now back in **System | Connectors**, after you have created the Connector, on the row of the Tenable scanner Connector, select **Test**. This will come back with a green box hopefully, showing that your test was successful. “Test Success.” Yay. Alright. But, if you did have a problem with for example the keys that you pasted in you'll immediately get an error letting you know that the authentication is invalid. For number 19, if no scans have been performed in the past 24 hours, the status of the Connector will show red.

**19:00-20:00**

If you go to cloud.Tenable.com and start a new scan, you have to wait for it to complete to have new data available. Once the data is available on cloud.Tenable.com, you are then waiting for the Connector poll interval. The next time the Connector does a poll, which is every 60 minutes, it will then be able to see this new scan data and pull it in. After the status is healthy or green on the row of the Tenable Connector, select the name of the Modular Sensor to see the configuration settings and detailed status. Next, in **Connector Details** in the **General** section, for step 21, you can view the configuration settings.

**20:00-21:00**

For example, you see the name of the Connector that is able to collect data is enabled, the type of Connector which is Tenable.io, whether it can respond, and we know that Tenable Connector does not have response actions, so that's NA. You see the tenant involved, which is the root tenant in this case, and that you could also see that it runs on the data processor, which is the Stellar Cyber platform, and the category, which is vulnerability scanner. For step 22 the status, you can view the last activity and last data received. This is good to know in case there's for some reason you're not receiving any data, but you could also see that you are getting consistent reports. For step 23 you have **Connector Ingestion Over Time**. This lets you view the ingestion details for security scan data and security scan reports just to make sure that you are receiving them over time.

**21:00-22:00**

For step 24 you have **Connector Documents** with our Interflow records. For step 25 you can select **View Data** on the upper right to view the Interflow records with the Connector name and the sensor monitoring index selected. After selecting **View Data** from the Connector **Details** page, a filter will automatically be applied to view Interflow records with statistics about the Connector in the sensor monitoring index. Once you have selected **View Data**, you're taken to the **Investigate | Threat Hunting** page. You'll see here that there is a search on Step 26 that says, “Searching for Connector name keyword, SCLABS Tenable Connector AND tenantid,”

**22:00-23:00**

which is the root tenant in this case. On step 27, you'll see that the indices is set to **Sensor Monitoring**. In step 28, you will see the Interflow are records related to the search and the indices selected. This is just letting you know that the connections are healthy because you are receiving Connector-related sensor monitoring Interflow records. For step 29, you go back to **System | Connectors**. With the Tenable Connector status, green or healthy, on the row of the Tenable Connector, select **View Events**, which is step 30. For step 31, after selecting **View Events**, a filter will be automatically applied.

**23:00-24:00**
You can then select **x** to clear that filter if you want to, but if you want to use this automatically create a filter, it will get you the data you are actually looking for. In step 32, the indices will be set to the **Scans** index. In step 33, you will see the Interflow records from the Tenable Nessus scan that have CVEs, or common vulnerabilities and exposures. This is the meat of it. When you are using the Tenable Connector, this is the data you want to see. Also, by using the Tenable Connector, you can view the assets from the bidirectional sharing with the Tenable Connector from the Tenable scans. When the Tenable scans different devices, that information gets reported back to the Stellar Cyber platform via the Connector.

**24:00-25:00**
Now you can view the assets in **Investigate** by selecting **Investigate**, then select **Asset Analytics**. On step three you can select **MAC Identified Assets**, and a MAC address is the layer two address for a device. An IP address is layer three. You can select **Filters** to view the filter sidebar. Step number five, you can type in `tenableio` as one word, and that will just help you see the assets that have also been identified by Tenable. For step number six, if you select the name of an asset, which is a host name,

**25:00-26:00**
in the **FRIENDLY NAME** column you get many more details about that asset. You'll see at step number seven there's an **Asset Details** tab. Next to that is the risk score. If it's a high risk asset, that number will be higher. You can see the detected name of the host. You can see the reputation. For example, if it's good or malicious or scanner the importance, the tenant that it's in, its location, if there are any tags associated with it again tags are created on the Stellar Cyber platform to help you identify and locate either Interflow records or assets themselves. You can see the vendor. In this case, the vendor is VMware because it's actually a virtual machine running windows

**26:00-27:00**
and the last time that it reported some information so the last scene. Lots of really good information about assets themselves. You also see the data source. The data sources in this case, where a Tenable.io recognized this asset. The Modular Sensor itself, uh found this asset. We have a Sophos firewall that found it and Active Directory as well. You can view the summary of Interflow records at step number nine and see for example, the risk, the Alerts, the vulnerabilities associated with that asset. In this case I will select the **Vulnerabilities** tab because I have seven of them and I can look at the CVEs or any other detailed information in the Interflow records for those vulnerabilities.

**27:00-28:00**
The Interflow records displayed are related only to that selected asset. That's really cool. Next, you can also view vulnerabilities in the threat hunting library by going to **Investigate | Threat Hunting** in step number one. For step number two, you can select **Threat Hunting Library**. These are pre-canned, essentially dashboards that give you lots of really cool information that are organized based on for example the vulnerabilities themselves or vulnerability scan or if you look at the top here there's AWS CloudTrail, or cloud email providers. It will give you charts related to those specific services or tools.

**28:00-29:00**
So on step number three, if I go down to **Vulnerabilities** I'll see that I have 1.6k vulnerabilities from the scanning itself. I have useful charts that show vulnerabilities over time, vulnerability scans over time, vulnerabilities by plug in families, vulnerabilities by risk score, vulnerabilities by top assets, vulnerabilities by top CVEs, vulnerabilities as Interflow records, lots of really cool information, just available in the charts. Now there are a couple of things that I think are good troubleshooting tips. One is if you look at the symptom, you get it during your test, you get a connection error unauthorized.

**29:00-29:13**
Please check auth. Well, as I mentioned before, it is very important to copy the exact access and security keys when creating the Connector. This error usually happens when you have the wrong key or you could have copied from the wrong key on the Tenable.io platform as well or maybe someone selected generate keys after you copied your key, which would then invalidate the keys that you have. Next, if the Tenable Connector shows a red status, this means–or at least in this case–you want to look at the message and what the message says right next to it. The message right next to this one is saying that there's been no data received in the last 24 hours.

**30:00-30:35**
What you could do is do a vulnerability scan on cloud.tenable.com, wait for the Tenable scan to complete, and then wait for the interval. It might be a 60-minute interval you have when you created the Connector, before it does the next pull to check if there is data. That's usually the case here. I hope you enjoyed the lesson. Thank you.

This lesson is complete. Thank you for watching.

## L24B-DEMO: Stellar Cyber Platform Connectors Example with Tenable.io (17m:45s)

**00:00-01:00**
This lesson continues with demonstrations for *Stellar Cyber Platform Connectors Example with Tenable.io (Cloud)*. This demonstration explains Stellar Cyber platform Connectors and provides an example of how to configure a Tenable.io Connector on the Stellar Cyber platform and on Tenable.io to collect vulnerability scan data and reports as well as collect assets learned by the Tenable vulnerability.

Hi. Great to have you back. Now on the previous module, I showed you how to make the Modular Sensor a Tenable vulnerability scanner. Now this is only visible though from Tenable.io or cloud.tenable.com.

**01:00-02:00**

If you want to view the scan, all the results are sent there. However, there's lots of good data that the scans will provide. For example, all the assets that are discovered during the scans. One of the things that we can do to grab that information is to configure a Connector. For this lesson, I will show you how to create a Connector to cloud.tenable.com. As I've shown you before, the best way to get started with the Connector is to go to the documentation. I can go here or go to docs.Stellar Cyber. I will go to the Knowledge Base home and search for a Tenable Connector. The Connector that we are going to be using is the Tenable.io Connector.

**02:00-03:00**
Again, when you go to a help page, one thing if you are searching for a word, it will highlight the word. I like to remove the highlighting. Here I can see the capabilities. With the Tenable.io Connector I can collect information. There is no response because there's nothing I need to do to the Tenable.io. There are no native Alerts mapped to what we collect. We're really going to collect assets. It says here it runs on the Data Processor. In other words, it runs on the Stellar Cyber platform. We configured an Active Directory Connector previously that was run on the Modular Sensor.

**03:00-04:00**
But here, since Tenable.io is in the cloud, then the Data Processor or the Stellar Cyber platform has better access to cloud.tenable.com. We will configure it on the platform itself. The interval is configurable. The default interval for the Tenable Connector is 60 minutes. We might change that for our testing. Make it smaller. The information will be in the scan's index. It's also showing me that the message class is security scan. When I do a search I can search for MSG_class:security scan or security scan reports. I can look for message_ origin.source of `tenableio` or `tenableio_reports`. There's also message_origin.vendor. All this will be very handy to have. Okay?

**04:00-05:00**
Some other things that are required are the Access key and Security key. I will show you how to get that information from my cloud.tenable.com account. Now I will make sure to change out those keys after I use them so they won't be valid for anyone else. First, I will start from the **System**. To create a Connector to Tenable.io, I will go to **System**, then **Connectors**. Here I have an Active Directory Connector. I will now create the Tenable Connector. Here the category I should have vulnerability scan. Here it is: **Vulnerability Scanner**.

**05:00-06:00**
I will select the **Tenable.io** Connector. There's also a Nessus scanner, but that's not what we're configuring here. We're configuring the Connector to Tenable.io. Next, the **Collect** option is the only option available to Tenable.io. Again, there is no response action for Tenable. We saw that in the documentation as well. Next, I will give a name. I will call this the "SC (for Stellar Cyber) Training Tenable.io Connector." Okay. This will be in my root tenant and it will run on the Data Processor. Again, the Data Processor is another name for the Stellar Cyber Open XDR platform.

**06:00-07:00**
Next, if there were some specific fields or any other types of logs I did not want to gather, I could make a filter here. But for now I will leave it as default to have nothing filtered. Go ahead and select

**Next**. Then here is where I need an Access Key and Security Key. Let me go to Tenable. Here I can see, by the way, that my last scan did complete. If I go to **Settings**. Here on Tenable I can go to my account profile. It's on the upper right-hand corner and view my profile.

**07:00-08:00**
Then in here, one of the options is **API Keys**. Now this is incredibly important right now. When you are configuring API keys, you select **Generate** down here. But if I'm using API keys already, and then I say **Generate** now, notice what happens. There's a warning message. This will replace any existing keys and unauthorize all applications currently using them. Are you sure you want to continue? In this environment, I have not used any API keys for anything else. So it's okay for me to continue. However, in your environment, you do have to make sure that you will not hurt anyone else. Right? You have to make sure that you don't disable someone else's service.

**08:00-09:00**
Again, my account is for me only so I'm not concerned. I do not have keys in use. I will go ahead and select **Generate**. **Continue** and then I have my access keys. I will make these available for my test. Then of course I will change out these keys by selecting **Generate** again as soon as this lab is over so that these keys will no longer function for anyone else. Sorry, but this is training material, and I do not want anyone else to access my Tenable. Here we go. Next, I will select my access key and make sure to grab the entire number. Okay. I've done this before where I haven't and that's also caused problems. I'll go back over here and paste in my access key.

**09:00-10:00**
Next, I will go back to Tenable and grab my Secret key. In the Secret key–oops, here we go–I will paste my Secret key here and set the interval to a smaller period of time. In my environment, this is a test environment, I'm going to switch it from a 60-minute polling interval. I will change this from a 60-minute interval to a 10-minute interval. Next, for **Asset Synchronization**, I will synchronize the assets collected from Stellar Cyber to Tenable.io and also from Tenable IO to Stellar Cyber. Next, there's the content. I want to grab the scans and the vulnerabilities. Okay. Now I can select **Next**.

**10:00-11:00**
The contactor configuration is complete. Let's see if this worked. Go ahead and **Submit**. Here's something that you see only on on-premises or private cloud deployments. You'll see that The Connector has been added successfully. Do you want to add it to the default data analyzer profile? The Data Analyzer is the Stellar Cyber platform. The answer to this question is yes. Yes, you do want to add it. If you are on a SaaS deployment, then you will not be asked this question. It will automatically be deployed to SaaS. Now that the configuration is complete, I see the status is not available. But I can select **Test**. Here I have, "Test Success!"

**11:00-12:00**
Again, if my key was wrong, I would not have, "Test Success." You will see here though that there's no data received in the last 24 hours. I can take a look at **Events**. There are no events at this time. Now I will then go to Tenable and start a new scan. Now from Tenable.io, I'll go ahead and create a new scan or actually just to keep it simple. I will modify the scan I already had and I will just change the

target host to a different host. Save and launch. Then I will get another scan. Again, this will take another 30 minutes to an hour.

**12:00-13:00**
If I go back over here, I can see if any other previous scans have been collected. It looks like there has. If I go back to the Tenable.io Connector, I see now the status is healthy. Now I can view the events related to the Tenable.io Connector. By selecting **View Events**, what it's actually taking me to is **Investigate | Threat Hunting**, and it adds a search automatically here. This is looking for the Tenable Connector and every event here is displayed from the Tenable Connector. If I look at one of these events here, I can see that they are just events. They are not Alerts. Alright? Notice there's no Alert score. There's no MITRE information here as well. Alright? But I do know I am receiving this traffic.

**13:00-14:00**
One of the other options as well if I go to **investigate** and **Asset Analytics** and search for **MAC Identified Assets** I can now look for any asset that has been discovered from Tenable.io. If I go to **Filters** and look at the data source, I can type Tenable. I think it's Tenable.io. There we go. All these host names are discovered from Tenable.io. If I scroll down on this list for example, this Windows server, I have discovered it already with my Active Directory Connector.

**14:00-15:00**
It's actually been discovered by my Sophos Connector. It's been discovered from the Modular Sensor by viewing the traffic, but it's also been discovered by Tenable.io. Each data source can give additional information about the asset. If I go down to here for example, I can take a look at Vulnerabilities. This would be from the vulnerability scan. If I look over here on the **Vulnerabilities** tab, here I see that I have some vulnerabilities. There's a CVE right here, which is Common Vulnerabilities and Exploits. If I select the Interflow record by selecting **More Info**, I can see what was reported by the Tenable.io Connector. If I go to **Details** and scroll down, I can see that the vulnerability output was that there's a medium-strength cipher.

**15:00-16:00**
This just gives you an idea of what each of the line items in the report has and it applies it to this asset. I can use that in my threat investigations. Really, really cool integration here with Tenable.io. The next thing I can do is go to **Investigate | Threat Hunting**. I can also look for Alerts. Before we looked for a message origin source, if you remember in the documentation there were some fields here: message origin source Tenable.io and Tenable.io Reports. Let me see what happens if I go to **Investigate | Threat Hunting**.

**16:00-17:00**
I'll close the filter I'll type in: `msg_origin.source:tenable.io`. Let's make sure that's correct. Yep, that is correct. I can also do underscore reports. Here, oh my god I have nothing. Why? The reason is because there is no Alert. However, if I go to **Scans**, now my message origin source shows me the Tenable. There are 35 Interflow records based on the message origin source of Tenable.io_reports. Now let's see what happens if I say just Tenable.io.

**17:00-17:45**
Now I have a lot more. Okay. If I look at these, again these will show me the CVEs. This is the output. I can see you know the vulnerabilities that have been found. Okay. If I look at the **Details** or the JSON, I can see the information about the source and the scan that was performed and see the CVEs related. Right? There's the CVE ID. I can also see it right from here as well. Okay. That's the integration with Tenable.io. That's pretty cool. Thank you for watching.

This lesson is complete. Thank you for watching.

# L25 - MODULAR SENSOR ACTIVE DIRECTORY (AD) Connector

## L25A-PRES: Modular Sensor Active Directory (AD) Connector (13m:22s)

**00:00-01:00**
The next lesson is *Active Directory Connector Details and Configuration.* This lesson, *Active Directory Connector Details and Configuration*, provides details and explanations of how to configure an Active Directory Connector on Modular Sensors to collect Windows event logs, user behavior and computer asset information, as well as perform response actions from the Stellar Cyber platform to disable users in Active Directory.

Hi everyone. Great to see you again. In this lesson, I will cover Modular Sensor, Active Directory Connector details, and configuration.

**01:00-02:00**
Stellar Cyber Active Directory Connectors run on Modular Sensors for Stellar Cyber to ingest data from Active Directory and add as Interflow records to store in the Stellar Cyber platform. Because Modular Sensors are deployed within an organization's network, the Modular Sensors have access using LDAP to the Active Directory servers. They can perform LDAP search to obtain information about users and computers within the organization. They can also use LDAP to set a parameter, for example, to disable a user with an Active Directory. This could be done from inside the platform and we'll discuss more about that later.

**02:00-03:00**
Stellar Cyber Active Directory Connectors provide key capabilities such as enriching event Interflow records with user and computer information from Active Directory and supporting Windows Event Collector. Now, I want to make sure that we discuss some prerequisites. Before getting started with configuring an Active Directory Connector that can collect data and perform response actions to disable and enable users in Active Directory, it is recommended you configure a dedicated user or group with specific permissions for the Connector. In this example, an account named stellarconnector will be used that is a domain user and a domain admin. Now for a very important note: To collect data using the Active Directory Connector, only domain user privileges are needed.

**03:00-04:00**
To provide response actions to disable the user, elevated permissions are needed, and that is why the user in this example has been assigned to the administrator group. This level of privilege is not required. But it is easy for lab environments. Please check with your Active Directory administrator to use the correct permissions needed for disabling and enabling user accounts in your Active Directory environment. If we take a look at a user in Active Directory, I have an account named stellarconnector at sclabs.ai. If I look at that picture right below that, that's the member of. Here you can see the user is a member of administrators and a member of domain users.

**04:00-05:00**
What I could have done instead of assigning the user to the administrators group is to delegate the permissions. One of the options in Active Directory is to select an organizational unit such in my case as sclabs.ai or the user's container. I can say delegate the user account control permissions. There's read user account control, and there's write user account control permissions. I can use those instead. Our documentation explains how to do that in the Active Directory Connectors documentation. To create an Active Directory Connector that will run on a Modular Sensor, in the Stellar Cyber platform, perform the following steps. Go to the **System** menu on the top and select **Connectors**.

**05:00-06:00**
On the **Connector Overview** page, select **Create**. Specify the following Connector parameters. For step number three, select **IdP** for identity provider. That will narrow the types you see on the next line. For the **Type**, select **Active Directory**. For the **Function** in this example, I will use **Collect** and **Respond**. Because there are different permissions that are required for **Collect** than there are for **Respond**, some organizations create a separate Connector for collect and a separate Connector for respond. This way, the elevated permissions are only used on the respond Connector that's used to disable users.

**06:00-07:00**
In this example, I will select both. Next, specify a name for the AD Connector. I like to use the host's name within the name of the Connector, mainly because some organizations have hundreds of Connectors because they have hundreds of different Active Directory servers at different sites. By putting the host name in the name I can clearly identify which Connector this is. Also, in this example, I have the **Tenant Name** for Root. I don't have any other customers and I don't have any other tenants, so everything will go into the root tenant. But if I did have other tenants, I could specify a Connector within a tenant.

**07:00-08:00**
Next, I'm going to run the AD Connector on a Modular Sensor. This is also where the naming of the Modular Sensor makes a big difference. If you have lots of different sites, please make sure that the name of your Modular Sensor represents the site or even the tenant that the Modular Sensor belongs in. Otherwise, you could possibly choose the wrong Modular Sensor. In my case, I have one so it's pretty easy. Select **Next**. Next, for step number 10, enter the **Active Directory Domain**. The domain

name is not case sensitive. Then enter the **Server Hostname**, which is the Active Directory server fully qualified domain name, or FQDN. You can also use an IP address.

**08:00-09:00**

This **Server Hostname** is also not case sensitive. In this example I've used both upper case and lower case to show you that it doesn't make a difference. Also in Windows, often you will see the domain names in uppercase. Next, for step number 12, the **Interval** by default is five minutes. I will leave it at that. This is how often the Active Directory Connector on the Modular Sensor obtains event logs and user information and computer information from the Active Directory server. For step number 13, enter the **User Name**. If performing response actions to disable a user, this user must have permissions for read and write user access control.

**09:00-10:00**

Those permissions can be delegated from within Active Directory. For ease of testing in a test environment, you can use a domain admin group. That's what I've used for the stellarconnector here. For step 14, enter the **Password** that is assigned to the user name that was entered in the step before. For step 15, enter the **Content Type**. By default, the Active Directory Connector collects user behavior information, and it will always do that. This option for **Content Type** here is to collect computer information as well that will be viewable in **Assets** in the Stellar Cyber platform. The Active Directory Connector configuration is complete.

**10:00-11:00**

For step number 16, select **Submit**. For step 17, on the row of the Active Directory Connector, select **Test.** Within a minute you should receive a green box with, "Test Success!" Then the status should turn green within five minutes as that is my polling interval. For step 20, after the status turns green, select the name of the Connector to get details and status information. From the **Connector Details**, you can verify the Connector is functioning and ingesting data.

**11:00-12:00**

Step 21, you can view the configuration settings. Here, I see the name and type and tenant category and that it is able to collect data from Active Directory and respond within Active Directory. It's run on the name of the sc-modular-sensor. That is good general configuration settings there. For step 22, there's the status. You can view the last activity and last data received. Here you know that the Connector is functioning, especially if the time is recent. For step 23, **Configuration**, you can view the configuration details. Here I see the Active Directory domain.

**12:00-13:21**

I see the **Server Hostname**, the **Interval** I configured, the **User Name** that was assigned to this Connector. I can see the **Protocol Type** that I assigned. In this case, it was LDAP. Also I have the **Content Type**, which is obtaining information from computers. But it also is grabbing user information that is happening by default. **Content Type** really should be users and computers, but users is by default, so it doesn't show up here. For step 24, there's **Ingestion**. You can see if you are getting ingestion data. That means the Connector is definitely working and healthy. You also have **Connector Documents**. This shows that data is being collected from Active Directory for computers

and users. This is really just the Connector communication logs and it's not related to actual events or Alerts. With all the data we see here, we know the Connector is successfully collecting data.

This lesson is complete. Thank you for watching.

# L26 - AD Connector COLLECTION OF ASSET & USER DETAILS

## L26A-PRES: AD Connector Collection of Asset & User Details (09m:19s)

**00:00-01:00**
The next lesson is *Active Directory Connector Collection of Asset and User Details.* This lesson, *Active Directory Connector Collection of Asset and User Details,* explains how to view the asset and user behavior information collected by the Active Directory Connector in the Stellar Cyber Open XDR platform.

Hi everyone. In this lesson, I will show you the assets and computers that are collected by the Active Directory Connector from Active Directory. Now that we've actually configured the Active Directory Connector appropriately, now we can check by going to **Investigate** then go to **Asset Analytics**.

**01:00-02:00**
From here, you can search for the assets collected from Active Directory. To do that for step number three, select **Filters**. That will expand the **Filters** bar. Then select **Data Source**, at which point you can enter active_directory. Here, the asset list is filtered to show only assets learned from Active Directory. Looking at the list I have here, under **Friendly Name**, those are my Active Directory servers that were learned from the Active Directory Connector. For step number five, you can select the hostname of an asset from the **Friendly Name** column to view the **Asset Details**.

**02:00-03:00**
For step number six, you can see the **Asset Details**, which contains the risk score. In this case it's 16.7, so this server is not a high risk. You can see the name, which is the hostname of the Active Directory server. I can see the **Tenant**. If I had multiple tenants, this would be very helpful because now I know which tenant it belongs to. I can see its **Location** and even the location on the map along with any **Tags** that might be assigned to it so I can find it quicker in searches. I can see the **Vendor** and lots of other good information including the operating system. Going a little bit further down I can view the **Data Sources**. Here I see that Active Directory is one of the data sources.

**03:00-04:00**
You could have multiple data sources reporting information for each asset. Here I actually have Tenable.io reporting information, the Modular Sensor itself from network traffic, and Sophos, which is a firewall. That's pretty cool. For step number eight, you can view a summary of the Interflow records for risk Alerts and vulnerabilities. As an example, I can see the latest risk score. See the Alerts over the past month, the Alerts, the vulnerabilities, commands that have been detected, logins that have been

detected, login failures, and files. On step number nine, I can scroll through the tabs from left to right to view the related Interflow records. For example, I have the **Alerts** tab selected.

**04:00-05:00**

Here, in the column, I see all the CVEs, which are the Common Vulnerabilities and Exposures. Each of those can be looked up on the internet to determine what they are or you go into **More Info** and the **Actions** column to view the detailed information. For step number 10, the Interflow records displayed are related to the selected asset only, so that's really cool. If I just want to see if this server has been having any threats involved, you can go here to view those threats. Similar to how the Active Directory Connector uses LDAP to collect information about assets, it also uses LDAP to search and obtain user information from Active Directory. To view the user information, you can go to **Investigate | User Behavior Analytics**.

**05:00-06:00**

For step number two, select the **Users** tab. For step number three, when a user and associated data is collected from the Active Directory Connector, the **Data Sources** column shows Active Directory. Note just like with assets, there might be multiple data sources reporting information about users. If you look at the last one, which is the administrator, both the Windows Server sensor, which is called Windows agent here and the Active Directory Connector are reporting information about administrator. For step number four, select the name of a user to get details collected by the Active Directory Connector from Active Directory that will be stored in the platform.

**06:00-07:00**

For step number five, **User Details** contains the name, job title, principal name, privilege level, last login, last password set, and much more. This information is very useful when you are running through the events and Alerts on the Stellar Cyber platform. Having access to the user information makes it much easier to identify who is involved and what is involved in identifying threats. For step number six, the **Data Source** is Active Directory. For step number seven, you can select **Show More** at the very bottom to get much more data about the user.

**07:00-08:00**

After selecting **Show More** on the previous step, you can see much more detailed information including the **Profile**, **Bad Password Count**, **CN**, the **Member Of** groups. All of this can help determine the level of risk especially when able to view permissions assigned to the user when identifying risks and threats. For step number nine, you can view a summary of the Interflow records for risks, Alerts, vulnerabilities, and more. Looking at the charts, I see the latest risk score. I see the Alerts associated with this user, commands that the user might have typed, the authentications, the login failures and files. If I go to step number 10, there's a set of tabs there as well.

**08:00-09:19**

I can scroll through them to view the Alerts, related assets, commands, authentications, login failures, and files. Step number 11 one of the ones I find really cool is the **Related Assets** tab. You can view the other servers that this user has been associated with and has logged into. If I go to step 12, if I select the three stacked dots next to an asset, I can get even a lot more information about the assets that

the user has logged into. For example, I can get the **Asset Details** view for that asset, I can do a VirusTotal lookup to see if there's anything bad about that asset that has been detected by other sources. I could even copy as a query so that I don't have to write down that host name, and I can use that copy as query to paste into the search bar, maybe to look for that and other items as well to help me investigate a threat.

This lesson is complete. Thank you for watching.

# L27 - AD Connector RESPONSE ACTIONS TO DISABLE A USER

## L27A-PRES: AD Connector Response Actions to Disable a User (17m:42s)

**00:00-01:00**
The next lesson is *Modular Sensor Active Directory Connector Response Actions to Disable a User.*
The next lesson is *Modular Sensor Active Directory Connector Response Actions to Disable a User*.

Hi everyone. In this lesson, I will provide an example of how to configure and use an Active Directory Connector with response actions. With the Stellar Cyber Active Directory Connector on the Modular Sensor, a response action can be performed automatically using automated threat hunting playbooks, or ATH playbooks, or manually by an analyst on the Stellar Cyber platform viewing an Interflow record.

**01:00-02:00**
When an analyst is alerted to a threat, an action can be taken from the Interflow record to disable a user in Active Directory. Viewing the Interflow record on the platform there will be an option to take actions. From there, you specify the Connector, which is on the Modular Sensor. Then you can select **Submit** to send an LDAP-modified query to change the user to disabled in Active Directory. That's pretty cool. I didn't actually have to call an administrator of Active Directory to do that task. But you can also specify duration. You don't have to disable the user permanently. You can say, "Hey look while I'm investigating, I think there's a significant possibility the user has a problem.

**02:00-03:00**
Therefore, I will disable that user for a certain amount of time." I could say ten minutes or an hour, just to make sure that the user will not be compromised any further or compromise any other machines any further. Alright. In this example, I'm showing a duration of five minutes, but that's just for a test. Response actions to disable a user with the Active Directory Connector could be taken from several different places in the Stellar Cyber Open XDR platform. I will give you a quick overview of each of the locations. For number one, you can go to **Investigate | User Behavior Analytics**, and select the **Users** tab. From the **Users** tab, there's an option in the **Actions** column to disable a user.

**03:00-04:00**

For step number two, for analysts that are investigating Interflow records, you can select **More Info** for Interflow records to get to select an **Actions** tab. On the **Actions** tab there's an option to disable a user there if the Interflow record has users within the record. For step number three, another location is from the **Respond | Actions | User Actions** menu. Here you can disable a user right from this screen. On step number four, if you are using Case management to analyze a Case, which is a collection of Alerts, there is a response section there where you can again disable a user if there are users involved in the Case.

**04:00-05:00**
For step number five, there's also the **Respond | Automation**, which is the ATH Playbooks inside an ATH playbook based on the condition, you can have it automatically trigger an action to disable a user. Or you create the ATH playbook, and then that playbook can manually be selected at any time as well. To perform a simple test to disable a user from the Stellar Cyber platform, using the Active Directory Connector on a Modular Sensor, go to **Investigate | User Behavior Analytics**, then select the **Users** tab. This is really important. Next to a user, do not disable the Active Directory Connector user.

**05:00-06:00**
In this case, mine is called stellarconnector. You definitely don't want to disable the stellarconnector because that is the user that is actually performing the response actions. Also, what's very important is you probably don't want to disable the administrator account unless of course that administrator is compromised, but especially if you're running a test, please just make sure to disable a test user. For step number four, you can select the Active Directory Connector. For step number five, for the **Duration** you can select **Forever**, but especially if it's a test environment, you can select **Limited**. At that point you can specify either the days, hours, or minutes.

**06:00-07:00**
After the duration is complete, the user will be re-enabled by the Stellar Cyber Connector. For step number seven, verify that the name is correct. That was the name that you clicked on to disable the user in Active Directory. It should be correct because that's what you selected. But just to make sure. Then for step number eight, select **Submit**. You should get a message that says the disabled user was created successfully. It just means that the disabled user action was created successfully. It does not mean that the user was disabled yet. To verify that, you go to the **Respond | Actions** menu.

**07:00-08:00**
From the top menu bar, select **Respond**, then **Actions**. For step number 10, you can select the **User Actions** tab. It defaults to the 100 most-recent actions, but you can go to the **User Actions** tab. From the **User Actions** tab, you will see that a user has been disabled or is being disabled. But you might look at it and go wait a minute. Like in my example, this time it's S-1-5-21- all the way ending in -1117. Well, depending on how the user was found, especially from Active Directory, you might just see the SID value when you are in the user behavior analytics, if you do select a user, one of the options is a **Real ID** tab. In the **Real ID** tab, you can see the SID.

**08:00-09:00**

In this case you'll notice that 117 was the user that was successfully disabled. All good here. If you need to undo the disable action, by the way, you do have a safety net. You can select **Revert**. If you disable the Active Directory Connector username, then you will not be able to revert. Next, when viewing Alerts that contain usernames from Active Directory, the Active Directory Connector on a Modular Sensor could be used to disable users from the Interflow record without having to log in to Active Directory. That's pretty cool. If you go to **Investigate** on step number one, **Threat Hunting**, you can verify the indices here. I have **Alerts** selected. You could also select **Windows Events**,

**09:00-10:00**
especially if you don't see any Alerts yet. When viewing Cases and Alerts, SOC analysts, based on the type of Alert, can disable users while they are investigating a threat. As the goal for this lesson to understand how to disable a user, search for Alerts with users to test the ability to disable users. In other words, if you just start going through the platform and selecting Alerts at random, you may not find an Alert that you can disable a user from. As step number three shows in the search, type the following: `smb_username:*`. That says look for any Alert with the SMB_field.

**10:00-11:00**
An SMB is a Windows username. There are other fields as well. Optionally, as step number four shows, there's the Interflow dictionary. If you are not sure what other usernames are involved in the Interflow records that you have collected, you can use the Interflow dictionary and use command f, which is a browser search or control f on Windows, command f on on Mac, to find all fields related to a certain word like username. You can type in `command f username`, and then you can scroll through looking for every single field name that has a username value in it. You can select that to see if you have any usernames collected and you can see the values of the usernames and then filter the records right from there.

**11:00-12:00**
The **Interflow Dictionary** is cool. I will provide a demonstration of that capability, but for now, SMB username works in my environment. For step number four, now that I've searched for SMB username:*, I will select one of the Interflow records that shows up in the list of documents that you'll see if you scroll down essentially on the **Investigate | Threat Hunting** page. For step number five, within the Interflow record, you can go to **Details**. You can now look for the SMB username to scroll down to find it. They are in alphabetical order. If you are using another name or you're not sure, select a record.

**12:00-13:00**
You can scroll down and look for anything that looks like a user and then see if you can use that field name instead. You need to know the field name for disabling a user. In this case SMB username will work fine. But you might also have something like desktop username, source IP username, target username so there there's other username fields that may appear in Interflow records related to Active Directory. For step number seven, select **Actions**. In the actions for the Alert, one of the options, again since the username is in the Alert and I have an Active Directory Connector, I do have the ability to disable the user associated with this Alert. I have other options as well, such as adding a filter or a tag or running a script.

**13:00-14:00**
I could even contain an endpoint depending on if I have an EDR associated with it. But for now, **Disable User** is what we are trying to show. Note, if you do not have **Disable User** as an option that means there are no username fields in the Alert Interflow record. You'll have to find another Alert with a username. In the **Disable User** configuration box, one is you can select the Connector. Now again if you have lots of Connectors, this is why it's important to make sure that you name your Connectors specifically related to the host they're going to connect to. In my case, I know exactly that this is my SCLABS-WINAD-San Jose training DC1.

**14:00-15:00**
It's a domain controller and it's the first one. The duration I'll set a time for will be five hours. Maybe I'm going to be investigating this Alert. I've already called the user to let them know that I'm disabling their account. It depends on what type of Alert it is. Now for the disable user, if you select the box next to **Disable User**, you can start typing in a username field. I know my username field is called SMB_username. I start to type `sm`. As soon as I do that I start seeing the list of entries that match. I can select SMB username in this case, and then select off the box. That will then show you that you have that SMB username selected.

**15:00-16:00**
Next, I can select **Submit**. After you select **Submit**, the green success box is displayed with a note to check the status in the **Respond | Actions** table. Again, the user is not disabled yet at this time. By viewing this table we can see the status. From the **Respond | Actions** menu, I see the table here. I can go to the **User Actions** tab. I see the username. It's the target user, which is the SMB username. I see that the disable user action succeeded. You can see the time and you can see where the Connector was run, which in this case, is the SC-Modular Sensor.

**16:00-17:00**
Alright. If you get here a little early, it might take about a minute for it to complete. You will see the progress in the **Status** column as you're waiting for it to succeed. In this case, the respond action for disabled user was successful, so I'm pretty happy about that. If I wasn't happy though, for example, I go, "Shoot, I should not have disabled that user." That's my boss and my boss was doing something very important. You can revert. Right? You do have a safety net that you can use. If you are an AD administrator, you can go to Active Directory and take a look at the user itself.

**17:00-17:42**
In my environment, I went to **Users**, and in Active Directory you'll see that Paul Levasseur has a little down arrow next to the name in the user's icon. That lets you know that that user is disabled. If you want to enable the user from Active Directory, I can right-click on the username and re-enable that user. Yes, this user was disabled successfully.

This lesson is complete. Thank you for watching.

## L27B-DEMO: AD Connector Response Actions to Disable a User (20m:05s)

**00:00-01:00**

This lesson continues with demonstrations for *Modular Sensor Active Directory Connector Configuration and Use*. This demonstration provides step-by-step details for configuring an Active Directory Connector on a Modular Sensor, setting the proper permissions in Active Directory for data collection and response actions using the Active Directory Connector and provides examples of viewing computer and user data, as well as disabling a user from an Interflow record in the Stellar Cyber Open XDR platform.

Hi everyone. In this lesson, I will show you how to configure an Active Directory Connector that will run on a Modular Sensor. With the Active Directory Connector, you will be able to collect assets which are computer information,

**01:00-02:00**

and you'll be able to gather user behavior analytics information, which are from the user accounts in Active Directory. Before getting started creating a Connector, what I like to do first is go to the documentation. I can search for Connector types, and functions. The first entry that appears is *Connector Types and Functions*. From the *Connector Types and Functions* documentation, I can see what the Connectors are capable of doing. If I scroll down to all Connectors, I can look at the first column is **Connector** that gives the Connector name.

**02:00-03:00**

The **Collect** column specifies that the Connector is able to obtain information from the system it works with. The **Respond** column means that the Connector can actually perform some type of response, such as disabling a user or containing an endpoint. **Alert Integration** lets me know if there's special machine learning integration or some rules that are in the Stellar Cyber platform that apply to this Connector. The **Indices** column specifies the indices the data collected goes into. **Runs On** means that the Connector can run on a Data Processor, which is another name for the Stellar Cyber platform, or a sensor, such as the Modular Sensor. The **Interval** is how often Connector pulls the system it is working with. These are the main settings that we are going to delve into right now.

**03:00-04:00**

If I scroll down a little bit, I'll find the **IdP** section. I see Active Directory. Again, IdP is identity provider. This category name will also be used in the configuration of the Connector. This shows the **Collect** column that the Active Directory can collect data from Active Directory. The **Respond** column shows that the Active Directory Connector can perform response actions, in this case to disable a user. Windows is the index or indices where the data collected by the Connector will go. The Active Directory Connector will run on a DP or sensor. The **Time Interval** is configurable, and the response or external action is the ability to disable a user.

**04:00-05:00**

Now from the **System | Connectors** page, I will create an Active Directory Connector by selecting **Create**. The category is IdP. That's what we read in the documentation. For the **Type**, we're going to choose Active Directory. Note you can create Connectors for other identity platforms as well. The **Function** will be for collecting and responding, so collect and respond. **Collect** means I'm obtaining user and computer and event logs. **Respond** as I am disabling a user. For the **Name** I like to use the hostname of my Windows AD server in the name of the Connector. Because I could have hundreds of Connectors in my organization. I might have hundreds of servers in my organization.

**05:00-06:00**
What I want to do is go to my Active Directory and make sure I'm using the right hostname. One of the easiest ways is just from a command prompt and Active Directory. I will obtain the hostname here and I'll go back. I will call this, "SC," for Stellar Cyber and I will paste the hostname, and then use the word, "Connector." I will install this in my root tenant. If I had other tenants with the Active Directory server, I would create it specifically for those tenants. I'm going to run it on this Modular Sensor. Even though I have log filters configured, I do not need any log filters for this Connector right now. Eventually if I find that there are Windows Events I do not want to import,

**06:00-07:00**
I can go ahead and add log filters here. Then select **Next**. Next, on the **Configuration** step, the Active Directory domain in my lab is sclabs.ai. The **Server Hostname** is the fully qualified domain name of the server or the IP address. You cannot just put the hostname itself. Now I copied the hostname. This can be uppercase or lowercase–it doesn't matter–SClabs.ai. Now this is the fully qualified domain name for the Active Directory server. The **Interval** is how often the Active Directory Connector obtains user information, computer information, and event logs from Active Directory.

**07:00-08:00**
The **User Name** is a user that I dedicate directly for this purpose, the purpose of having a Connector to obtain information. Now one of the best ways to do this is to go to Active Directory and create yourself an account. Here I go to **Tools** on my Active Directory server, **Active Directory Users and Computers**. I have created a user called Stellar Connector. This is just a domain user. Now in order to collect information from Active Directory, this user only needs to be a domain user. I can see this by going to **Member Of**.

**08:00-09:00**
If I go to **Member Of**, I will see that this user is only a domain user. To create a response action from the Connector to Active Directory, the user needs to have elevated permissions. The user needs to have permissions for read and write account control. I can do that by going to sclabs.ai, which is my organization. I can right-click that and specify **Delegate Control**. From here it brings up a wizard where I can specify the permissions that the user can have. I'll go ahead and select **Users** and **Groups**. I will add and select my Stellar Connector user. I just hit **Return** and when you hit return it'll find it. Then I go ahead and select **Next**.

**09:00-10:00**

At this point I will create a custom task to delegate. I then will only delegate for the following objects in the folder. That will be all the way at the bottom. It'll be the user objects. Then I select **Next**. For the permissions I can remove the general and I will select a property-specific permission and I'm looking for user account control. This will be alphabetical from the second column. I'm going to scroll down nice and neat and find l m n o p q r s t u and find let's find user. There it is right there:

**10:00-11:00**

Read userAccountControl and Write userAccountControl. I'll select **Next**, and that's it. If I scroll down here it says for that domain sclabs.ai. The Stellar Connector user will have the following permissions: Read userAccountControl and Write userAccountControl. For the following object types: User. Perfect. That's all I need. Now back on the platform, I will specify my username, which is stellarconnector. I will enter my secret password. This should be a really good and complex password. The protocol type is LDAP that I will use.

**11:00-12:00**

You could use LDAPS, which is a secure or more secure version of LDAP and kerberos. In my environment I will use LDAP as my Modular Sensor is on the same network as my Active Directory server. Now by default, the Active Directory Connector collects user behavior information and event information. But if I want to also collect computer information that will be in the **Investigate Assets**, then go ahead and select the content type of computers as well. This is really computers and user information. Select **Next**. I will select **Submit**. My Active Directory Connector has been configured, but it's not functioning yet. Notice the status right now.

**12:00-13:00**

It's not available, but what I can do is try to test. This takes about 30 seconds to a minute. I'll wait right here. Here I see, "Test Success!" I see an error. That is okay. That's really just saying there's been no data received in the past 24 hours. Within the first five minutes though this Connector will turn green. Okay. It's been a couple of minutes, and now my status is healthy. If I take a look now at the Connector, I will see that the name of the Connector is here. I see that it's run on the Modular Sensor. It has both collect and respond capabilities.

**13:00-14:00**

It shows me the last activity, which is a minute ago. It also gives me the configuration settings such as the **Active Directory Domain**, and the **Server Hostname** and the time **Interval** for how often it collects and the **User Name** associated with this Connector. The fact that I'm using LDAP and the content type is also pulling users and computers. If I look over here, I can see that here in the legend as well. It's collecting computer information and Active Directory user profiles information. When you look down here, this is letting you know that the Connector is communicating with the Active Directory server and it's obtaining AD computer information and AD user information. These aren't event records. These are really Connector records. Okay. Now I'll go back to **Investigate | Threat Hunting**. What I'll see here is this.

**14:00-15:00**

I have the **Time Interval** set for the last 30 minutes. If I deselect **Alerts**, remember that when we looked at the documentation, the Connector traffic is going to the Windows Events. I see here in the last 30 minutes I now have 15,000 Windows Events I have collected. If I look at one of these events again these aren't Alerts, these are the events themselves. Looking at the details, I can see the event that was collected from the Active Directory server. Over time if I get Alerts, the Alerts also might be associated with the Connector. But if I look over here I also have the username. I've collected some username details. If I go to **Investigate | User Behavior Analytics** I can look at the list of users in the last 30 minutes.

**15:00-16:00**
In this case, it did pull administrator and a system account. If I go back in time a little bit further now because I've actually been using this Active Directory Connector before I installed it, I had it installed previously. It was able to collect more information for users as well. My Connector will do the same thing in a couple of minutes. But if I take a look at my username, I can see that it was able to grab my **Principal Name**, my **Job Title**, my **Company**. I grabbed all the information from my user account, and you'll see the data source is Active Directory. If I go down a little bit further, I'll also see the number of times that it's located **Authentications** and **Alerts**. Records associated with authentications and an Alert over here as well.

**16:00-17:00**
If I look at the Alert, I see that there was an external firewall deny action. Here I can see the Alert. If I go to **Details**, I can get information as to what was involved, and my Macbook was involved, and the pfSense firewall actually collected this information. What's really cool about that is pfSense's firewall has no concept of users, but the Connector to Active Directory was able to obtain my user information and correlate that with this Alert. Along with the enrichment that we have, the geolocation, we have the IP type–private or public–and a whole bunch of other information, this makes it much easier to identify threats or even if I need to disable a user.

**17:00-18:00**
Because this user and this Connector exist, and I have a problem identified such as this right here. If I found that this Alert was serious, I could select the Connector. I can specify **Duration**. Maybe I'm investigating so I only want to disable it for one hour. You have to use the name of the field that had the username. Again, let me go back one more time so this is clear. I'm going to cancel out of this right now. Let's go back to **Details**. Let's look where my username was detected. My username was assigned to source IP username (srcip_username) right here. This is the field. If I go to **Actions**, **Disable User**,

**18:00-19:00**
go to the **Connector, Duration** one hour, **Disable User** will be the source IP username. I started typing that in and it found it. Then I'm going to go ahead and **Submit**. If you look up here it says, "Request to disable user is in progress. Monitor progress in the **Respond | Actions** table." If I go to **User Actions** I'll see the username right here and I'll see that it's in progress. I've done this in the past before. But here I'll wait till it's not in progress anymore. I'll just refresh the screen a couple of times. Still in progress. This is actually using LDAP to disable the user.

**19:00-20:05**
It's doing an LDAP modify from the Modular Sensor to the Connector. I see the user was disabled. Now if I go to Active Directory, I can see that my user has a down arrow, which lets me know this account is disabled. Have no fear, if I want to re-enable this account, I can go to **All Tasks** as an administrator and specify **Enable Account**. Also though, from the platform itself, I can also revert my action here as well. Now the user was reverted successfully, then this account will be re-enabled again. Now going back to my Active Directory server, If I take a look at my user account, Paul Levasseur, there's no longer a down arrow. Therefore, the revert was processed successfully. That's it! I hope you enjoyed the lesson. Thank you.

This lesson is complete. Thank you for watching.

# L28 - MODULAR SENSOR BUFFERING

## L28A-PRES: Modular Sensor Buffering (04m:22s)

**00:00-01:00**
The next lesson is *Buffering*. This lesson, *Buffering*, provides a summary of the buffering capabilities of the Modular Sensor that are used in an event where connectivity is temporarily lost between a Modular Sensor and the Stellar Cyber Open XDR platform and how data is trickled back once the connection is restored.

Hi all! In this lesson, I will just do a recap of the buffering that's done by Modular Sensors. In each of the individual lessons for each part, I discussed the buffering option, but now I just want to put them together in one place as more of a review than anything else.

**01:00-02:00**
Looking at the Modular Sensor with the Log Forwarder, network traffic metadata, and IDS features on the Modular Sensors, buffering is enabled by default to temporarily store Interflow records locally. If there is a connection issue to the Stellar Cyber platform, Modular Sensors temporarily hold the buffered Interflow records. When connections are restored to the Stellar Cyber platform, buffered Interflow records resume being forwarded at a trickle to prevent overwhelming the network connection. Again, just to be clear, the Modular Sensor, when the connection comes back up, will still be doing its ordinary duties. It will be normalizing the logs and network traffic that it's receiving into Interflow records to send to the Stellar Cyber platform. But it also has previous Interflow records buffered.

**02:00-03:00**
Instead of blasting all those at once to the Stellar Cyber platform, it will start trickling in the Interflow records to the Stellar Cyber platform until the buffer has been depleted. The amount of time logs are buffered depends on the configured buffer size. Buffering options are available in the Modular Sensor Profile in the following modules. Number one is the Log Forwarder. In the Modular Sensor Profile for the Log Forwarder, there's an option for buffering. This will save data locally where there's

a connection issue with the DP and send the data later. Again, the DP is called the Data Processor. That's just the term for the Stellar Cyber platform.

**03:00-04:00**
The next option is for Network Traffic. In the Network Traffic section in the Modular Sensor profile. There's an option to save data locally when there is a connection issue with the DP, it sends the data later. Again you see the buffer size. For IDS, again, there's an option for buffering. It saves data locally when there is a connection issue with the DP, and sends the data later. Again, here there is a buffer size. The buffer sizes are related to the amount of traffic that usually comes in Network traffic is typically a lot more. Log traffic is usually less, and then IDS is even less than that. The buffer sizes are related to the type of traffic being received. Just to make it really clear,

**04:00-04:22**
if connectivity is lost to the Stellar Cyber platform, the data will be buffered based on the size specified in the configuration until the connection is restored, and the data from the buffers is trickled back to the Stellar Cyber platform.

This lesson is complete. Thank you for watching.

# L29 - MODULAR SENSOR LESSONS SUMMARY

## L29A-PRES: Modular Sensor Lessons Summary (02m:48s)

**00:00-01:00**
The next lesson is *Modular Sensor Summary.* This lesson, *Modular Sensor Summary,* provides a brief summary of Modular Sensors within the Stellar Cyber Open XDR Architecture and ends the *Stellar Cyber Modular Sensor Concepts and Configuration* module.

Hi everyone. We have now completed the training module for *Modular Sensor Concepts and Configuration.* Looking at the slide once again, the Stellar Cyber Open XDR Architecture Overview, the “Big Picture,” we have now covered everything on the left-hand side related to Modular Sensors. We talked about the Log Forwarder, log filtering, how network traffic is captured from SPAN ports or TAP devices.

**01:00-02:00**
We talked about app identification with deep packet inspection, how it creates session state and correlates processes and how it collects metadata and how to optimize the ingestion with data filtering. We've covered the IDS module on the Modular Sensors along with the Malware Sandbox. I've provided examples of how to configure the Modular Sensors as an Aggregator. I've covered the Tenable Nessus scanner that's built into the Modular Sensor and how to initiate scans from Tenable.io into the local environment using Modular Sensors, as well as I've demonstrated how to use Connectors to Tenable to obtain a scan details and asset information. I've also explained and demonstrated how to create Active Directory Connectors from the Modular Sensor

**02:00-02:47**
that can obtain event logs from Windows as well as computer assets and user information. I've shown you where to configure buffering so that if the connectivity to the Open XDR platform is lost due to maybe an internet outage that the Stellar Cyber Modular Sensor can buffer that data locally, and then when the connection comes back up can start trickling those Interflow records back to the Stellar Cyber Open XDR platform. Thank you for participating in the lessons in this module. I hope to see you in other modules as well. Happy learning!

This lesson is complete. Thank you for watching.